

Chemical Bonding and Molecular Structure

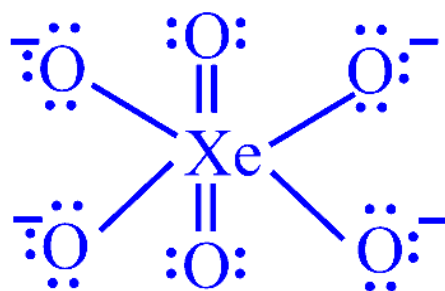
Question1

According to Lewis theory, the total number of σ bond-pairs and lone pair of electrons around the central atom of XeO_6^{4-} ion is ____ .

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Answer: 6

Solution:



In the ion XeO_6^{4-} , xenon (Xe) is the central atom and it is bonded to 6 oxygen atoms.

Each Xe–O bond has one bond pair of electrons, and in Lewis theory each single bond is counted as one σ bond.

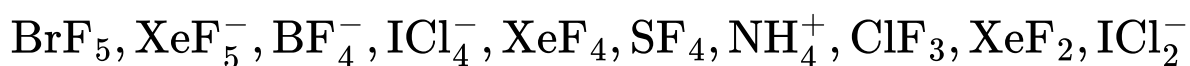
So, the number of σ bond-pairs around Xe is 6.

In this structure, xenon uses all its available electron pairs for bonding with the 6 oxygen atoms, so there is no electron pair left as a lone pair on Xe.

$$\therefore \sigma = 6 \quad \text{Lone pairs} = 0$$

Question2

Consider the following species:

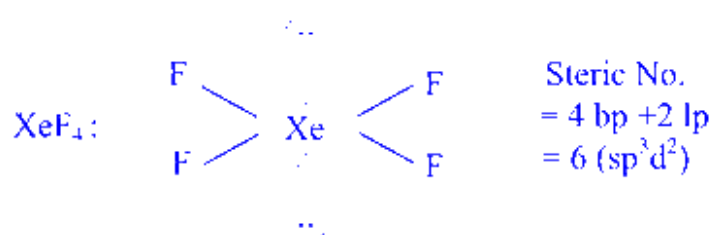
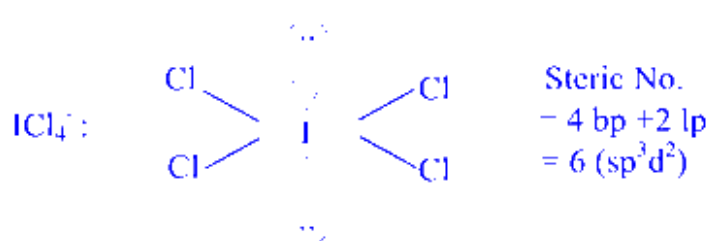
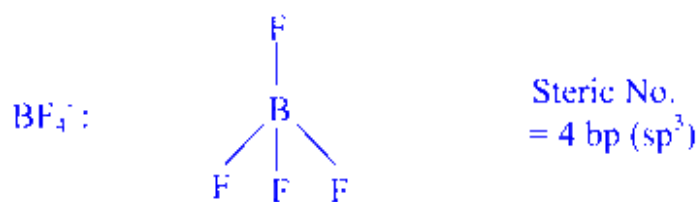
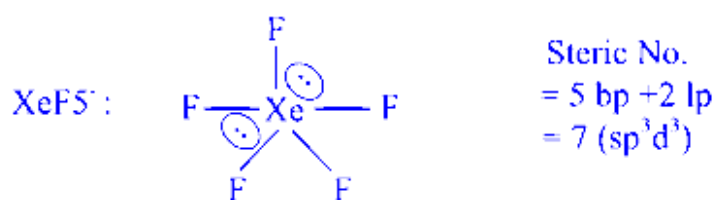
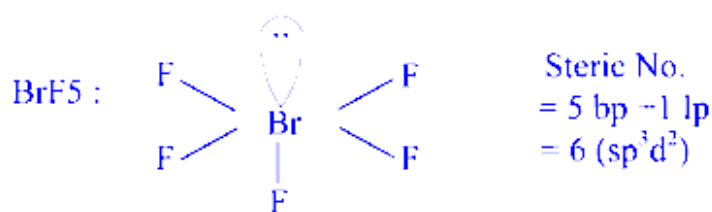


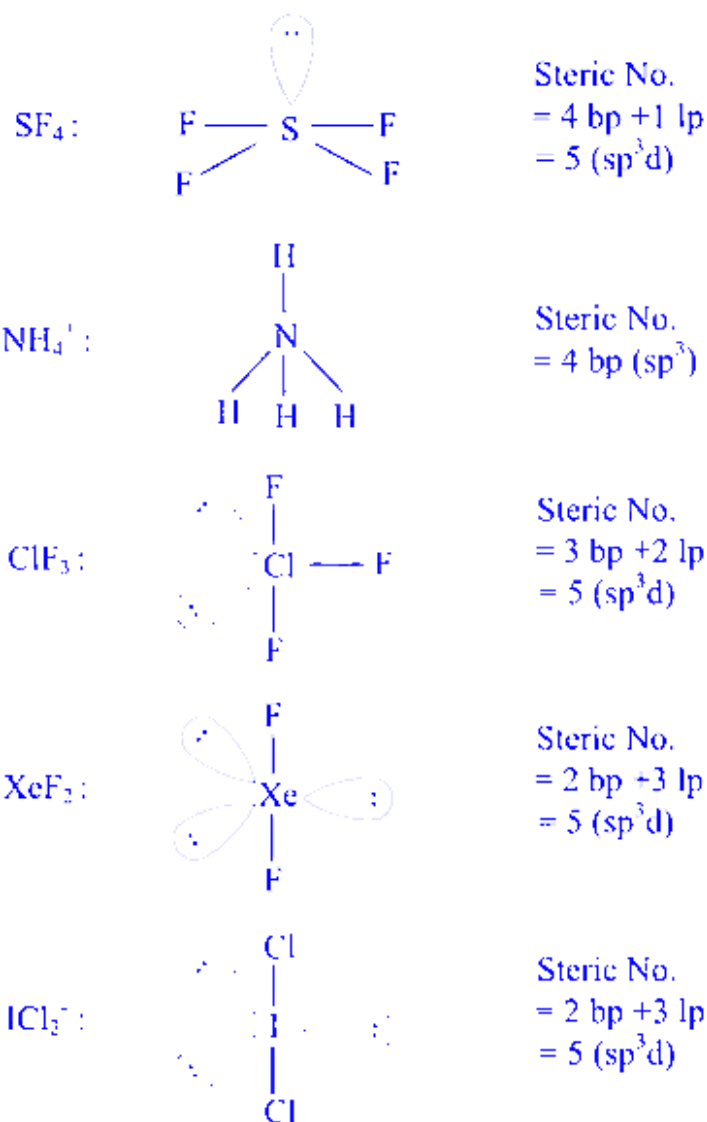
Number of species having $\text{sp}^3 \text{d}$ hybridized central atom is ____ .

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Answer: 4

Solution:





Question3

Given below are two statements :

Statement I :

The number of species among

SF_4 , NH_4^+ , $[\text{NiCl}_4]^{2-}$, XeF_4 , $[\text{PtCl}_4]^{2-}$, SeF_4 and $[\text{Ni}(\text{CN})_4]^{2-}$, that have tetrahedral geometry is 3 .

Statement II :

In the set [NO_2 , BeH_2 , BF_3 , AlCl_3], all the molecules have incomplete octet around central atom.

In the light of the above statements, choose the correct answer from the options given below:

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Options:

A.

Both Statement I and Statement II are true

B.

Statement I is true but Statement II is false

C.

Statement I is false but Statement II is true

D.

Both Statement I and Statement II are false

Answer: C

Solution:

Statement I: Count species with **tetrahedral molecular geometry**:

- $\text{SF}_4 \rightarrow$ seesaw (AX_4E), **not** tetrahedral
- $\text{NH}_4^+ \rightarrow$ **tetrahedral**
- $[\text{NiCl}_4]^{2-}$ (Ni^{2+} , d^8 with weak-field Cl^-) $\rightarrow$ **tetrahedral**
- $\text{XeF}_4 \rightarrow$ square planar, **not** tetrahedral
- $[\text{PtCl}_4]^{2-}$ (Pt^{2+} , d^8) $\rightarrow$ square planar, **not** tetrahedral
- $\text{SeF}_4 \rightarrow$ seesaw (AX_4E), **not** tetrahedral
- $[\text{Ni}(\text{CN})_4]^{2-}$ (Ni^{2+} , d^8 with strong-field CN^-) $\rightarrow$ square planar, **not** tetrahedral

So tetrahedral = **2**, not 3 $\Rightarrow$ **Statement I is false.**

Statement II: Check incomplete octet around central atom:

- NO_2 : odd-electron molecule; N does **not** complete octet
- BeH_2 : Be has only 4 electrons around it
- BF_3 : B has only 6 electrons

- AlCl_3 : Al has only 6 electrons in monomeric form (though it dimerizes to Al_2Cl_6 to complete octet)

Thus the statement is taken as **true** $\Rightarrow$ **Statement II is true.**

Correct option: C — Statement I is false but Statement II is true.

Question4

Given below are two statements :

Statement I : The correct order in terms of bond dissociation enthalpy is $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$.

Statement II : The correct trend in the covalent character of the metal halides is $[\text{SnCl}_4 > \text{SnCl}_2]$, $[\text{PbCl}_4 > \text{PbCl}_2]$ and $[\text{UF}_4 > \text{UF}_6]$.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both **Statement I** and **Statement II** are true

B.

Both **Statement I** and **Statement II** are false

C.

Statement I is false but **Statement II** is true

D.

Statement I is true but **Statement II** is false

Answer: D

Solution:

Statement I: True

Bond dissociation enthalpy generally decreases down the group because bond length increases (bond becomes weaker). But F_2 is an exception: its $F - F$ bond is **unusually weak** due to strong **lone pair–lone pair repulsions** on the small fluorine atoms. Hence,

$$D(\text{Cl}_2) > D(\text{Br}_2) > D(\text{F}_2) > D(\text{I}_2)$$

So **Statement I is correct.**

Statement II: False

Using **Fajans' rule (NCERT)**: covalent character increases when the cation has **higher charge and smaller size** (greater polarising power).

- SnCl_4 vs SnCl_2 : Sn^{4+} has higher charge and smaller size than Sn^{2+}
 $\Rightarrow \text{SnCl}_4$ is more covalent than SnCl_2 (True)
- PbCl_4 vs PbCl_2 : similarly, Pb^{4+} polarises more than Pb^{2+}
 $\Rightarrow \text{PbCl}_4 > \text{PbCl}_2$ in covalent character (True)
- UF_4 vs UF_6 : here U^{6+} has **higher charge** than U^{4+} , so it has **greater polarising power**
 $\Rightarrow \text{UF}_6$ should be more covalent than UF_4

But the statement claims $\text{UF}_4 > \text{UF}_6$, which is **incorrect**.

So **Statement II is false.**

Correct Option: D

Statement I is true but Statement II is false

Question 5

The correct increasing order of C–H(A), C–O(B), C=O(C) and C≡N(D) bonds in terms of covalent bond length is :

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Options:

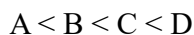
A.

$D < C < B < A$

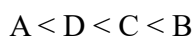
B.

$D < C < A < B$

C.



D.



Answer: D

Solution:

Bond length is **inversely related to bond order** (as per NCERT idea):

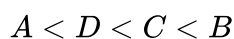
triple bond < double bond < single bond (shorter to longer).

Also, for **single bonds**, bond length increases when the bonded atom is larger (so $C-O$ is longer than $C-H$).

Now compare:

- $C-H$ (A): single bond, but H is very small $\Rightarrow$ **short**
- $C \equiv N$ (D): **triple bond** $\Rightarrow$ **very short**
- $C=O$ (C): **double bond** $\Rightarrow$ short, but longer than triple
- $C-O$ (B): **single bond** with larger O $\Rightarrow$ **longest**

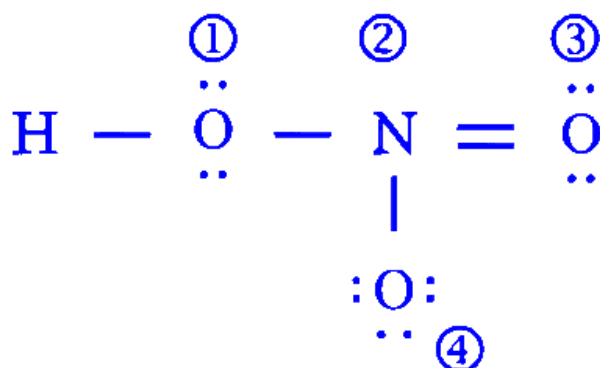
So increasing order of bond length is:



Correct option: D

Question6

The formal charges on the atoms marked as (1) to (4) in the Lewis representation of HNO_3 molecule respectively are



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Options:

A.

$0, 0, -1, +1$

B.

$0, -1, 0, +1$

C.

$0, +1, 0, -1$

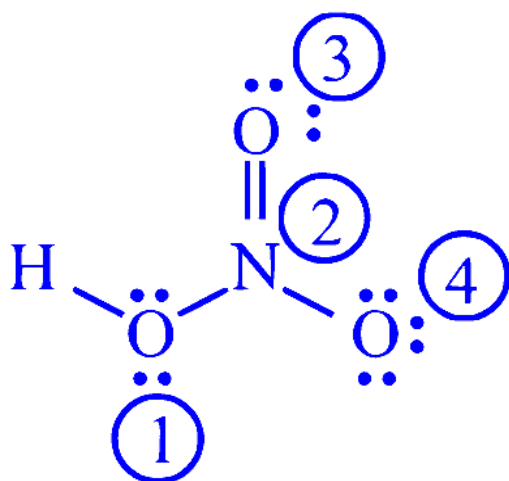
D.

$+1, 0, 0, -1$

Answer: C

Solution:

Consider the structure of HNO_3



1 : -0

2 : $-(+1)$

3 : -0

4 : $-(-1)$

$$\text{Formal charge} = \text{valence e's} - \text{non bonding e's} - \left(\frac{\text{bonding electrons}}{2} \right)$$

Question 7

Two p-block elements X and Y form fluorides of the type EF_3 . The fluoride compound XF_3 is a Lewis acid and YF_3 is a Lewis base. The hybridizations of the central atoms of XF_3 and YF_3 respectively are

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Options:

A.

sp^2 and sp^3

B.

Both sp^3

C.

Both sp^2

D.

sp^3 and sp^2

Answer: A

Solution:

XF_3 acts as a Lewis acid, so it must have an incomplete octet (it can accept an electron pair).

A common p-block fluoride with formula EF_3 that is a Lewis acid is BF_3 .

In BF_3 , boron forms three σ bonds with fluorine and has no lone pair, so it has only 6 electrons around it. Therefore, it is electron-deficient and behaves as a Lewis acid.

With three bond pairs around boron, the hybridization is sp^2 .

YF_3 acts as a Lewis base, so it must have a lone pair to donate.

A common p-block fluoride with formula EF_3 that is a Lewis base is NF_3 .

In NF_3 , nitrogen forms three σ bonds with fluorine and has one lone pair, so it can donate that lone pair and behave as a Lewis base.

With four electron pairs (3 bond pairs + 1 lone pair) around nitrogen, the hybridization is sp^3 .

So, $\text{XF}_3 = \text{BF}_3$ has hybridization sp^2 and $\text{YF}_3 = \text{NF}_3$ has hybridization sp^3 .

$\text{XF}_3 = \text{BF}_3; \text{sp}^2$

$\text{YF}_3 = \text{NF}_3; \text{sp}^3$

Question8

Among H_2S , H_2O , NF_3 , NH_3 and CHCl_3 , identify the molecule (X) with lowest dipole moment value. The number of lone pairs of electrons present on the central atom of the molecule (X) is :

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Options:

A.

3

B.

2

C.

0

D.

1

Answer: D

Solution:

Molecule	Dipole moment
H ₂ S	0.95
H ₂ O	1.85
NF ₃	0.23 (minimum)
NH ₃	1.47
CHCl ₃	1.04

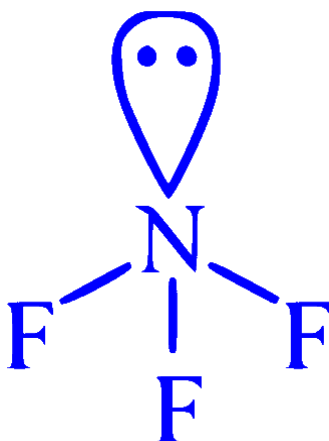
From the given values, the dipole moment is the smallest for NF₃ (it is 0.23), so molecule (X) is NF₃.

In NF₃, the central atom is nitrogen (N).

Nitrogen has 5 valence electrons. It makes 3 N–F bonds, so 3 electrons are used for bonding.

The remaining 2 valence electrons stay on nitrogen as one lone pair.

So, the number of lone pairs on the central atom in (X) = NF₃ is 1.



Number of lone pair on central atom = 1

Question9

Identify the molecule (X) with maximum number of lone pairs of electrons (obtained using Lewis dot structure) among HNO₃, H₂SO₄, NF₃ and O₃. Choose the correct bond angle made by the central atom of the molecule (X).

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Options:

A.

107°

B.

102°

C.

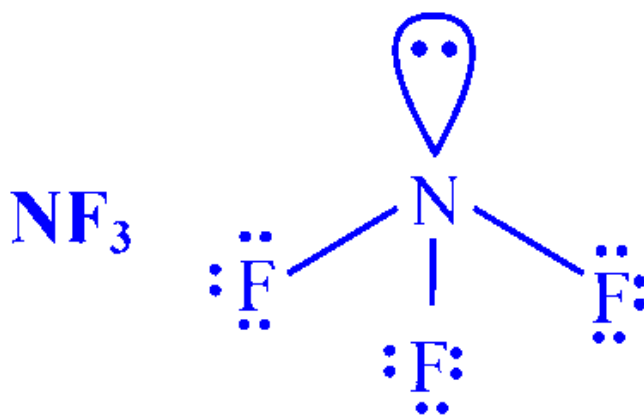
120°

D.

116°

Answer: B

Solution:



Hybridization : sp^3

Number of lone pair in $NF_3 = 10$

Bond angle in $NF_3 \approx 102^\circ$

Question10

Which statements are NOT TRUE about $XeO_2 F_2$?

A. It has a see-saw shape.

B. Xe has 5 electron pairs in its valence shell in XeO_2F_2 .

C. The $\text{O} - \text{Xe} - \text{O}$ bond angle is close to 180° .

D. The $\text{F} - \text{Xe} - \text{F}$ bond angle is close to 180° .

E. Xe has 16 valence electrons in XeO_2F_2 .

Choose the *correct* answer from the options given below :

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Options:

A.

B, D and E Only

B.

B and D Only

C.

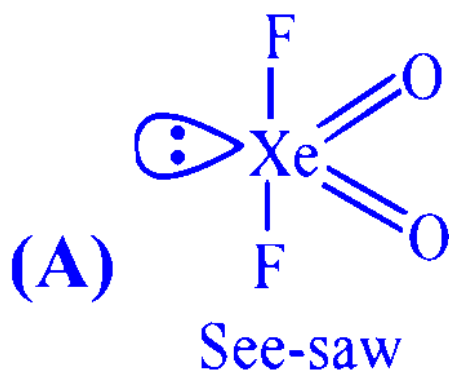
B, C and E Only

D.

A and D Only

Answer: C

Solution:



(B) This statement is not true because Xe has 7 electron pairs in its valence shell in this molecule.

(C) This statement is not true because the $O - Xe - O$ bond angle is close to 120° , not 180° .

(D) This statement is true because the $F - Xe - F$ bond angle is close to 180° .

(E) This statement is not true because Xe has 14 valence electrons in XeO_2F_2 , not 16.

Question11

Among the following, the **CORRECT** combinations are

A. $IF_3 \rightarrow$ T-shaped ($sp^3 d$)

B. $IF_5 \rightarrow$ Square pyramidal ($sp^3 d^2$)

C. $IF_7 \rightarrow$ Pentagonal bipyramidal ($sp^3 d^3$)

D. $ClO_4^- \rightarrow$ Square planar ($sp^2 d$)

Choose the correct answer from the options given below:

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Options:

A.

B, C and D Only

B.

A, B, C and D

C.

A, B and C Only

D.

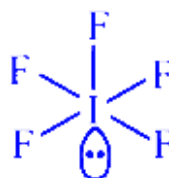
A and B Only

Answer: C

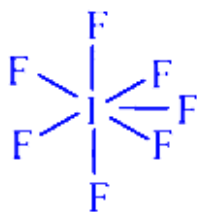
Solution:



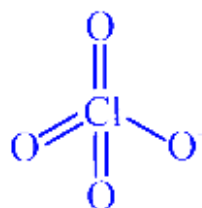
Hybridisation sp^3d
(T-shape)



Hybridisation sp^3d^2
(Square pyramidal)



Hybridisation sp^3d^3
(Pentagonal bipyramidal)



Hybridisation sp^3
(Tetrahedral)

Question12

Given below are statements about some molecules/ions.

Identify the **CORRECT** statements.

- A. The dipole moment value of NF_3 is higher than that of NH_3 .
- B. The dipole moment value of BeH_2 is zero.
- C. The bond order of O_2^{2-} and F_2 is same.
- D. The formal charge on the central oxygen atom of ozone is -1 .
- E. In NO_2 , all the three atoms satisfy the octet rule, hence it is very stable.

Choose the correct answer from the options given below:

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Options:

A.

B & C Only

B.

B, C & D Only

C.

A, C & D Only

D.

A, B, C, D & E

Answer: A

Solution:

Check each statement:

A. $\mu(\text{NF}_3) > \mu(\text{NH}_3)$

Both are trigonal pyramidal. In NH_3 , the bond dipoles and lone pair dipole add to give a **larger** net dipole moment. In NF_3 , the N – F bond dipoles oppose the lone pair direction, so net dipole is **small**.

So, $\mu(\text{NF}_3) < \mu(\text{NH}_3)$.

A is incorrect.

B. Dipole moment of BeH_2 is zero

BeH_2 is **linear**: $\text{H} - \text{Be} - \text{H}$ (NCERT: sp hybridised). The two $\text{Be} - \text{H}$ bond dipoles are equal and opposite, hence cancel.

So, $\mu = 0$.

B is correct.

C. Bond order of O_2^{2-} and F_2 is same

Using MO theory:

- O_2^{2-} has 14 valence electrons (like F_2). For peroxide ion, bond order becomes

$$\text{B.O.} = 1$$

- For F_2 , bond order is also

$$\text{B.O.} = 1$$

So they are the same.

C is correct.

D. Formal charge on central oxygen of ozone is -1

In one resonance form: $\text{O} = \text{O} - \text{O}$ (central O has one lone pair, one double bond, one single bond)

Formal charge on central O:

$$\text{F.C.} = 6 - \left(2 + \frac{6}{2}\right) = 6 - (2 + 3) = +1$$

So it is $+1$, not -1 .

D is incorrect.

E. In NO_2 , all atoms satisfy octet rule, hence very stable

NO_2 has an **odd number of electrons** (17 valence electrons), so it is an **odd-electron molecule** and nitrogen does **not** complete its octet. Hence it is quite reactive (dimerises to N_2O_4).

E is incorrect.

Correct statements: B and C only

☒ Option A

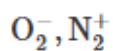
Question13

Pair of species among the following having same bond order as well as paramagnetic character will be-

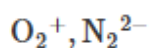
JEE Main 2026 (Online) 24th January Evening Shift

Options:

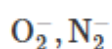
A.



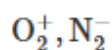
B.



C.



D.



Answer: D

Solution:

Species	Bond order	Magnetic Nature
O_2^+	2.5	Paramagnetic
O_2^-	1.5	Paramagnetic
O_2^+	2.5	Paramagnetic
N_2^-	2.5	Paramagnetic
N_2^{2-}	2	Paramagnetic

Question14

Given below are two statements :

Statement I : The number of species among BF_4^- , SiF_4 , XeF_4 and SF_4 , that have unequal E — F bond lengths is two. Here, E is the central atom.

Statement II : Among O_2^- , O_2^{2-} , F_2 and O_2^+ , O_2^- has the highest bond order.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Both Statement I and Statement II are false

B.

Statement I is true but Statement II is false

C.

Statement I is false but Statement II is true

D.

Both Statement I and Statement II are true

Answer: A

Solution:

Statement I

1. BF_4^-

Central atom B has 4 bond pairs and 0 lone pairs $\Rightarrow$ tetrahedral.

All four B – F bonds are equivalent $\Rightarrow$ **equal bond lengths.**

2. SiF_4

Central atom Si has 4 bond pairs and 0 lone pairs $\Rightarrow$ tetrahedral.

All four Si – F bonds equivalent $\Rightarrow$ **equal bond lengths.**

3. XeF_4

Xe has 4 bond pairs and 2 lone pairs (AX_4E_2) $\Rightarrow$ square planar.

All four Xe – F bonds equivalent by symmetry $\Rightarrow$ **equal bond lengths.**

4. SF₄

S has 4 bond pairs and 1 lone pair (AX₄E) ⇒ see-saw shape.

There are **two axial** and **two equatorial** S – F bonds, and they are not equivalent (axial bonds are longer).

⇒ **unequal bond lengths**.

So, number of species with unequal E – F bond lengths = 1 (only SF₄), **not 2**.

⇒ **Statement I is false**.

Statement II

Bond order (NCERT MO theory):

$$\text{Bond order} = \frac{N_b - N_a}{2}$$

Using known MO results for these diatomic species:

- O₂⁺ has bond order 2.5
- O₂ has bond order 2
- O₂⁻ has bond order 1.5
- O₂²⁻ has bond order 1
- F₂ has bond order 1

Highest bond order among O₂⁻, O₂²⁻, F₂, O₂⁺ is O₂⁺, not O₂⁻.

⇒ **Statement II is false**.

Correct Option

Both Statement I and Statement II are false.

Option A

Question 15

Match List - I with List - II according to shape.

List – I	List – II
A. XeO ₃	I. BrF ₅
B. XeF ₂	II. NH ₃
C. XeO ₂ F ₂	III. [I ₃] ⁻
D. XeOF ₄	IV. SF ₄

Choose the correct answer from the options given below :

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Options:

A.

A-II, B-III, C-I, D-IV

B.

A-III, B-II, C-IV, D-I

C.

A-II, B-III, C-IV, D-I

D.

A-II, B-I, C-III, D-IV

Answer: C

Solution:

XeF_2 and I_3^- have 2 bond pairs and 3 lone pairs on the central atom (total 5 electron pairs). In this case, the molecular shape becomes **linear**.

XeOF_4 and BrF_5 have 5 bond pairs and 1 lone pair on the central atom (total 6 electron pairs). This gives a **square pyramidal** shape.

XeO_2F_2 and SF_4 have 4 bond pairs and 1 lone pair on the central atom (total 5 electron pairs). This leads to a **see-saw** shape.

XeO_3 and NH_3 have 3 bond pairs and 1 lone pair on the central atom (total 4 electron pairs). So, the shape is **pyramidal**.

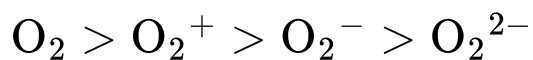
Question16

Given below are two statements :

Statement (I): The correct sequence of bond lengths in the following species is :



Statement (II): The correct sequence of number of unpaired electrons in the following species is :



In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2026 (Online) 2nd April Morning Shift

Options:

A.

Both **Statement I** and **Statement II** are true

B.

Both **Statement I** and **Statement II** are false

C.

Statement I is true but **Statement II** is false

D.

Statement I is false but **Statement II** is true

Answer: C

Solution:

Note :

(1) Bond strength $\propto$ Bond order

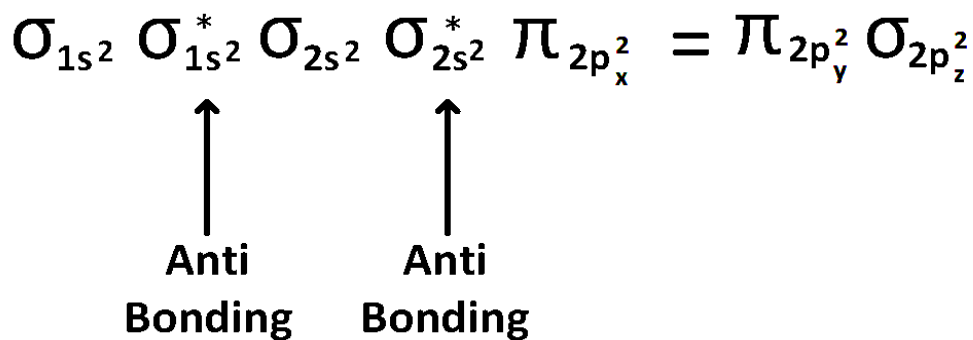
(2) Bond length $\propto \frac{1}{\text{Bond order}}$

$$(3) \text{ Bond order} = \frac{1}{2} [N_b - N_a]$$

N_b = No of electrons in bonding molecular orbital

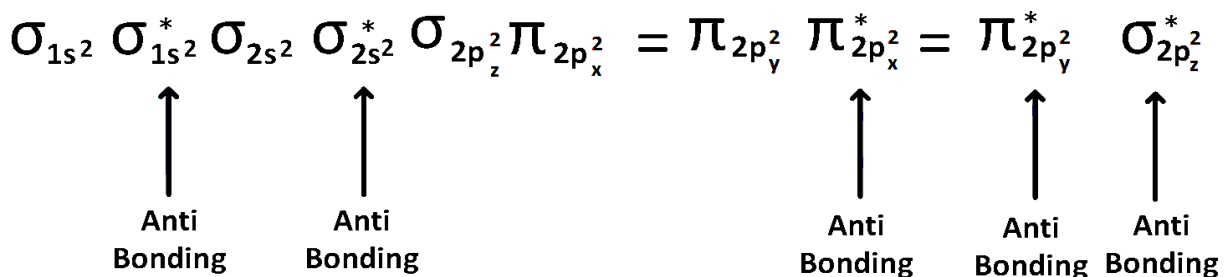
N_a = No of electrons in anti bonding molecular orbital

(4) upto 14 electrons, molecular orbital configuration is



Here N_a = Anti bonding electron = 4 and N_b = 10

(5) After 14 electrons to 20 electrons molecular orbital configuration is - - -



Here $N_a = 10$

and $N_b = 10$

In O atom 8 electrons present, so in O_2 , $8 \times 2 = 16$ electrons present.

Then in O_2^+ no of electrons = 15

in O_2^- no of electrons = 17

in O_2^{2-} no of electrons = 18

$\therefore$ Molecular orbital configuration of O_2 (16 electrons) is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^1}^* = \pi_{2p_y^1}^*$$

$$\therefore N_a = 6$$

$$N_b = 10$$

$$\therefore BO = \frac{1}{2} [10 - 6] = 2$$

Here 2 unpaired electron present.

Molecular orbital configuration of O_2^+ (15 electrons) is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^1}^* = \pi_{2p_y^0}^*$$

$$\therefore N_b = 10$$

$$N_a = 5$$

$$\therefore \text{BO} = \frac{1}{2} [10 - 5] = 2.5$$

Here 1 unpaired electron present.

Molecular orbital configuration of O_2^- (17 electrons) is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^2}^* = \pi_{2p_y^1}^*$$

$$\therefore N_b = 10$$

$$N_a = 7$$

$$\therefore \text{BO} = \frac{1}{2} [10 - 7] = 1.5$$

Here 1 unpaired electron present.

Molecular orbital configuration of O_2^{2-} (18 electrons) is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^2}^* = \pi_{2p_y^2}^*$$

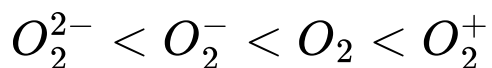
$$\therefore N_b = 10$$

$$N_a = 8$$

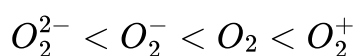
$$\therefore \text{BO} = \frac{1}{2} [10 - 8] = 1$$

Here 0 unpaired electron present.

As Bond strength $\propto$ Bond order so, correct order is



So, Correct order of Bond Order is

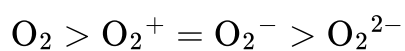


Bond length $\propto \frac{1}{\text{Bond order}}$

So, Correct order of Bond Length is



$\therefore$ The correct sequence of number of unpaired electrons is :



Question17

SF₄ is isostructural with :



Choose the correct answer from the options given below :

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Options:

A.

C Only

B.

C and E Only

C.

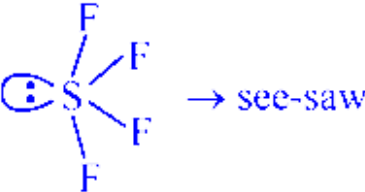
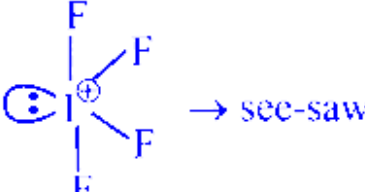
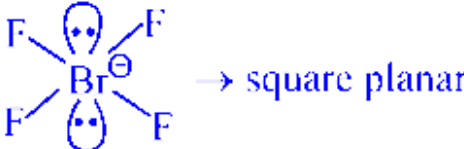
A and D Only

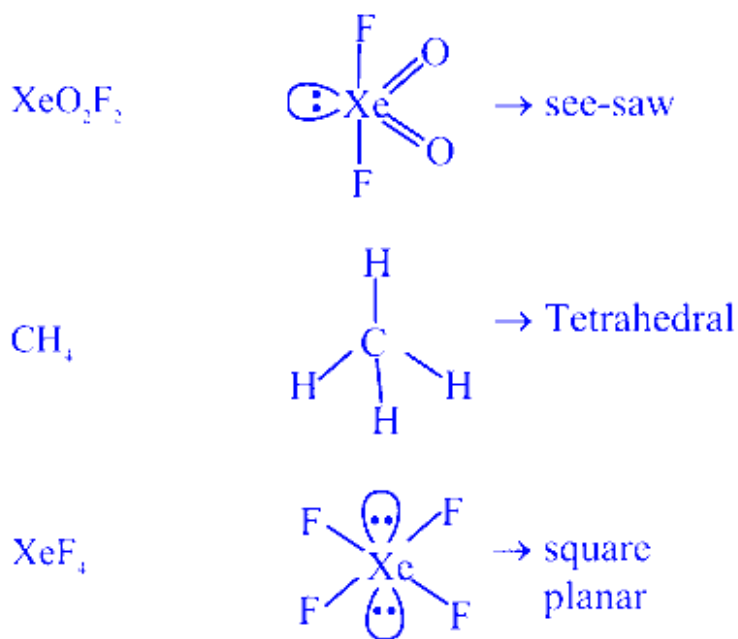
D.

B and E Only

Answer: B

Solution:

Species	Structure / Shape
SF_4	 $\rightarrow$ see-saw
IF_4^+	 $\rightarrow$ see-saw
BrF_4^-	 $\rightarrow$ square planar



Question18

Which of the following pictorial diagram most correctly represents the π^* (π antibonding) molecular orbital between two atoms if the internuclear axis is taken to be in the z -direction ($\xrightarrow{\text{z-axis}}$) ?

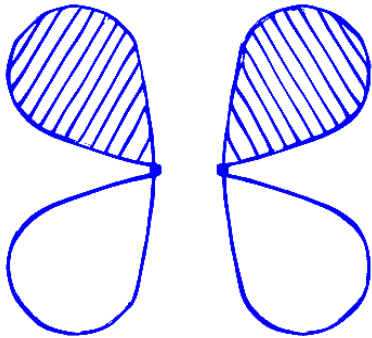
JEE Main 2026 (Online) 4th April Evening Shift

Options:

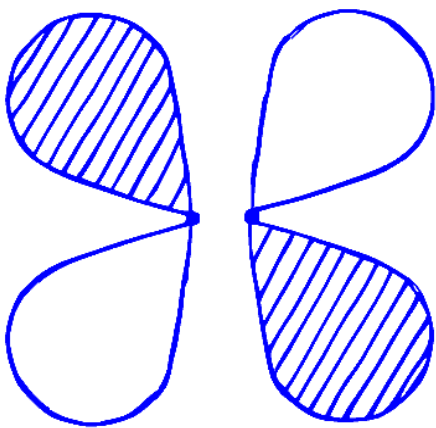
A.



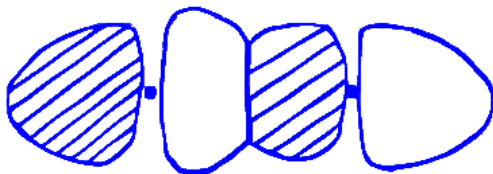
B.



C.

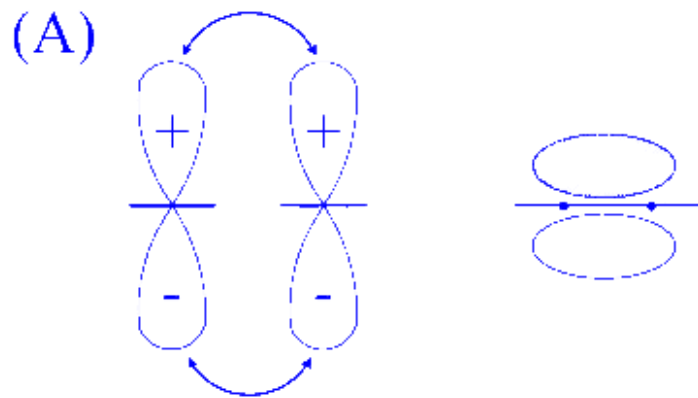


D.

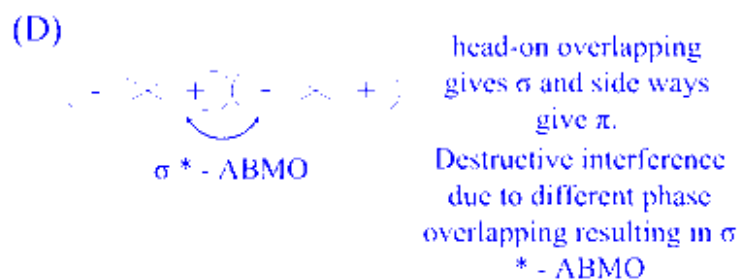
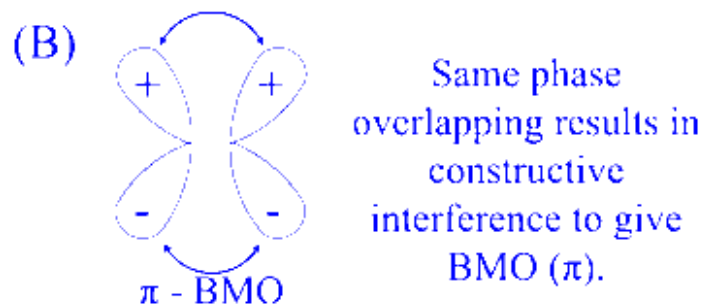


Answer: C

Solution:



constructive Interference due to same phase overlapping gives $\pi - BMO$.



Question19

The covalent radii of atoms A and B are r_A and r_B , respectively. The covalent bond length and total length of AB molecule are respectively

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Options:

A.

$$(r_A + r_B), 2(r_A + r_B)$$

B.

$$\frac{1}{2}(r_A + r_B), (r_A + r_B)$$

C.

$$(r_A + r_B), (r_A + r_B)$$

D.

$$2(r_A + r_B), \frac{1}{2}(r_A + r_B)$$

Answer: A

Solution:

1. Finding the Covalent Bond Length:

In chemistry, the covalent bond length is defined as the average distance between the centers of the nuclei of two bonded atoms in a molecule.

When atoms A and B form a single covalent bond, we can approximate the distance between their nuclei by simply adding their individual covalent radii.

Therefore, if the covalent radius of atom A is r_A and the covalent radius of atom B is r_B , the internuclear distance, which is the bond length of the AB molecule, is:

$$\text{Bond Length} = r_A + r_B$$

2. Finding the Total Length of the AB Molecule:

To find the total length of the entire molecule, we need to consider the full physical extent of both atoms. Imagine the two atoms as spheres placed side by side on a straight axis.

- The center of atom A to its outermost left edge is equal to its radius, r_A .
- The distance from the center of atom A to the center of atom B is the bond length, which we already established is $(r_A + r_B)$.
- The center of atom B to its outermost right edge is equal to its radius, r_B .

The total length of the molecule is the distance from the outermost left edge of atom A to the outermost right edge of atom B . We can find this by adding the three segments together:

$$\text{Total Length} = (\text{radius of } A) + (\text{internuclear distance}) + (\text{radius of } B)$$

$$\text{Total Length} = r_A + (r_A + r_B) + r_B$$

$$\text{Total Length} = 2r_A + 2r_B$$

$$\text{Total Length} = 2(r_A + r_B)$$

So, the covalent bond length is $(r_A + r_B)$ and the total length of the full AB molecule is $2(r_A + r_B)$.

Comparing our findings with the given options, this corresponds perfectly to Option A.

Correct Answer:

Option A

$$(r_A + r_B), 2(r_A + r_B)$$

Question 20

The pairs among

$A = [\text{SO}_3^{2-}, \text{CO}_3^{2-}]$, $B = [\text{O}_2^{2-}, \text{F}_2]$, $C = [\text{CN}^-, \text{CO}]$, $D = [\text{NH}_3, \text{H}_3\text{O}^+]$
and $E = [\text{MnO}_4^{2-}, \text{CrO}_4^{2-}]$ that do not have similar Lewis dot structure are

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Options:

A.

A, B and E

B.

A and E

C.

B, C and D

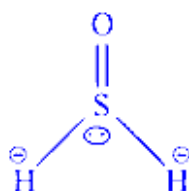
D.

C and D

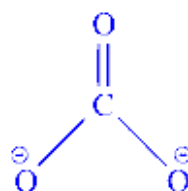
Answer: B

Solution:

(A)



S does not follow octet rule.



C follow octet rule.

(B)



O follow octet rule.



F follow octet rule.

(C)

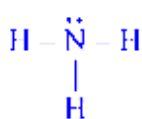


C, N follows octet rule

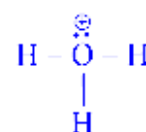


C, O follows octet rule

(D)



N follows octet rule.



O follows octet rule.

$$(E) \text{MnO}_4^{2-} = 7 + 4 \times 2 + 2 = 4 + 10 = 17e^-$$

$$\text{CrO}_4^{2-} = 6 + 4 \times 2 + 2 = 6 + 10 = 16e^-$$

since e^- count is not equal, Lewis dot structure will be different.

Question 21

Given below are two statements :

Statement I : $\text{F}_2\text{O} < \text{H}_2\text{O} < \text{Cl}_2\text{O}$ is the correct trend in terms of bond angle.

Statement II : SiF_4 , SnF_4 and PbF_4 are ionic in nature.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

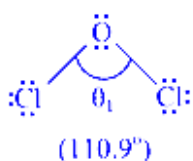
Statement I is true but Statement II is false

D.

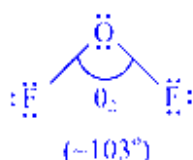
Statement I is false but Statement II is true

Answer: C

Solution:

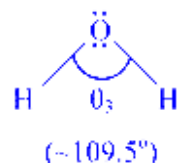


O is 2nd period element and Cl is a 3rd period element. Due to lp-lp repulsion of Cl atoms, the bond angle increases and don't show ideal tetrahedral angle. This is known as NBEPR. (Non-bonding electron pair repulsion)



Here, NBEPR does not exist as F is a 2nd group element. F is a more electronegative element which has more p % character according to Bent's rule.

$$\%p \text{ Character} \propto \frac{1}{\%s \text{ character}} \propto \frac{1}{\theta}$$



H is less electronegative than F and % s character is more. % s-character $\propto \theta$.

Order: $\theta_1 > \theta_3 > \theta_2$

SiF_4 is covalent while SnF_4 and PbF_4 are ionic due to large electronegative difference between them.

Question22

Given below are two statements :

Statement I : The number of compounds among SO_2 , SO_3 , SF_4 , SF_6 and H_2S in which sulphur does not obey the Octet rule is 3 .

Statement II : Among $[H_2O, ClF_3, SF_4]$, $[NH_3, BrF_5, SF_4]$, $[BrF_5, ClF_3, XeF_4]$ and $[XeF_4, ClF_3, H_2O]$, the number of sets in which all the molecules have one lone pair of electrons on the central atom is 1 .

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

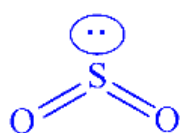
Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: D

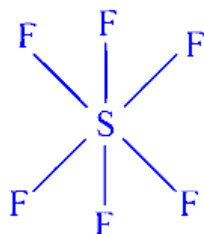
Solution:



Total e^- surrounding S
= 2 X $4e^-$ from O + 1 lp
= $10e^-$



Total e^- surrounding S
= 3 X $4e^-$ from O
= $12e^-$

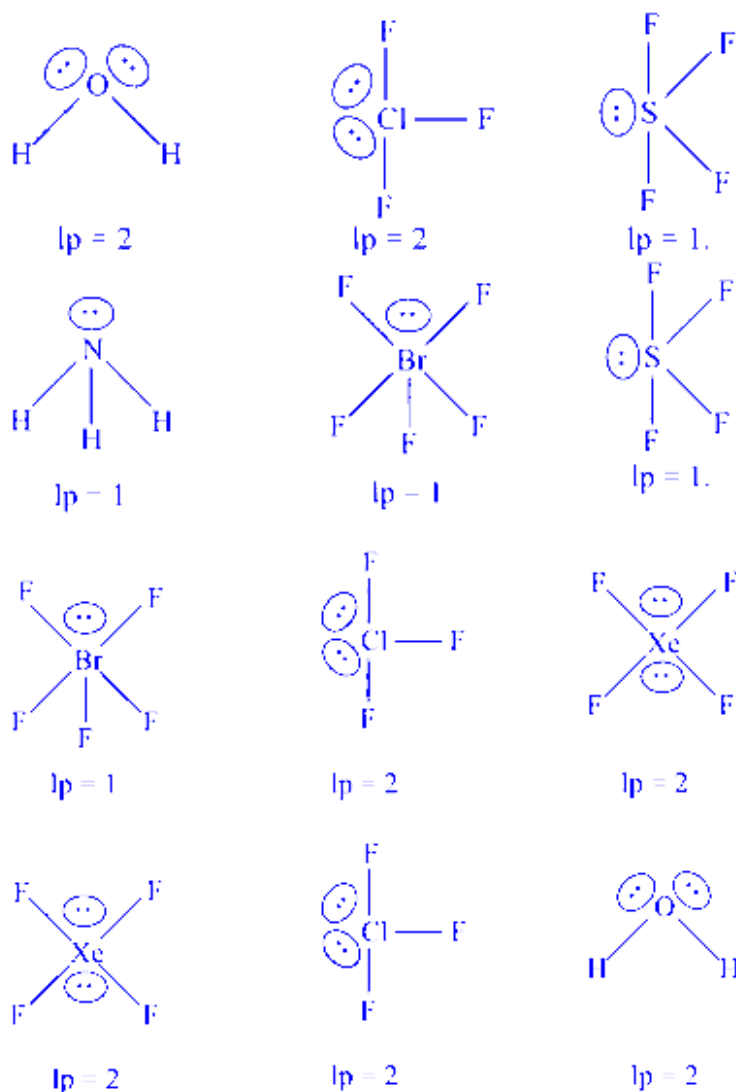


Total e^- surrounding S
= 2 X $6e^-$ from F
= $12e^-$



Total e^- surrounding S
= 2 X $2e^-$ from H + 2 lp
= $8e^-$

Hence, 4 compounds does not follow octet rule for sulphur.



Only 1 set has all molecules with one lp.

Question23

Bromine trifluoride autoionizes to form BrF_2^+ and BrF_4^- . The shapes of the cation and anion are respectively _____ and _____.

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Options:

A.

bent, square planar

B.

linear, square planar

C.

bent, see-saw

D.

linear, tetrahedral

Answer: A

Solution:

First, let us recall the autoionization of bromine trifluoride:



We have to determine the shapes of the cation and the anion.

- Central atom: Br
- Valence electrons of Br = 7
- Each F contributes one bond $\rightarrow$ 2 Br–F bonds = 2 pairs of bonding electrons

So, total electrons around Br for shape determination:

$$7 - 2(\text{by bonding}) - 1(\text{positive charge}) = 4$$

These 4 electrons form **2 lone pairs** on Br.

Hence, the total number of electron pairs around Br =

$$2 (\text{bonding}) + 2 (\text{lone pairs}) = 4 \text{ electron domains.}$$

According to VSEPR theory:

- 4 electron pairs $\rightarrow$ tetrahedral electron geometry
- With 2 bonding and 2 lone pairs $\rightarrow$ **bent shape**

Thus, BrF_2^+ is **bent**.

- Central atom: Br
- Valence electrons of Br = 7
- Each F contributes one bond $\rightarrow$ 4 Br–F bonds = 4 pairs of bonding electrons
- Additional one electron due to negative charge $\rightarrow +1$

Hence total electrons on central bromine:

$$7 + 1 - 4 = 4$$

which means **2 lone pairs** on Br.

Thus, the total number of electron pairs =

$$4 (\text{bonding}) + 2 (\text{lone pairs}) = 6 \text{ electron domains.}$$

According to VSEPR theory:

- 6 electron domains $\rightarrow$ octahedral electron geometry
- With 4 bonding and 2 lone pairs $\rightarrow$ **square planar shape**

Shapes are:

- $\text{BrF}_2^+ \rightarrow$ **bent**
- $\text{BrF}_4^- \rightarrow$ **square planar**

Correct Option: A) bent, square planar

Question24

The number of molecules/ions that show linear geometry among the following is _____.

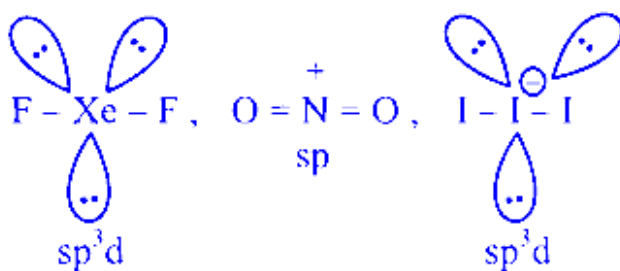
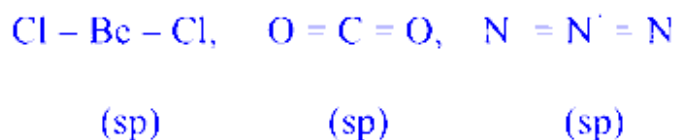
$\text{SO}_2, \text{BeCl}_2, \text{CO}_2, \text{N}_3^-, \text{NO}_2, \text{F}_2\text{O}, \text{XeF}_2, \text{NO}_2^+, \text{I}_3^-, \text{O}_3$

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Answer: 6

Solution:

Linear species are



Question25

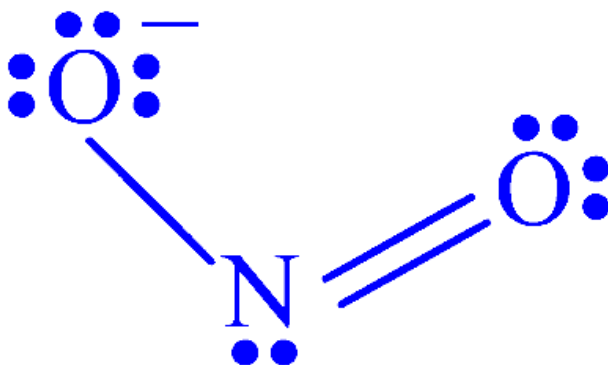
Total number of non-bonded electrons present in NO_2^- ion based on Lewis theory is _____.

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Answer: 12

Solution:

NO_2^- ion



Structure (Lewis Structure)

Nitrogen (N) has 2 non-bonding electrons. The total valence electrons of N is 5. Out of 5, 3 electrons are bonding electrons and 2 electrons are non-bonding electrons.

Double bonded oxygen has 4 non-bonding electrons. Oxygen valence electrons : 6

Out of 6, 2 are bonding electrons and 4 are non-bonding electrons.

Negatively charged oxygen has 6 non-bonding electrons. The negative charge is due to the extra electron present. So, the total number of non-bonded electrons present in NO_2^- ion based on Lewis theory is

$$2(\text{of N}) + 4(\text{of O}) + 6(\text{of O}^-)$$

$$= 2 + 4 + 6 = 12$$

Question26

Arrange the following compounds in increasing order of their dipole moment :

HBr, H₂ S, NF₃ and CHCl₃

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Options:

A. HBr < H₂ S < NF₃ < CHCl₃

B. H₂ S < HBr < NF₃ < CHCl₃

C. NF₃ < HBr < H₂ S < CHCl₃

D. CHCl₃ < NF₃ < HBr < H₂S

Answer: C

Solution:

We need to compare the experimental (or well-established) dipole moments of the given molecules: HBr, H₂S, NF₃, and CHCl₃.

1. NF₃

Structure: Trigonal pyramidal (like NH₃), but each bond is more polar toward fluorine.

Net dipole moment: Quite small, about 0.23–0.24 *D* (the bond dipoles partly oppose the lone-pair contribution).

2. HBr

Structure: Simple diatomic.

Dipole moment: About 0.78 *D*.

3. H₂S

Structure: Bent (like H₂O) but S is less electronegative than O, and the S–H bond angle is wider than the O–H angle in water.

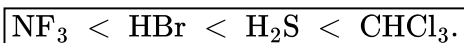
Dipole moment: About 0.95 *D*.

4. CHCl₃ (chloroform)

Structure: Tetrahedral around carbon with three Cl and one H.

Dipole moment: About 1.01 *D*.

Ordering from smallest to largest dipole moment



Hence, the correct choice is:



Question 27

Match the List - I with List - II

	List - I (Classification of molecules based on octet rule)		List - II (Example)
(A)	Molecules obeying octet rule	(I)	NO, NO ₂
(B)	Molecules with incomplete octet	(II)	BCl ₃ , AlCl ₃
(C)	Molecules with incomplete octet with odd electron	(III)	H ₂ SO ₄ , PCl ₅
(D)	Molecules with expanded octet	(IV)	CCl ₄ , CO ₂

Choose the correct answer from the options given below:

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Options:

A. A-IV, B-I, C-III, D-II

B. A-II, B-IV, C-III, D-I

C. A-IV, B-II, C-I, D-III

D. A-III, B-II, C-I, D-IV

Answer: C

Solution:

(A) A → IV

(B) $B \rightarrow II$

(C) $C \rightarrow I$

(D) $D \rightarrow III$

Question28

Which of the following statement is true with respect to H_2O , NH_3 and CH_4 ?

A. The central atoms of all the molecules are sp^3 hybridized.

B. The $H - O - H$, $H - N - H$ and $H - C - H$ angles in the above molecules are 104.5° , 107.5° and 109.5° , respectively.

C. The increasing order of dipole moment is $CH_4 < NH_3 < H_2O$.

D. Both H_2O and NH_3 are Lewis acids and CH_4 is a Lewis base.

E. A solution of NH_3 in H_2O is basic. In this solution NH_3 and H_2O act as Lowry-Bronsted acid and base respectively.

Choose the correct answer from the options given below:

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Options:

A. A, D and E Only

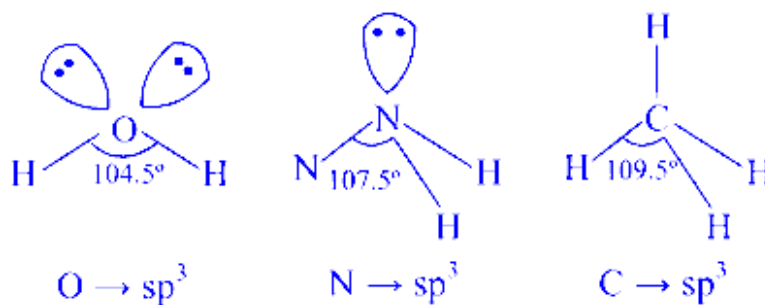
B. C, D and E Only

C. A, B and C Only

D. A, B, C and E Only

Answer: C

Solution:



Dipole moment $H_2O > NH_3 > CH_4$

H_2O & NH_3 are Lewis Bases

NH_3 act as Lowry- Bronsted base

Hence, A, B & C are correct

Question 29

Which of the following linear combination of atomic orbitals will lead to formation of molecular orbitals in homonuclear diatomic molecules [internuclear axis in z -direction] ?

A. $2p_z$ and $2p_x$

B. $2s$ and $2p_x$

C. $3d_{xy}$ and $3d_{x^2-y^2}$

D. $2s$ and $2p_z$

E. $2p_z$ and $3d_{x^2-y^2}$

Choose the correct answer from the options given below:

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Options:

A. D Only

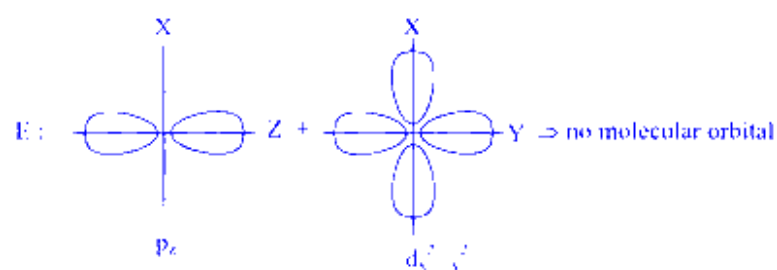
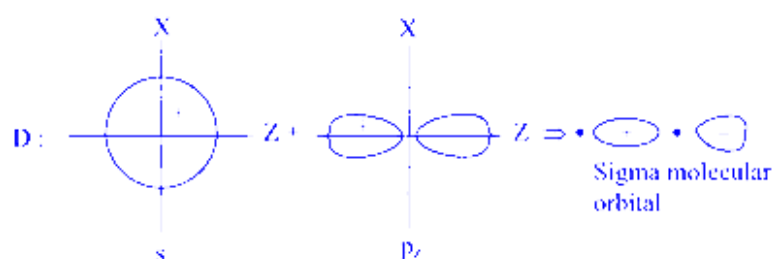
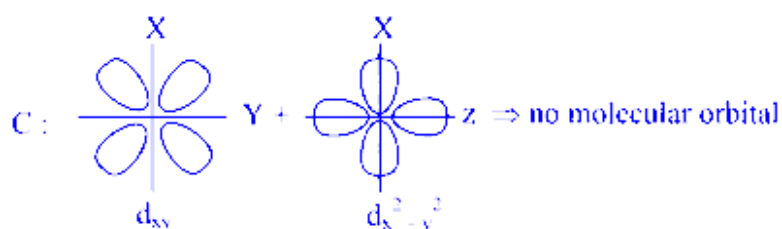
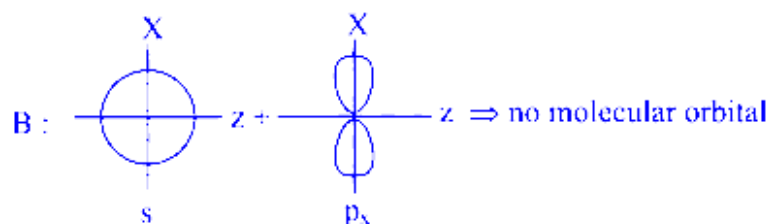
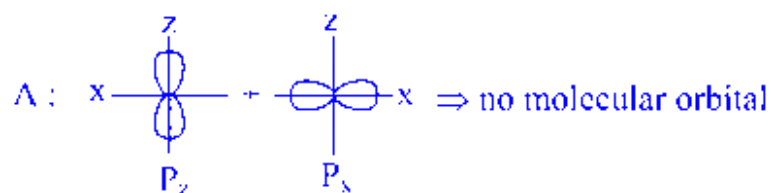
B. E Only

C. C and D Only

D. A and B Only

Answer: A

Solution:



Question30

Given below are two statements :

Statement (I) : Experimentally determined oxygen-oxygen bond lengths in the O_3 are found to be same and the bond length is greater

than that of a $\text{O} = \text{O}$ (double bond) but less than that of a single ($\text{O} - \text{O}$) bond.

Statement (II) : The strong lone pair-lone pair repulsion between oxygen atoms is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond ($\text{O} = \text{O}$) but more than that of a single bond ($\text{O} - \text{O}$).

In the light of the above statements, choose the correct answer from the options given below :

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Options:

- A. Statement I is false but Statement II is true
- B. Both Statement I and Statement II are true
- C. Statement I is true but Statement II is false
- D. Both Statement I and Statement II are false

Answer: C

Solution:

Due to resonance bond length is identical in ozone. Therefore statement I is true and statement II is false.

Question31

The molecules having square pyramidal geometry are

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Options:

- A. BrF_5 & PCl_5

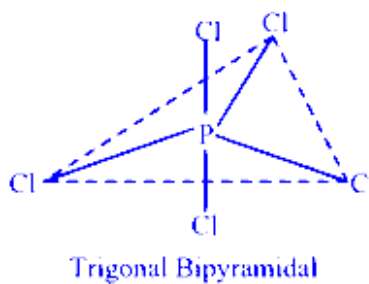
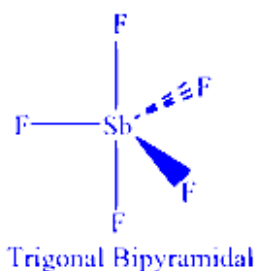
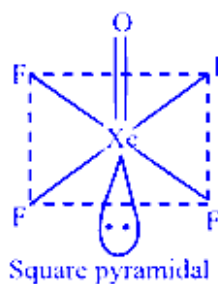
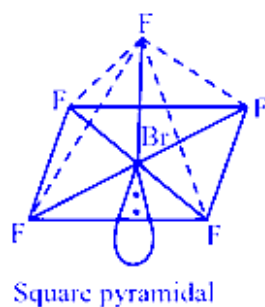
B. SbF_5 & PCl_5

C. BrF_5 & XeOF_4

D. SbF_5 & XeOF_4

Answer: C

Solution:



BrF_5 : Square pyramidal

XeOF_4 : Square pyramidal

SbF_5 : Trigonal bipyramidal

PCl_5 : Trigonal bipyramidal

Question32

Consider ' n ' is the number of lone pair of electrons present in the equatorial position of the most stable structure of ClF_3 . The ions from the following with ' n ' number of unpaired electrons are

A. V^{3+}

B. Ti^{3+}

C. Cu^{2+}

D. Ni^{2+}

E. Ti^{2+}

Choose the correct answer from the options given below:

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Options:

A. B and D Only

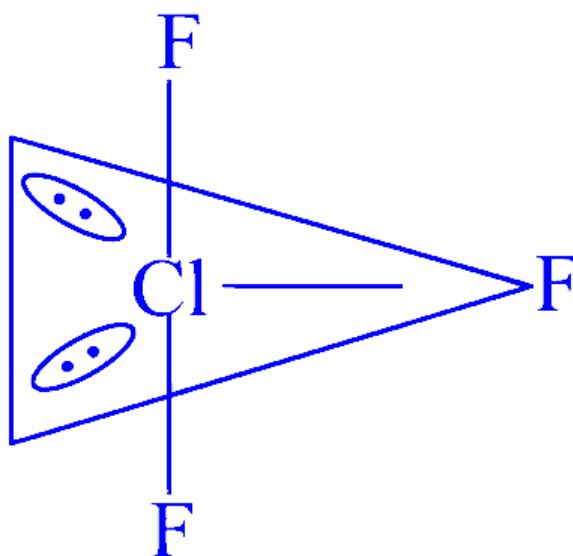
B. A, D and E Only

C. B and C Only

D. A and C Only

Answer: B

Solution:



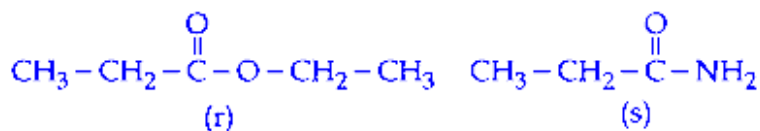
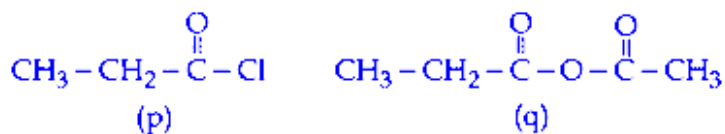
$n = 2$ (No of lone pair present in equatorial plane)

(Unpaired e⁻)

- (A) V⁺³ : [Ar]3 d² 2
(B) Ti³⁺ : [Ar]3 d¹ 1
(C) Cu⁺² : [Ar]3 d⁹ 1
(D) Ni⁺² : [Ar]3 d⁸ 2
(E) Ti⁺² : [Ar]3 d² 2
-

Question33

Consider the following molecules:



The correct order of rate of hydrolysis is :

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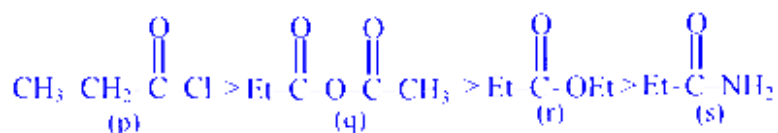
Options:

- A. $p > q > r > s$
B. $r > q > p > s$
C. $p > r > q > s$
D. $q > p > r > s$

Answer: A

Solution:

Rate of hydrolysis $\propto$ Leaving group ability



Question34

Among SO_2 , NF_3 , NH_3 , XeF_2 , ClF_3 and SF_4 , the hybridization of the molecule with nonzero dipole moment and highest number of lone-pairs of electrons on the central atom is

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Options:

A. sp^3

B. dsp^2

C. $\text{sp}^3 \text{d}^2$

D. $\text{sp}^3 \text{d}$

Answer: D

Solution:

Molecule	Hybridisation	Dipole Moment	Lone pair on the central atom
SO_2	sp^2	Non-zero	1
NF_3	sp^3	Non-zero	1
NH_3	sp^3	Non-zero	1
XeF_2	sp^3	Non-zero	3
ClF_3	$\text{sp}^3 \text{d}$	Non-zero	2
SF_4	$\text{sp}^3 \text{d}$	Non-zero	1

Question35

A molecule with the formula AX_4Y has all its elements from p-block. Element A is rarest, monoatomic, non-radioactive from its group and has the lowest ionization enthalpy value among A, X and Y. Elements X and Y have first and second highest electronegativity values respectively among all the known elements. The shape of the molecule is:

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Options:

- A. Pentagonal planar
- B. Square pyramidal
- C. Trigonal bipyramidal
- D. Octahedral

Answer: B

Solution:

Given A is rarest, monoatomic, non-radioactive p-block element and form AX_4Y type of molecule.

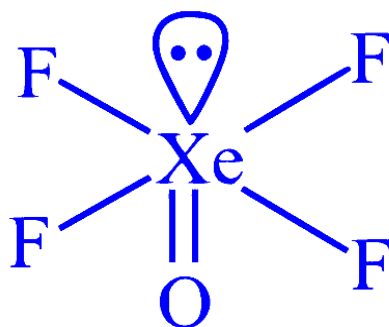
∴ It is concluded that it is Xe

It is given the electronegativity of A is less than X & Y

It is given the electronegativity of X&Y is highest and second highest respectively among all element.

∴ X & Y are F & O

∴ Compound is consider as $XeOF_4$ with square pyramidal shape.



Question36

Which among the following molecules is (a) involved in $sp^3 d$ hybridization, (b) has different bond lengths and (c) has lone pair of electrons on the central atom?

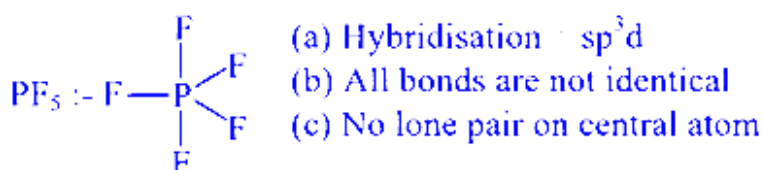
JEE Main 2025 (Online) 2nd April Evening Shift

Options:

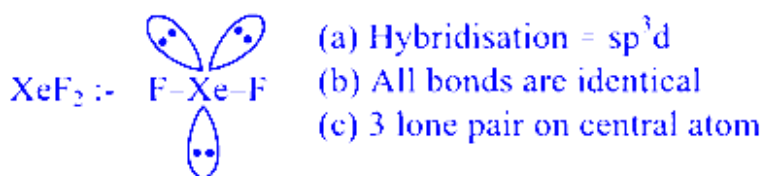
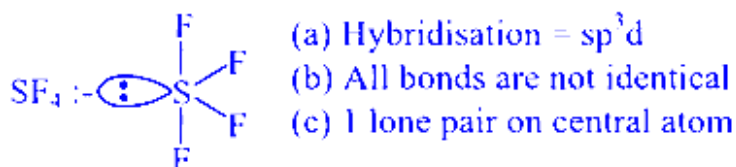
- A. XeF_4
- B. XeF_2
- C. PF_5
- D. SF_4

Answer: D

Solution:



XeF_4 :- $sp^3 d^2$ Hybridisation



Question37

Given below are two statements:

Statement I : Wet cotton clothes made of cellulose based carbohydrate takes comparatively longer time to get dried than wet nylon polymer based clothes.

Statement II : Intermolecular hydrogen bonding with water molecule is more in nylon-based clothes than in the case of cotton clothes.

In the light of above statements, choose the correct answer from the options given below

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Options:

- A. Both Statement I and Statement II are false
- B. Both Statement I and Statement II are true
- C. Statement I is false but Statement II is true
- D. Statement I is true but Statement II is false

Answer: D

Solution:

The correct choice is Option D: Statement I is true but Statement II is false.

Explanation:

Statement I (True)

Cotton fibers are made of cellulose, which has a large number of -OH groups. These hydroxyls form strong hydrogen bonds with water, so cotton holds onto moisture and dries more slowly than nylon.

Statement II (False)

Nylon contains only a few amide -NH and C=O groups per repeat unit, so it forms fewer hydrogen bonds with water than cellulose does. In other words, cotton actually has **more** intermolecular H-bonding with water than nylon.

Question38

Which of the following molecule(s) show/s paramagnetic behavior?

A. O_2

B. N_2

C. F_2

D. S_2

E. Cl_2

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. A & C only

B. A & E Only

C. A & D Only

D. B Only

Answer: C

Solution:

No. of unpaired e^-

(A)	O_2	2
(B)	N_2	0
(C)	F_2	0
(D)	S_2	2
(E)	Cl_2	0

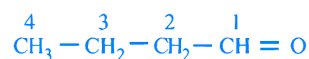
If species contain unpaired electron then it is paramagnetic.

So A & D are paramagnetic.

Question39

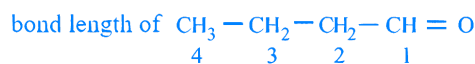
Given below are two statements.

Statement I: The dipole moment of $\overset{4}{CH_3} - \overset{3}{CH} = \overset{2}{CH} - \overset{1}{CH} = O$ is greater than



the dipole moment of $\overset{4}{CH_3} - \overset{3}{CH} = \overset{2}{CH} - \overset{1}{CH} = O$ is greater than $\overset{4}{CH_3} - \overset{3}{CH_2} - \overset{2}{CH_2} - \overset{1}{CH} = O$

Statement II: $C_1 - C_2$ bond length of $\overset{4}{CH_3} - \overset{3}{CH} = \overset{2}{CH} - \overset{1}{CH} = O$ is greater than $C_1 - C_2$



In the light of the above statements, choose the correct answer from the options given below:

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. Both Statement I and Statement II are true

B. Statement I is true but Statement II is false

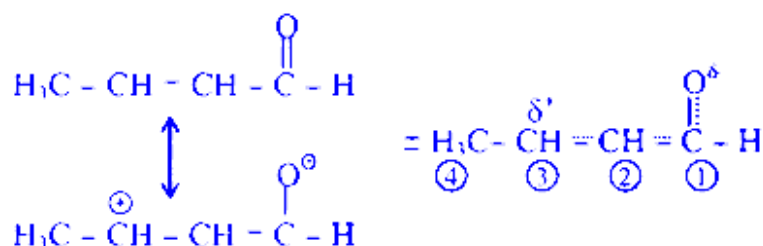
C. Statement I is false but Statement II is true

D. Both Statement I and Statement II are false

Answer: B

Solution:

Statement - I :

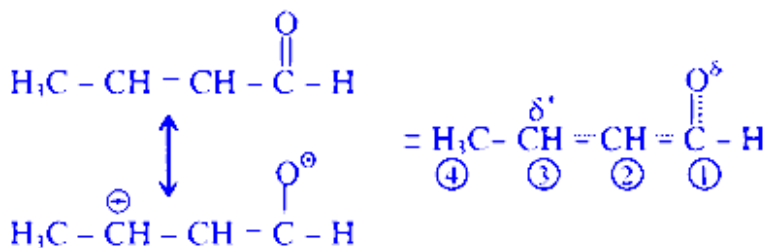


$$\mu = q \times d$$

More charges and more distance between charges than other compound so more dipole moment.

Statement-I is true.

Statement-II :



$\text{C}_1 - \text{C}_2$ bond has partial double bond character that means lesser bond length than $\text{C}_1 - \text{C}_2$ bond of other compound.

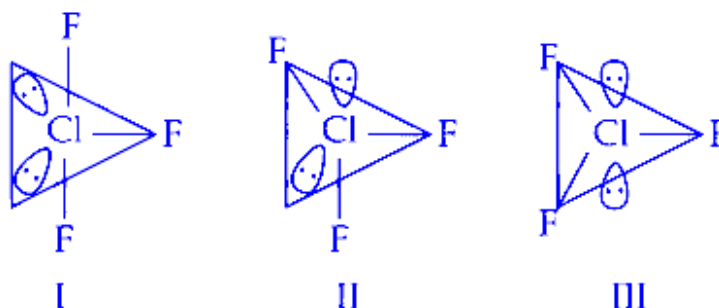
Statement-II is false.

Question40

Given below are two statements:

Statement (I): For ClF_3 , all three

possible structures may be drawn as follows.



Statement (II): Structure III is most stable, as the orbitals having the lone pairs are axial, where the lp – bp repulsion is minimum.

In the light of the above statements, choose the most appropriate answer from the options given below:

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Options:

A.

Both Statement I and Statement II are correct

B.

Both Statement I and Statement II are incorrect

C.

Statement I is correct but Statement II is incorrect

D.

Statement I is incorrect but Statement II is correct

Answer: C

Solution:

Statement 1 is correct.

Statement 2 is incorrect since in sp^3 d hybridization; lone pair cannot occupy axial position.

Question41

Match the List I with List II.

	List - I Molecule/ion		List - II Bond pair : lone pair (on the central atom)
(A)	ICl_2^-	(I)	4 : 2
(B)	H_2O	(II)	4 : 1
(C)	SO_2	(III)	2 : 3
(D)	XeF_4	(IV)	2 : 2

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. A-IV, B-III, C-II, D-I

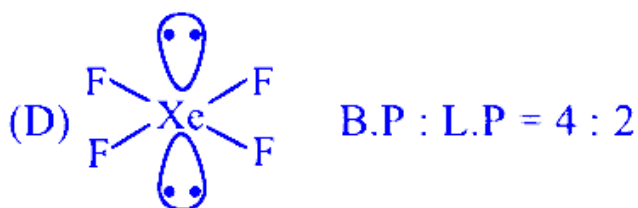
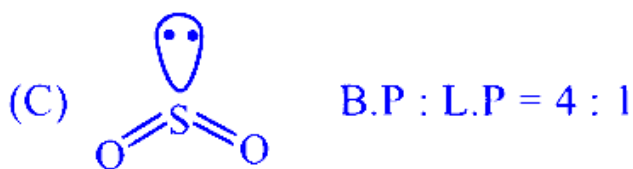
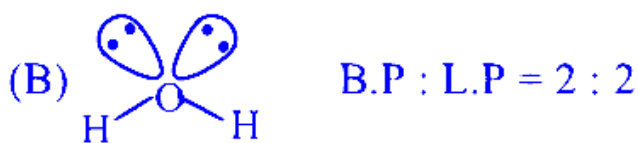
B. A-III, B-IV, C-I, D-II

C. A-II, B-I, C-IV, D-III

D. A-III, B-IV, C-II, D-I

Answer: D

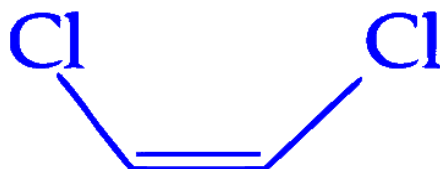
Solution:



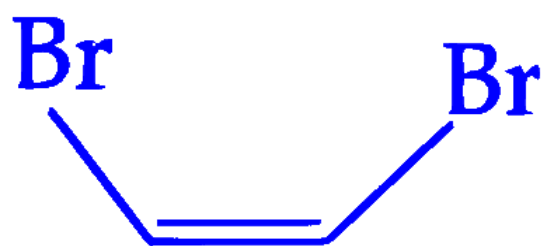
Question42

Given below are two statements :

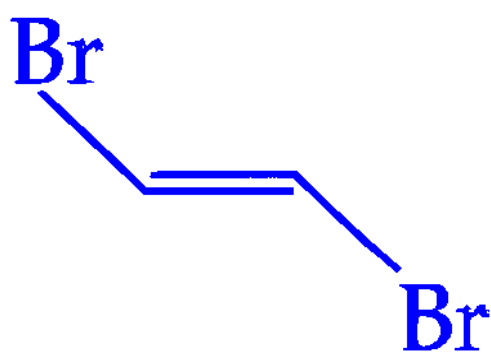
Statement (I) :



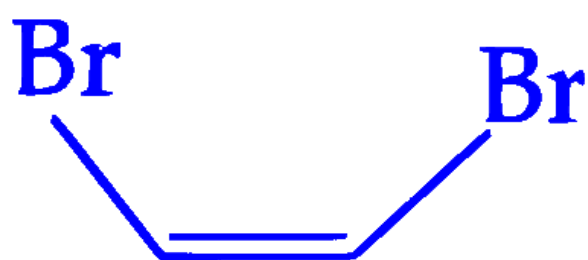
is more polar than



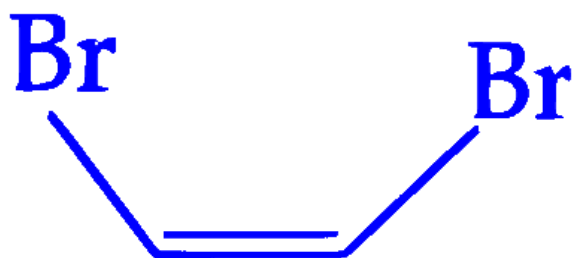
Statement (II) : Boiling point of



is lower than



but it is more polar than



.

In the light of the above statements, choose the most appropriate answer from the options given below :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

Both Statement I and Statement II are correct

B.

Statement I is incorrect but Statement II is correct

C.

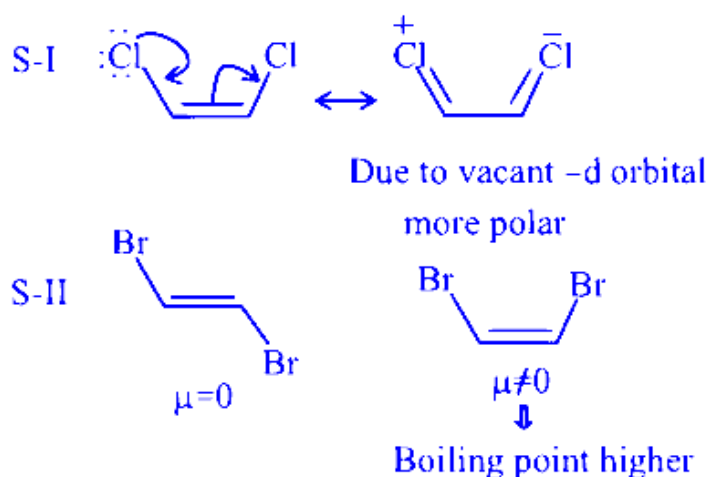
Both Statement I and Statement II are incorrect

D.

Statement I is correct but Statement II is incorrect

Answer: D

Solution:



Question43

In SO_2 , NO_2^- and N_3^- , the hybridizations at the central atom are respectively :

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Options:

A.

sp^2 , sp^2 and sp^2

B.

sp , sp^2 and sp

C.

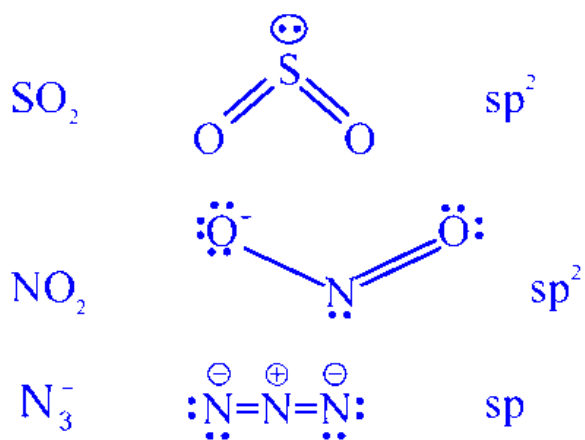
sp^2 , sp and sp

D.

sp^2 , sp^2 and sp

Answer: D

Solution:



Question44

Choose the polar molecule from the following :

[27-Jan-2024 Shift 1]

Options:

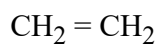
A.



B.



C.

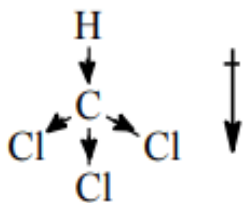


D.



Answer: D

Solution:



$$\mu \neq 0$$

CHCl_3 is polar molecule and rest all molecules are non-polar

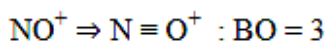
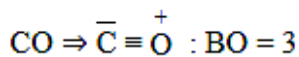
Question45

Sum of bond order of CO and NO^+ is _____

[27-Jan-2024 Shift 1]

Answer: 6

Solution:



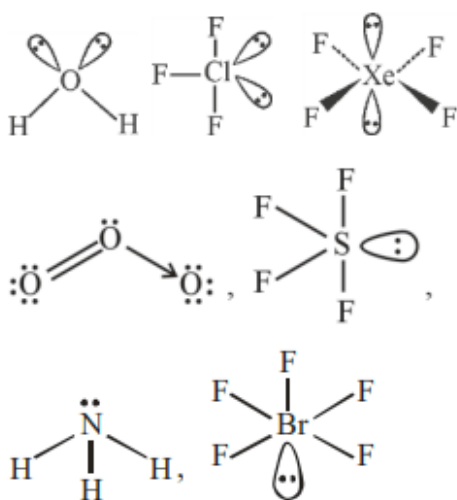
Question46

Number of compounds with one lone pair of electrons on central atom amongst following is O_3 , H_2O , SF_4 , ClF_3 , NH_3 , BrF_5 , XeF_4 ____

[29-Jan-2024 Shift 1]

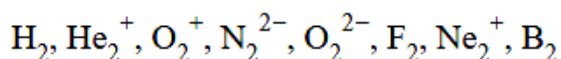
Answer: 4

Solution:



Question47

The number of species from the following which are paramagnetic and with bond order equal to one is _____



[29-Jan-2024 Shift 1]

Answer: 1

Solution:

Magnetic behaviour		Bond order
H_2	Diamagnetic	1
O_2^+	Paramagnetic	0.5
He_2^+	Paramagnetic	2.5
N_2^{2-}	Paramagnetic	2
O_2^{2-}	Diamagnetic	1
F_2	Diamagnetic	1
Ne_2^+	Paramagnetic	0.5
B_2	Paramagnetic	1

Question48

The total number of molecules with zero dipole moment among CH_4 , BF_3 , H_2O , HF , NH_3 , CO_2 and SO_2 is _____

[29-Jan-2024 Shift 2]

Answer: 3

Solution:

Molecules with zero dipole moment = CO_2 , CH_4 , BF_3

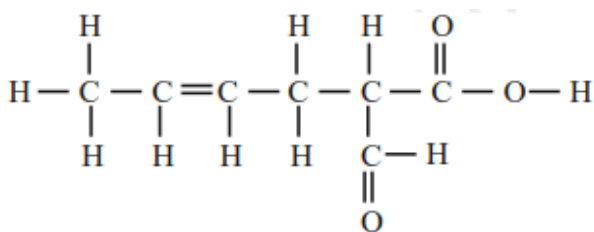
Question49

The total number of 'Sigma' and Pi bonds in 2formylhex-4-enoic acid is

[29-Jan-2024 Shift 2]

Answer: 22

Solution:



22 bonds

Question50

The total number of anti bonding molecular orbitals, formed from 2 s and 2p atomic orbitals in a diatomic molecule is ____

[29-Jan-2024 Shift 2]

Answer: 4

Solution:

Antibonding molecular orbital from 2 s = 1

Antibonding molecular orbital from 2p = 3

Total = 4

Question51

Match List-I with List-II

List-I Molecule	List-II Shape
(A) BrF_5	(I) T-shape
(B) H_2O	(II) See saw
(C) ClF_3	(III) Bent
(D) SF_4	(IV) Square pyramidal

[30-Jan-2024 Shift 1]

Options:

A.

(A)-I, (B)-II, (C)-IV, (D)-III

B.

(A) -II, (B)-I, (C)-III, (D)-IV

C.

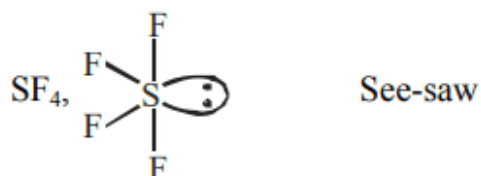
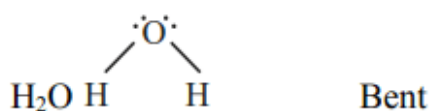
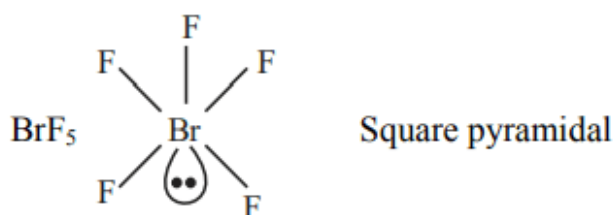
(A)-III, (B)-IV, (C)-I, (D)-II

D.

(A)-IV, (B)-III, (C)-I, (D)-II

Answer: D

Solution:



Question52

The total number of molecular orbitals formed from 2 s and 2p atomic orbitals of a diatomic molecule

[30-Jan-2024 Shift 1]

Answer: 8

Solution:

Two molecular orbitals $\sigma 2s$ and $\sigma^* 2s$.

Six molecular orbitals $\sigma 2p_z$ and $\sigma^* 2p_z$.

$\pi 2p_x$, $\pi 2p_y$ and $\pi^* 2p_x$, $\pi^* 2p_y$

Question53

Given below are two statements:

Statement-I: Since fluorine is more electronegative than nitrogen, the net dipole moment of NF_3 is greater than NH_3 .

Statement-II: In NH_3 , the orbital dipole due to lone pair and the dipole moment of NH bonds are in opposite direction, but in NF_3 the orbital dipole due to lone pair and dipole moments of N-F bonds are in same direction.

In the light of the above statements. Choose the most appropriate from the options given below.

[30-Jan-2024 Shift 2]

Options:

A.

Statement I is true but Statement II is false.

B.

Both Statement I and Statement II are false.

C.

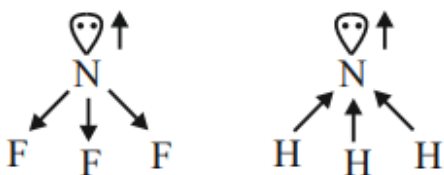
Both statement I and Statement II is are true.

D.

Statement I is false but Statement II is are true.

Answer: B

Solution:



Question54

The linear combination of atomic orbitals to form molecular orbitals takes place only when the combining atomic orbitals

A. have the same energy

B. have the minimum overlap

C. have same symmetry about the molecular axis

D. have different symmetry about the molecular axis

Choose the most appropriate from the options given below:

[31-Jan-2024 Shift 1]

Options:

A.

A, B, C only

B.

A and C only

C.

B, C, D only

D.

B and D only

Answer: B

Solution:

* Molecular orbital should have maximum overlap

* Symmetry about the molecular axis should be similar

Question55

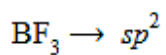
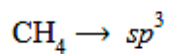
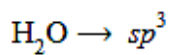
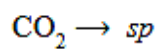
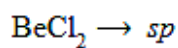
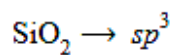
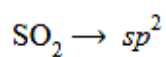
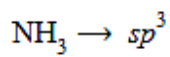
The number of species from the following in which the central atom uses sp^3 hybrid orbitals in its bonding is _____

NH_3 , SO_2 , SiO_2 , $BeCl_2$, CO_2 , H_2O , CH_4 , BF_3

[31-Jan-2024 Shift 1]

Answer: 4

Solution:



Question 56

Which of the following is least ionic?

[31-Jan-2024 Shift 2]

Options:

A.



B.



C.

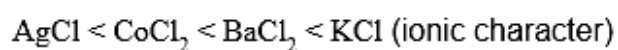


D.



Answer: B

Solution:



Reason: Ag^+ has pseudo inert gas configuration.

Question57

A diatomic molecule has a dipole moment of 1.2D. If the bond distance is 1Å, then fractional charge on each atom is _____ $\times 10^{-1}$ esu.

(Given 1D = 10^{-18} esu cm)

[31-Jan-2024 Shift 2]

Answer: 0

Solution:

$$\mu = 1.2D = q \times d$$

$$\Rightarrow 1.2 \times 10^{-10} \text{ esu } \text{\AA} = q \times 1 \text{\AA}$$

$$\therefore q = 1.2 \times 10^{-10} \text{ esu}$$

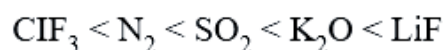
Question58

Arrange the bonds in order of increasing ionic character in the molecules. LiF, K₂O, N₂, SO₂ and ClF₃

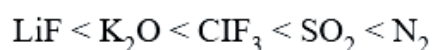
[1-Feb-2024 Shift 1]

Options:

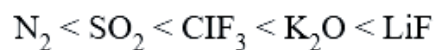
A.



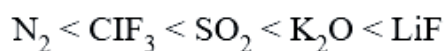
B.



C.



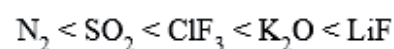
D.



Answer: C

Solution:

Increasing order of ionic character



Ionic character depends upon difference of electronegativity (bond polarity).

Question59

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : PH_3 has lower boiling point than NH_3 .

Reason (R) : In liquid state NH_3 molecules are associated through vander waal's forces, but PH_3 molecules are associated through hydrogen bonding.

In the light of the above statements, choose the most appropriate answer from the options given below:

[1-Feb-2024 Shift 1]

Options:

A.

Both (A) and (R) are correct and (R) is not the correct explanation of (A)

B.

(A) is not correct but (R) is correct

C.

Both (A) and (R) are correct but (R) is the correct explanation of (A)

D.

(A) is correct but (R) is not correct

Answer: D

Solution:

Unlike NH_3 , PH_3 molecules are not associated through hydrogen bonding in liquid state. That is why the boiling point of PH_3 is lower than NH_3 .

Question60

The number of molecules/ion/s having trigonal bipyramidal shape is

PF_5 , BrF_5 , PCl_5 , $[\text{PtCl}_4]^{2-}$, BF_3 , $\text{Fe}(\text{CO})_5$

[1-Feb-2024 Shift 1]

Answer: 3

Solution:

PF_5 , PCl_5 , $\text{Fe}(\text{CO})_5$; Trigonal bipyramidal

BrF_5 ; square pyramidal

$[\text{PtCl}_4]^{2-}$; square planar

BF_3 ; Trigonal planar

Question61

Given below are two statements :

Statement (I) : A π bonding MO has lower electron density above and below the inter-nuclear axis.

Statement (II) : The π^* antibonding MO has a node between the nuclei.

In the light of the above statements, choose the most appropriate answer from the options given below:

[1-Feb-2024 Shift 2]

Options:

A.

Both Statement I and Statement II are false

B.

Both Statement I and Statement II are true

C.

Statement I is false but Statement II is true

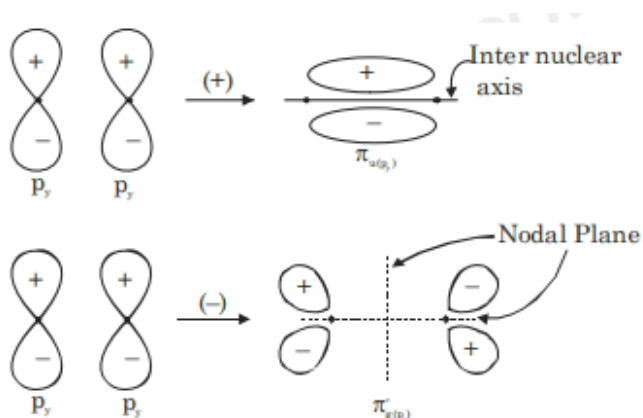
D.

Statement I is true but Statement II is false

Answer: C

Solution:

A π bonding molecular orbital has higher electron density above and below inter nuclear axis



Question62

Select the compound from the following that will show intramolecular hydrogen bonding.

[1-Feb-2024 Shift 2]

Options:

A.

H_2O

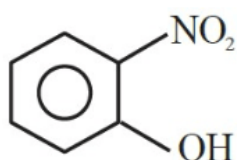
B.

NH_3

C.

$\text{C}_2\text{H}_5\text{OH}$

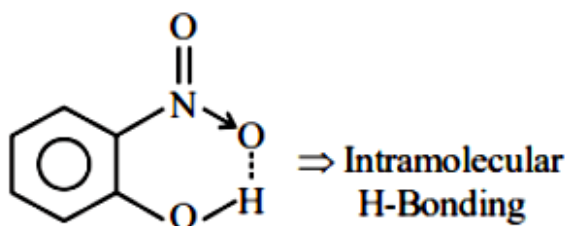
D.



Answer: D

Solution:

H_2O , NH_3 , $\text{C}_2\text{H}_5\text{OH} \Rightarrow$ Intermolecular H-Bonding



Question63

What is the number of unpaired electron(s) in the highest occupied molecular orbital of the following species : N_2 ; N_2^+ ; O_2 ; O_2^+ ?

[24-Jan-2023 Shift 2]

Options:

A. 0, 1, 2, 1

B. 2, 1, 2, 1

C. 0, 1, 0, 1

D. 2, 1, 0, 1

Answer: A

Solution:

N_2

$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \quad \frac{\sigma 2p_z^2}{\text{HOMO}}$$

$$N_2^+ - \sigma 1s^2 \sigma^* 1s^2 \sigma^* 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \quad \frac{\sigma 2p_z^1}{\text{HOM}}$$

$$O_2 - \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2$$

$$\pi 2p_x^2 = \pi 2p_y^2$$

$$\pi^* 2p_x^1 = \pi^* 2p_y^1 (\text{HOMO})$$

$$O_2^+ - \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2$$

$$\pi^* 2p_x^1 = \pi^* 2p_y^0 (\text{HOMO})$$

$$N_2 \Rightarrow 0 \text{ unpaired } e^- \text{ in HOMO}$$

$$N_2^+ \Rightarrow 1 \text{ unpaired } e^- \text{ in HOMO}$$

$$O_2 \Rightarrow 2 \text{ unpaired } e^- \text{ in HOMO}$$

$$O_2^+ \Rightarrow 1 \text{ unpaired } e^- \text{ in HOMO}$$

Question64

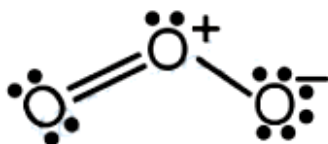
The total number of lone pairs of electrons on oxygen atoms of ozone is

[25-Jan-2023 Shift 1]

Answer: 6

Solution:

(Total no, of lone pairs on oxygen atoms = 6)



Question65

Statement I :- Dipole moment is a vector quantity and by convention it is depicted by a small arrow with tail on the negative centre and head pointing towards the positive centre.

Statement II :- The crossed arrow of the dipole moment symbolizes the direction of the shift of charges in the molecules.

In the light of the above statements, choose the most appropriate answer from the options given below :-

[25-Jan-2023 Shift 2]

Options:

- A. Both Statement I and Statement II are correct.
- B. Statement I is incorrect but Statement II is correct.
- C. Both Statement I and Statement II are incorrect.
- D. Statement I is correct but Statement II is incorrect.

Answer: D

Solution:

Statement II : The crossed arrow symbolises the direction of the shift of electron density in the molecule.

Question66

The number of given orbitals which have electron density along the axis is _____

$p_x, p_y, p_z, d_{xy}, d_{yz}, d_{xz}, d_z, d_{x^2-y^2}$

[25-Jan-2023 Shift 2]

Answer: 5

Solution:

p_x, p_y, p_z, d_z & $d_{x^2-y^2}$ are axial orbitals.

Question 67

According to MO theory the bond orders for O_2^{2-} , CO and NO^+ respectively, are
[29-Jan-2023 Shift 2]

Options:

- A. 1, 3 and 3
- B. 1, 3 and 2
- C. 1, 2 and 3
- D. 2, 3 and 3

Answer: A

Solution:

Theory based.

Question 68

Match List I with List II:

List I (Complexes)	List II (Hybridisation)
(A) $[\text{Ni}(\text{CO})_4]$	I. sp^3
(B) $[\text{Cu}(\text{NH}_3)_4]^{2+}$	II. dsp^2
(C) $[\text{Fe}(\text{NH}_3)_6]^{2+}$	III. sp^3d^2
(D) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	IV. $d^2 sp^3$

[30-Jan-2023 Shift 2]

Options:

A. A - II, B - I, C - III, D - IV

B. A - I, B - II, C - III, D - IV

C. A - II, B - I, C - IV, D - III

D. A - I, B - II, C - IV, D - III

Answer: D

Solution:

For $[\text{Fe}(\text{NH}_3)_6]^{2+}$, $\Delta_0 < P$, hence the pairing of electrons does not occur in t_{2g} . Therefore complex is outer orbital and its hybridisation is sp^3d^2 .

List I (Complexes)	List II (Hybridisation)
$[\text{Ni}(\text{CO})_4]$	sp^3
$[\text{Cu}(\text{NH}_3)_4]^{2+}$	dsp^2
$[\text{Fe}(\text{NH}_3)_6]^{2+}$	sp^3d^2
$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	sp^3d^2

Question 69

Match List I with List II

List I	List II
A. XeF_4	I. See - saw
B. SF_4	II. Square planar
C. NH_4^+	III. Bent T- shaped
D. BrF_3	IV. Tetrahedral

Choose the correct answer from the options given below :
[31-Jan-2023 Shift 1]

Options:

A. A-IV, B-III, C-II, D-I

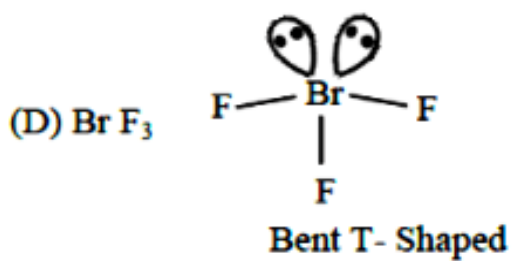
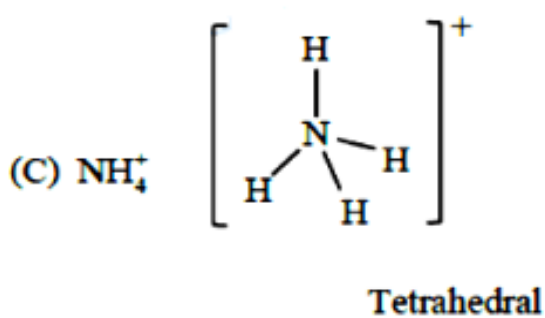
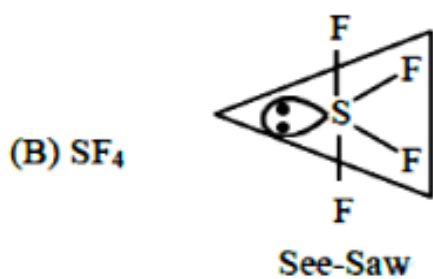
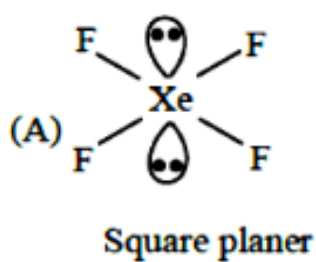
B. A-II, B-I, C-III, D-IV

C. A-IV, B-I, C-II, D-III

D. A-II, B-I, C-IV, D-III

Answer: D

Solution:



Question70

For OF_2 molecule consider the following:

(A) Number of lone pairs on oxygen is 2 .

(B) FOF angle is less than 104.5° .

(C) Oxidation state of O is -2 .

(D) Molecule is bent ' V ' shaped.

(E) Molecular geometry is linear.

Correct options are:

[30-Jan-2023 Shift 1]

Options:

A. C, D, E only

B. B, E, A only

C. A, C, D only

D. A, B, D only

Answer: D

Solution:



- Two lone pair on oxygen
 - Molecule is ' v ' shaped
 - Bond angle is less than 104.5° (102°)
 - O.S. of ' O ' is $+2$
-

Question 71

Match List I with List II

LIST-I (molecules/ions)	LIST-II (No. of lone pairs of e^- on central atom)
(A) IF_7	I. Three
(B) ICl_4^-	II. One
(C) XeF_6	III. Two
(D) XeF_2	IV. Zero

Choose the correct answer from the options given below:
[30-Jan-2023 Shift 1]

Options:

- A. A - II, B - III, C - IV, D - I
- B. A - IV, B - III, C - II, D - I
- C. A - II, B - I, C - IV, D - III
- D. A - IV, B - I, C - II, D - III

Answer: B

Solution:

IF_7	zero lone pair
ICl_4^-	two lone pair
XeF_6	one lone pair
XeF_2	three lone pair

Question 72

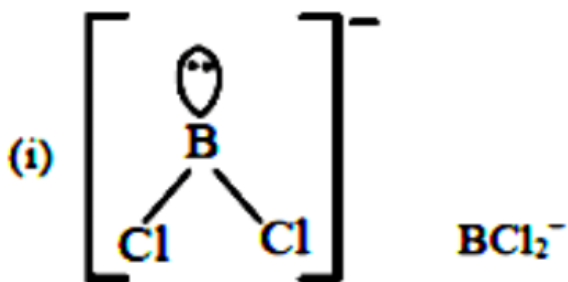
Amongst the following, the number of species having the linear shape is _____.

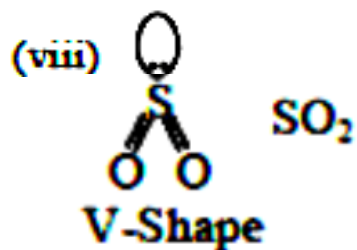
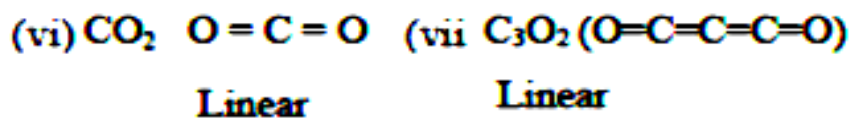
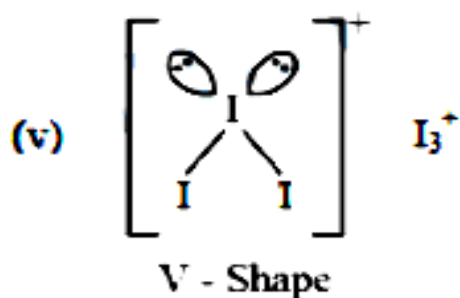
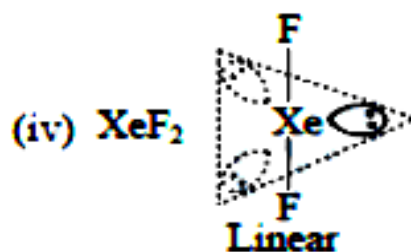
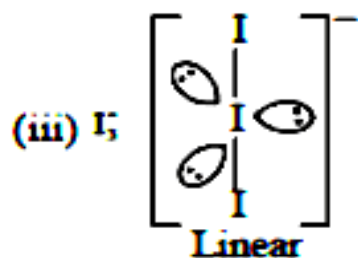
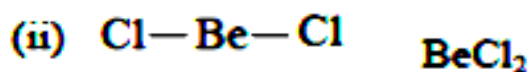
XeF_2 , I_3^+ , C_3O_2 , I_3^- , CO_2 , SO_2 , BeCl_2 and $\text{BCl}_2^\ominus$

[31-Jan-2023 Shift 2]

Answer: 5

Solution:





Question73

The number of species from the following which have square pyramidal structure is

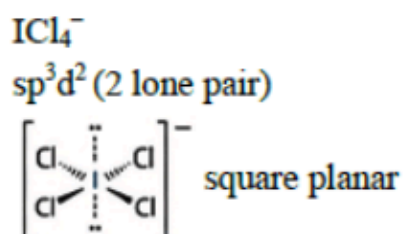
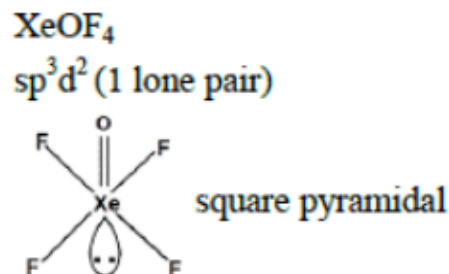
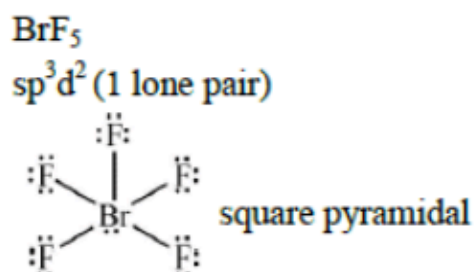
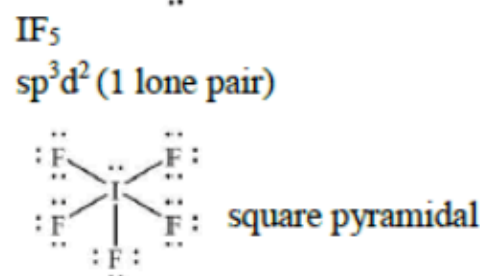
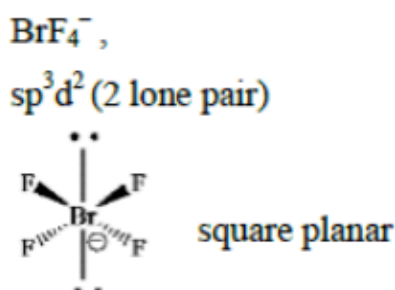


[6-Apr-2023 shift 1]

Answer: 3

Solution:

PF_5
 sp^3d (0 lone pair)
Trigonal bipyramidal



Question74

The number of species having a square planar shape from the following is _____

XeF_4 , SF_4 , SiF_4 , BF_4^- , BrF_4^- , $[\text{Cu}(\text{NH}_3)_4]^{2+}$,

$[\text{FeCl}_4]^{2-}$, $[\text{PtCl}_4]^{2-}$

[6-Apr-2023 shift 2]

Answer: 4

Solution:

XeF_4 , BrF_4^{-1} , $[\text{Cu}(\text{NH}_3)_4]^{+2}$, $[\text{PtCl}_4]^{-2}$ has square planar shape.

Question75

In an ice crystal, each water molecule is hydrogen bonded to _____ neighbouring molecules.

[6-Apr-2023 shift 2]

Answer: 4

Solution:

In ice each water molecule is hydrogen bonded with four other water molecules.

Question76

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Butan-1-ol has higher boiling point than ethoxyethane.

Reason R : Extensive hydrogen bonding leads to stronger association of molecules. In the light of the above statements, choose the correct answer from the options given below:

[8-Apr-2023 shift 1]

Options:

A. Both A and R are true but R is not the correct explanation of A

B. Both A and R are true and R is the correct explanation of A

C. A is false but R is true

D. A is true but R is false

Answer: B

Solution:

At comparable molecular mass, alcohol has higher b.p. than ether due to H-bond, because H-bond leads to stronger associated of molecules.

Question77

The number of following factors which affect the percent covalent character of the ionic bond is _____
[8-Apr-2023 shift 1]

Options:

A. Polarising power of cation

B. Extent of distortion of anion

C. Polarisability of the anion

D. Polarising power of anion

Answer: C

Solution:

Percent covalent character of the ionic bond

- (1) Polarising power of cation
 - (2) Extent of distortion of anion
 - (3) Polarisability of the anion
-

Question78

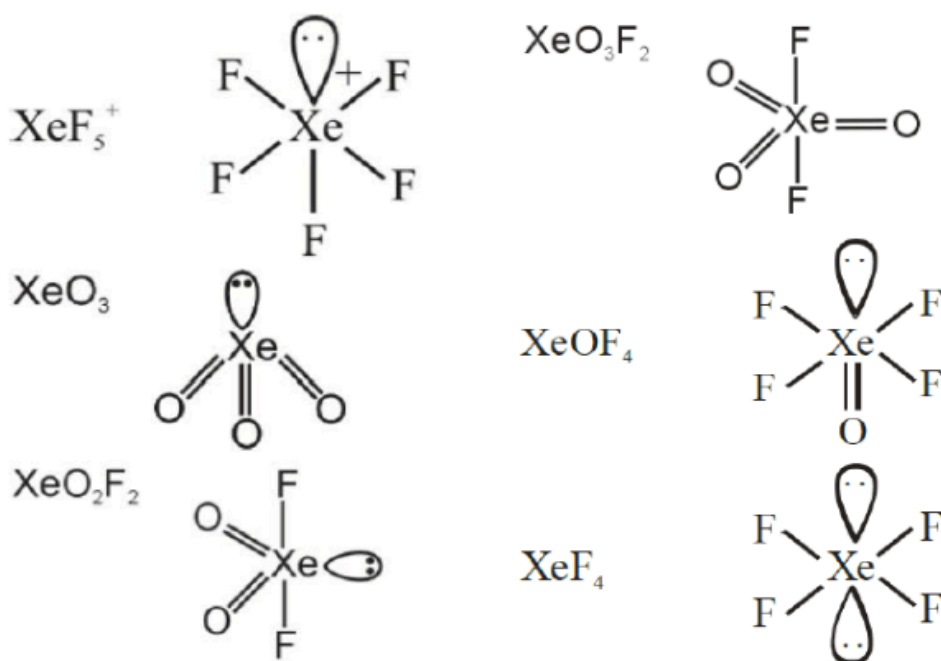
The number of species from the following carrying a single lone pair on central atom Xenon is

_____ XeF_5^+ , XeO_3 , XeO_2F_2 , XeF_5^- , XeO_3F_2 , XeOF_4 , XeF_4

[8-Apr-2023 shift 2]

Answer: 4

Solution:



Question79

The pair from the following pairs having both compounds with net non-zero dipole moment is

[10-Apr-2023 shift 1]

Options:

A. cis-butene, trans-butene

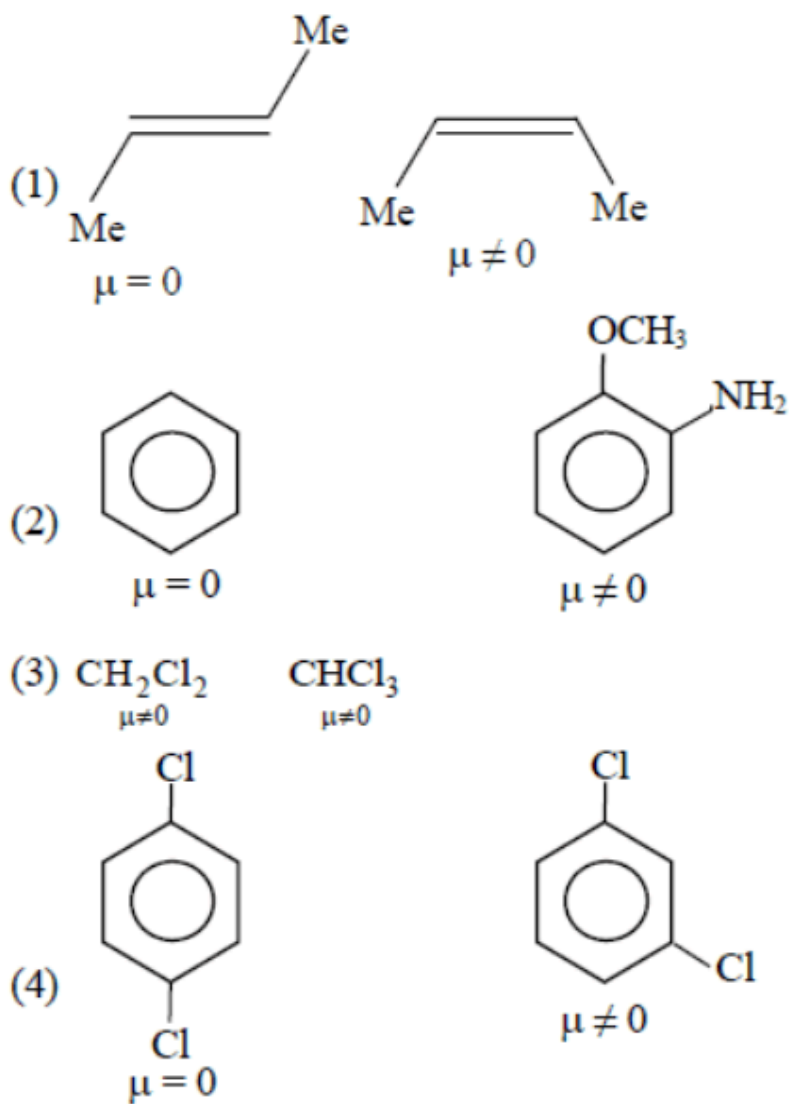
B. Benzene, anisidine

C. CH_2Cl_2 , CHCl_3

D. 1,4-Dichlorobenzene, 1,3-Dichlorobenzene

Answer: C

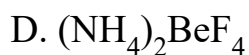
Solution:



Question80

The compound which does not exist is
[10-Apr-2023 shift 1]

Options:



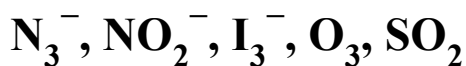
Answer: C

Solution:

Sodium superoxide is not stable

Question81

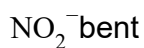
The number of bent-shaped molecule/s from the following is



[10-Apr-2023 shift 1]

Answer: 3

Solution:



O₃ bent
SO₂ bent

Question82

The sum of lone pairs present on the central atom of the interhalogen IF₅ and IF₇ is _____

[10-Apr-2023 shift 1]

Answer: 1

Solution:

IF₅ = 1 lone pair

IF₇ = 0 lone pair

1 + 0 = 1

Question83

Match list I with list II

List I Species	List II Geometry/ Shape
A. H ₃ O ⁺	I. Tetrahedral
B. Acetylide anion	II. Linera
C. NH ₄ ⁺	III. Pyramidal
D. ClO ₂ ⁻	IV. Bent

Choose correct answer from the options given below:

[11-Apr-2023 shift 1]

Options:

A. A-III, B-IV, C-I, D-II

B. A-III, B-IV, C-II, D-I

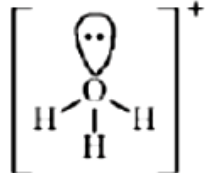
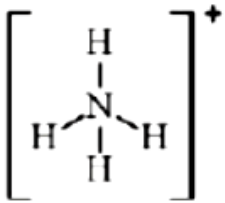
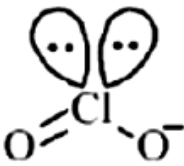
C. A-III, B-I, C-II, D-IV

D. A-III, B-II, C-I, D-IV

Answer: D

Solution:

Molecule/Ion Hybridisation Shape

H_3O^+	sp^3	Pyramidal	
Acetylide	sp	linear	$\text{C} \equiv \text{C}$
NH_4^+	sp^3	tetrahedral	
ClO_2^-	sp^3	Bent	

Question84

Which one of the following pairs is an example of polar molecular solids?

[11-Apr-2023 shift 2]

Options:

- A. $\text{SO}_2(\text{ s}), \text{CO}_2(\text{ s})$
- B. $\text{SO}_2(\text{ s}), \text{NH}_2(\text{ s})$
- C. $\text{MgO}(\text{ s}), \text{SO}_2(\text{ s})$
- D. $\text{HCl}(\text{ s}), \text{AlN}(\text{ s})$

Answer: B

Solution:

SO_2 and NH_3 are polar molecules. They are constituent particles of polar molecular solids.

Question85

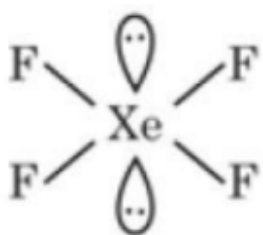
The maximum number of lone pairs of electrons on the central atom from the following species is _____

ClO_3^- , XeF_4 , SF_4 and I_3^-

[11-Apr-2023 shift 2]

Answer: 3

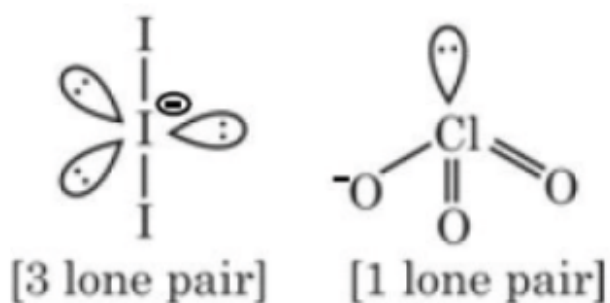
Solution:



[2 lone pair]



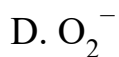
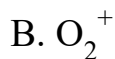
[1 lone pair]



Question86

The bond order and magnetic property of acetylide ion are same as that of
[12-Apr-2023 shift 1]

Options:



Answer: C

Solution:



Both are diamagnetic because both have absence of unpaired electron.

Question87

Given below are two statements :

Statement I : SbCl_5 is more covalent than SbCl_3

Statement II : The higher oxides of halogens also tend to be more stable than lower ones.

In the light of the above statements, choose the most appropriate answer from the options given below

[12-Apr-2023 shift 1]

Options:

A. Statement I is correct but statement II is incorrect

B. Both statement I and statement II are incorrect

C. Both statement I and statement II are correct

D. Statement I is incorrect but statement II is correct

Answer: C

Solution:

I $\rightarrow \text{SbCl}^{+5}$ is more covalent due to sb in higher 0.5 more covalent due to more charge.

II $\rightarrow$ Higher oxides of halogen tend to be more stable because higher oxidation states are less reactive and also the size of the atoms are more higher so they are less reactive.

Question88

In which of the following processes, the bond order increases and paramagnetic character changes to diamagnetic one?

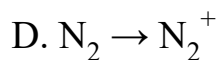
[13-Apr-2023 shift 1]

Options:

A. $\text{O}_2 \rightarrow \text{O}_2^+$

B. $\text{O}_2 \rightarrow \text{O}_2^{2-}$

C. $\text{NO} \rightarrow \text{NO}^+$



Answer: C

Solution:

NO is paramagnetic with $\text{BO} = 2.5$, NO^+ is diamagnetic with $\text{BO} = 3$

Question89

ClF_5 at room temperature is a:

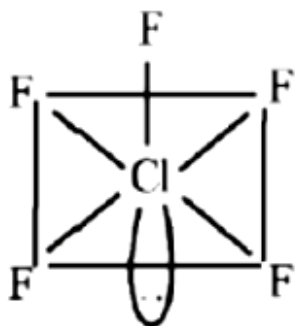
[13-Apr-2023 shift 1]

Options:

- A. Colourless liquid with square pyramidal geometry
- B. Colourless gas with trigonal bipyramidal geometry
- C. Colourless gas with square pyramidal geometry
- D. Colourless liquid with trigonal bipyramidal geometry

Answer: A

Solution:



ClF_5 is colourless liquid.

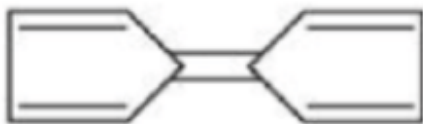
Question90

Among the following compounds, the one which shows highest dipole moment is

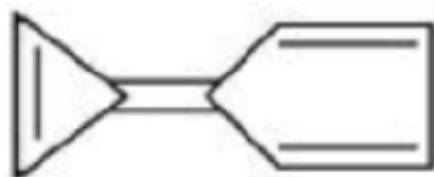
[13-Apr-2023 shift 1]

Options:

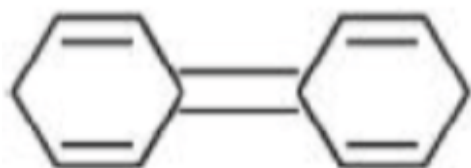
A.



B.



C.



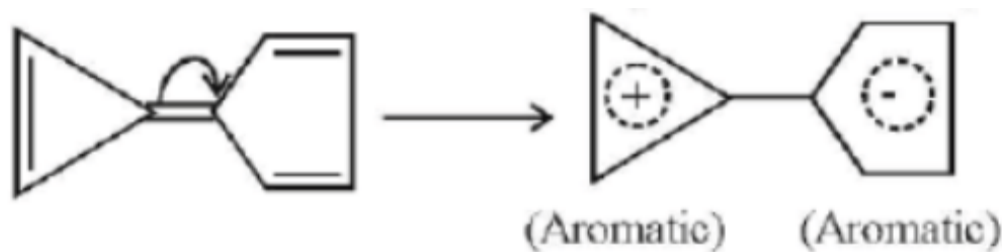
D.



Answer: B

Solution:

Among the given compounds, the following compound has the highest dipole moment because both the +ve and -ve ends acquire aromaticity.



Question91

Match List I with List II

LIST I	LIST II
A. Weak intermolecular farces of attraction	I. Hexamethylenediamine + adipic
B. Hydrogen bonding	II. $\text{AlEt}_3 + \text{TiCl}_4$
C. Heavily branched polymer	III. 2-chloro-1,3-butadiene
D. High density polymer	IV. Phenol + formaldehyde

Choose the correct answer from the options given below :
[13-Apr-2023 shift 2]

Options:

- A. A-IV, B-I, C-III, D-II
- B. A-III, B-I, C-IV, D-II
- C. A-II, B-IV, C-I, D-III
- D. A-IV, B-II, C-III, D-I

Answer: B

Solution:

- Hexamethylenediamine on reaction with adipic acid forms Nylon 6, 6 which shows H-bonding due to presence of amide group.
- $\text{AlEt}_3 + \text{TiCl}_4$ is Ziegler-Natta catalyst used to prepare high density polyethylene.
- 2-chloro-1, 3-butadiene (chloroprene) is monomer of neoprene which is a rubber (an elastomer)
- Phenol - formaldehyde forms Bakelite which is heavily branched (cross-linked) polymer

Question92

Given below are two statements :

Statement I: SO_2 and H_2O both possess V-shaped structure.

Statement II : The bond angle of SO_2 less than that of H_2O

In the light of the above statements, choose the most appropriate answer from the options given below :

[13-Apr-2023 shift 2]

Options:

- A. Both Statements I and Statement II are incorrect
- B. Both Statement I and Statements II are correct
- C. Statement I is correct but Statement II is incorrect
- D. Statements I is incorrect but Statement II is correct

Answer: C

Solution:

Both are bent in shape.

Bond angle of SO_2 (sp^2) is greater than that of H_2O (sp^3) due to higher repulsion of multiple bonds.

Question93

Consider the following statements:

(A) NF_3 molecule has a trigonal planar structure.

(B) Bond length of N_2 is shorter than O_2

(C) Isoelectronic molecules or ions have identical bond order

(D) Dipole moment of HS is higher than that of water molecule.

Choose the correct answer from the options given below:

[15-Apr-2023 shift 1]

Options:

- A. (A) and (B) are correct

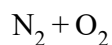
B. (C) and (D) are correct

C. (B) and (C) are correct

D. (A) and (D) are correct

Answer: C

Solution:



$$\text{B.O. } 3 > 2$$

$$\text{B.L. } 3 < 2$$

Isoelectronic have identical bond order

Question 94

The correct order of bond orders of C_2^{2-} , N_2^{2-} and O_2^{2-} is, respectively
[24-Jun-2022-Shift-2]

Options:

A. $\text{C}_2^{2-} < \text{N}_2^{2-} < \text{O}_2^{2-}$

B. $\text{O}_2^{2-} < \text{N}_2^{2-} < \text{C}_2^{2-}$

C. $\text{C}_2^{2-} < \text{O}_2^{2-} < \text{N}_2^{2-}$

D. $\text{N}_2^{2-} < \text{C}_2^{2-} < \text{O}_2^{2-}$

Answer: B

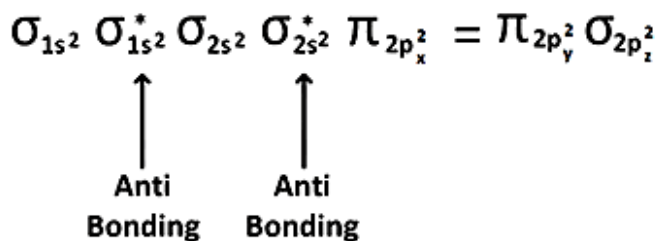
Solution:

$$(1) \text{ Bond order} = \frac{1}{2}[\text{N}_b - \text{N}_a]$$

N_b = No of electrons in bonding molecular orbital

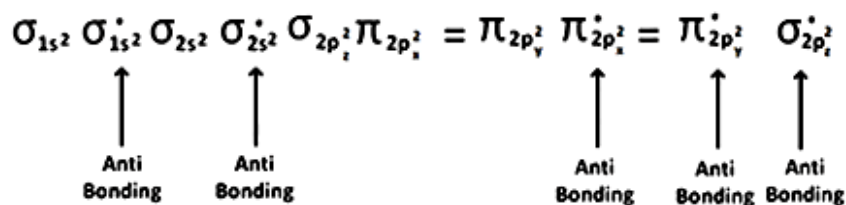
N_a = No of electrons in anti bonding molecular orbital

(2) upto 14 electrons, molecular orbital configuration is



Here N_a = Anti bonding electron = 4 and N_b = 10

(3) After 14 electrons to 20 electrons molecular orbital configuration is -



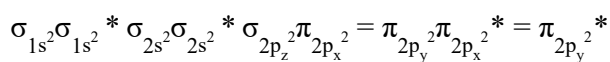
Here N_a = 10

and N_b = 10

In O atom 8 electrons present, so in O_2 , $8 \times 2 = 16$ electrons present.

in O_2^{2-} no of electrons = 18

(A) Molecular orbital configuration of O_2^{2-} (18 electrons) is



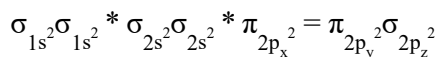
$$\therefore N_b = 10$$

$$N_a = 8$$

$$\therefore BO = \frac{1}{2}[10 - 8] = 1$$

(B) C_2^{2-} has 14 electrons.

Molecular orbital configuration of C_2^{2-} is



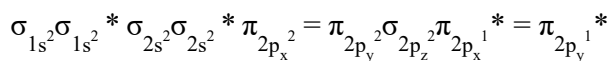
$$\therefore N_a = 4$$

$$N_b = 10$$

$$\therefore BO = \frac{1}{2}[10 - 4] = 3$$

(C) N_2^{2-} has 16 electrons.

Molecular orbital configuration of N_2^{2-} is

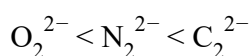


$$\therefore N_a = 6$$

$$N_b = 10$$

$$\therefore BO = \frac{1}{2}[10 - 6] = 2$$

The correct order of bond orders of C_2^{2-} , N_2^{2-} and O_2^{2-}



Question95

Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO Theory, by removal of an electron?

(A) NO

(B) N₂

(C) O₂

(D) C₂

(E) B₂

**Choose the most appropriate answer from the options given below:
[25-Jun-2022-Shift-1]**

Options:

A. (A), (B), (C) only

B. (B), (C), (E) only

C. (A), (C) only

D. (D) only

Answer: C

Solution:

If an electron is removed from the anti-bonding orbital, then it will tend to increase the bond order. The HOMO in NO and O₂ is antibonding molecular orbital.

Hence, in NO and O₂ bond order will increase on loss of electron.

Question96

Amongst BeF₂, BF₃, H₂O, NH₃, CCl₄ and HCl, the number of molecules with non-zero net dipole moment is_____

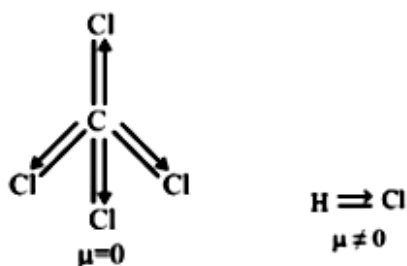
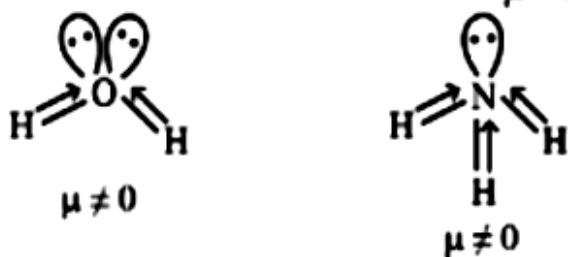
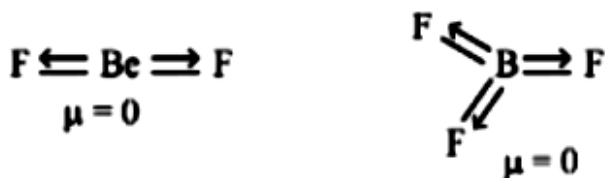
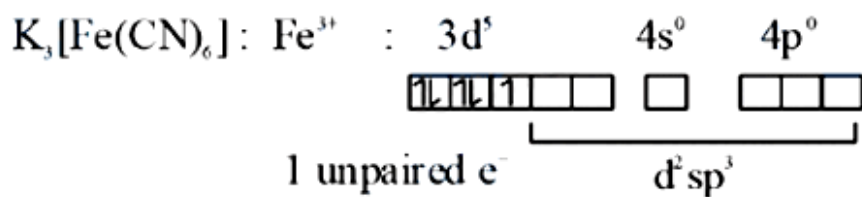
[25-Jun-2022-Shift-2]

Answer: 3

Solution:

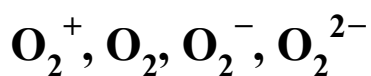
BeF_2 , BF_3 and $\text{CCl}_4 \Rightarrow \mu_{\text{net}} = 0$

H_2O , NH_3 and $\text{HCl} \Rightarrow \mu_{\text{net}} \neq 0$



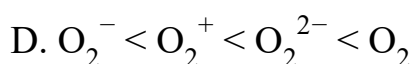
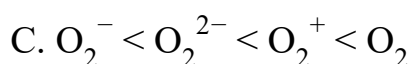
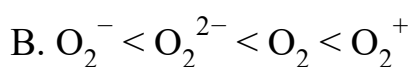
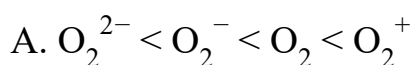
Question97

Consider the ions/molecule



For increasing bond order the correct option is :
 [26-Jun-2022-Shift-1]

Options:



Answer: A

Solution:

(1) Bond strength $\propto$ Bond order

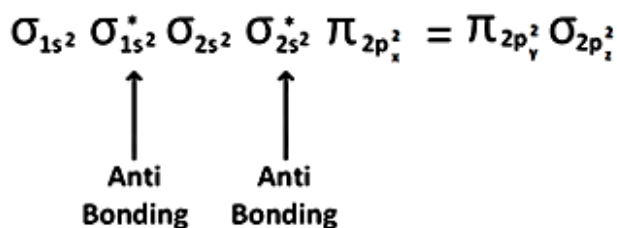
(2) Bond length $\propto \frac{1}{\text{Bond order}}$

(3) Bond order = $\frac{1}{2}[N_b - N_a]$

N_b = No of electrons in bonding molecular orbital

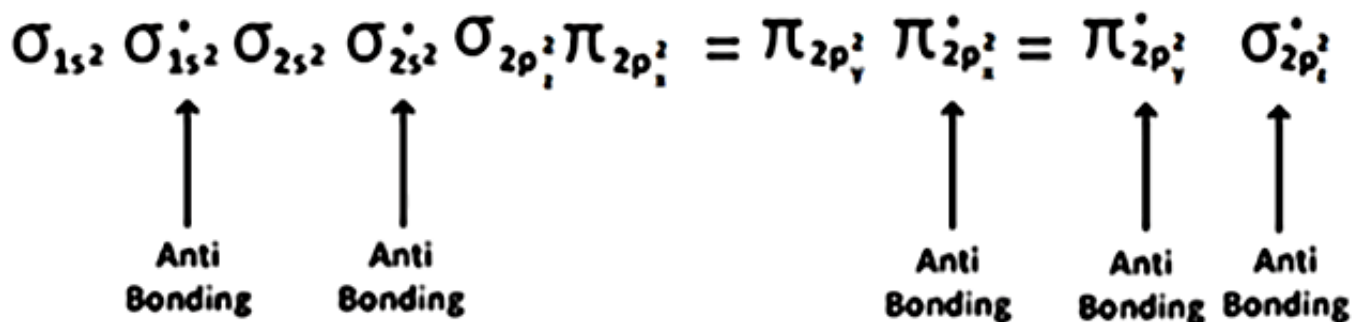
N_a = No of electrons in anti bonding molecular orbital

(4) upto 14 electrons, molecular orbital configuration is



Here N_a = Anti bonding electron = 4 and N_b = 10

(5) After 14 electrons to 20 electrons molecular orbital configuration is - - -



Here N_a = 10

and N_b = 10

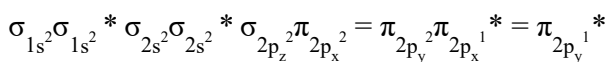
In O atom 8 electrons present, so in O_2 , $8 \times 2 = 16$ electrons present.

Then in O_2^+ no of electrons = 15

in O_2^- no of electrons = 17

in O_2^{2-} no of electrons = 18

$\therefore$ Molecular orbital configuration of O_2 (16 electrons) is

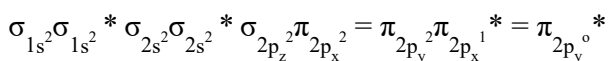


$\therefore N_a = 6$

$N_b = 10$

$\therefore BO = \frac{1}{2}[10 - 6] = 2$

Molecular orbital configuration of O_2^+ (15 electrons) is

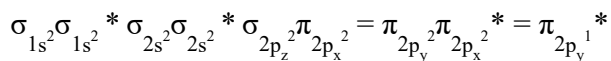


$\therefore N_b = 10$

$N_a = 5$

$\therefore BO = \frac{1}{2}[10 - 5] = 2.5$

Molecular orbital configuration of O_2^- (17 electrons) is

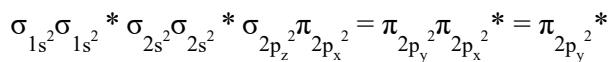


$$\therefore N_b = 10$$

$$N_a = 7$$

$$\therefore BO = \frac{1}{2}[10 - 7] = 1.5$$

Molecular orbital configuration of O_2^{2-} (18 electrons) is

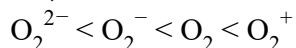


$$\therefore N_b = 10$$

$$N_a = 8$$

$$\therefore BO = \frac{1}{2}[10 - 8] = 1$$

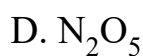
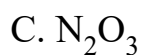
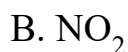
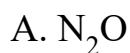
So, correct order of Bond order is



Question98

The oxide which contains an odd electron at the nitrogen atom is
[26-Jun-2022-Shift-2]

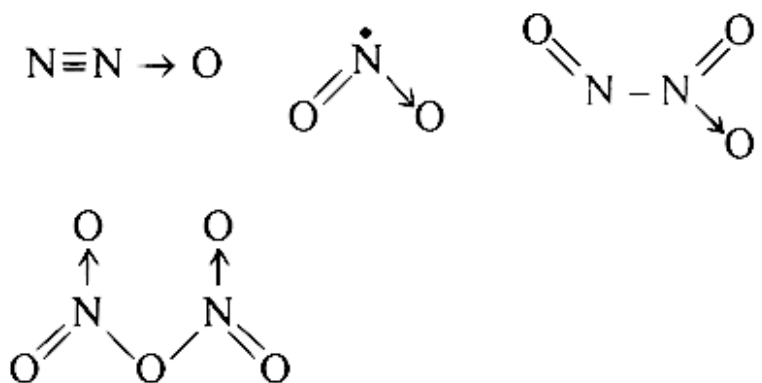
Options:



Answer: B

Solution:

The oxide of nitrogen which contains odd electron is NO_2



Question99

Amongst SF_4 , XeF_4 , CF_4 and H_2O , the number of species with two lone pairs of electrons is _____
 [26-Jun-2022-Shift-2]

Answer: 2

Solution:

Number of lone pair on central atom for H_2O and XeF_4 is equal to 2.

Question100

Based upon VSEPR theory, match the shape (geometry) of the molecules in List-I with the molecules in List-II and select the most

appropriate option.

List - I (Shape)		List - II (Molecules)	
(A)	T-shaped	(I)	XeF_4
(B)	Trigonal planar	(II)	SF_4
(C)	Square planar	(III)	ClF_3
(D)	See-saw	(IV)	BF_3

[27-Jun-2022-Shift-1]

Options:

A. (A) – (I), (B) – (II), (C) – (III), (D) – (IV)

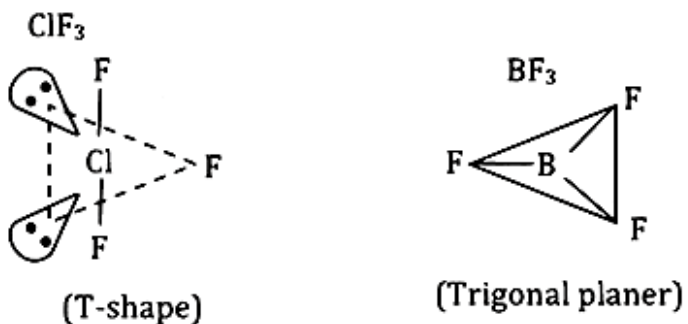
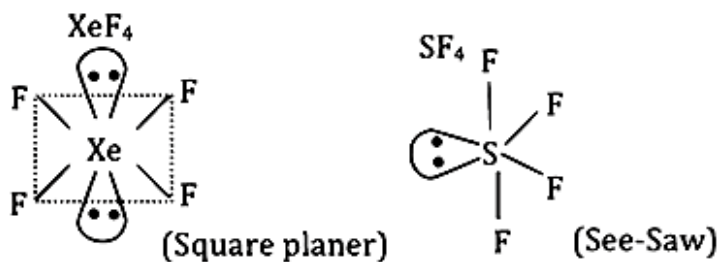
B. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

C. (A) – (III), (B) – (IV), (C) – (I), (D) – (I)

D. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)

Answer: B

Solution:



Question101

Identify the incorrect statement for PCl_5 from the following.

[27-Jun-2022-Shift-2]

Options:

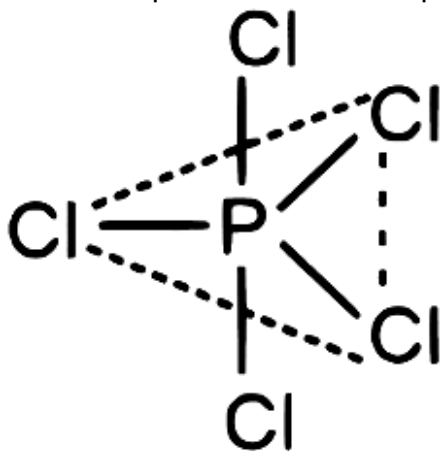
- A. In this molecule, orbitals of phosphorous are assumed to undergo sp^3d hybridization.
- B. The geometry of PCl_5 is trigonal bipyramidal.
- C. PCl_5 has two axial bonds stronger than three equatorial bonds.
- D. The three equatorial bonds of PCl_5 lie in a plane.

Answer: C

Solution:

PCl_5

All three equatorial bonds in a plane



sp^3d hybridization

Trigonal bipyramidal

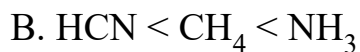
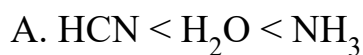
Axial bonds are weaker than equatorial bonds.

Question102

The correct order of increasing intermolecular hydrogen bond strength is

[27-Jun-2022-Shift-2]

Options:



Answer: C

Solution:

Due to the high difference in electronegativity of H and N the H-bond strength of NH_3 is highest. There is no H bond in CH_4 .

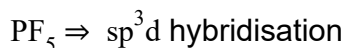


Question103

The hybridization of P exhibited in PF_5 is sp^xd^y . The value of y is ____
[28-Jun-2022-Shift-1]

Answer: 1

Solution:



(5 sigma bonds, zero lone pair on central atom)

Value of y = 1

Question104

In the structure of SF_4 , the lone pair of electrons on S is in.
[28-Jun-2022-Shift-2]

Options:

- A. equatorial position and there are two lone pair - bond pair repulsions at 90° .
- B. equatorial position and there are three lone pair - bond pair repulsions at 90° .
- C. axial position and there are three lone pair - bond pair repulsion at 90° .
- D. axial position and there are two lone pair - bond pair repulsion at 90° .

Answer: A

Solution:

$\text{SF}_4 \rightarrow \text{sp}^3\text{d}$ hybridisation.

The lone pair of electrons on S is in an equatorial position and there are two lone pair-bond pair repulsions at 90° .

Question105

Arrange the following in the decreasing order of their covalent character:

- (A) LiCl
- (B) NaCl
- (C) KCl
- (D) CsCl

**Choose the most appropriate answer from the options given below :
[29-Jun-2022-Shift-1]**

Options:

- A. (A) > (C) > (B) > (D)
- B. (B) > (A) > (C) > (D)
- C. (A) > (B) > (C) > (D)
- D. (A) > (B) > (D) > (C)

Answer: C

Solution:

Covalent character $\propto$ polarising power of cation
Correct decreasing order of covalent character
 $\text{LiCl} > \text{NaCl} > \text{KCl} > \text{CsCl}$

Question106

Consider the species CH_4 , NH_4^+ and BH_4^- . Choose the correct option with respect to the these species.

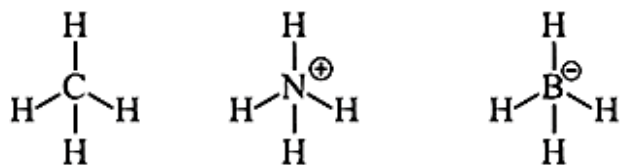
[29-Jun-2022-Shift-2]

Options:

- A. They are isoelectronic and only two have tetrahedral structures.
- B. They are isoelectronic and all have tetrahedral structures.
- C. Only two are isoelectronic and all have tetrahedral structures.
- D. Only two are isoelectronic and only two have tetrahedral structures.

Answer: B

Solution:



All are tetrahedral and each have 10 electrons.

Question107

Number of lone pair(s) of electrons on central atom and the shape BrF_3 molecule respectively, are

[29-Jun-2022-Shift-2]

Options:

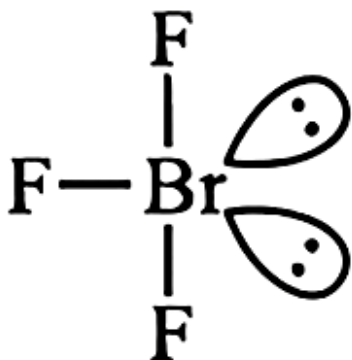
- A. 0, triangular planar.
- B. 1, pyramidal.

C. 2, bent T-shape.

D. 1, bent T-shape.

Answer: C

Solution:



Steric no. = 5(sp^3d), lone pair = 2
Bent T shape.

Question108

Among the following species

N_2 , N_2^+ , N_2^- , N_2^{2-} , O_2 , O_2^+ , O_2^- , O_2^{2-}

the number of species showing diamagnetism is

[25-Jul-2022-Shift-1]

Answer: 2

Solution:

Those species which have unpaired electrons are called paramagnetic species.

And those species which have no unpaired electrons are called diamagnetic species.

(1) N_2 has 14 electrons.

Molecular orbital configuration of N_2

$$= \sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \pi_{2p_x^2} = \pi_{2p_y^2} \sigma_{2p_z^2}$$

Here no unpaired electron present, so it is diamagnetic.

(2) Molecular orbital configuration of N_2^+ (13 electrons)

$$= \sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \pi_{2p_x^2} = \pi_{2p_y^2} \sigma_{2p_z^1}$$

Here in N_2^+ , 1 unpaired electron present, so it is paramagnetic.

(3) N_2^{2-} has 16 electrons.

Molecular orbital configuration of N_2^{2-} is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^1}^* = \pi_{2p_y^1}^*$$

Here 2 unpaired electron present, so it is paramagnetic.

(4) N_2^- has 15 electrons.

Molecular orbital configuration of N_2^- is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^1}^* = \pi_{2p_y^0}^*$$

Here 1 unpaired electron present, so it is paramagnetic.

(a) O_2^{2-} has 18 electrons.

Molecular orbital configuration of O_2^{2-} is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^2}^* = \pi_{2p_y^2}^*$$

Here is no unpaired electron so it is diamagnetic.

(b) O_2^- has 17 electrons.

Molecular orbital configuration of O_2^{2-} is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^2}^* = \pi_{2p_y^1}^*$$

Here 1 unpaired electron present, so it is paramagnetic.

(c) O_2 has 16 electrons.

Molecular orbital configuration of O_2 is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^1}^* = \pi_{2p_y^1}^*$$

Here 2 unpaired electron present, so it is paramagnetic.

(d) O_2^+ has 15 electrons.

Molecular orbital configuration of O_2^+ is

$$\sigma_{1s^2} \sigma_{1s^2}^* \sigma_{2s^2} \sigma_{2s^2}^* \sigma_{2p_z^2} \pi_{2p_x^2} = \pi_{2p_y^2} \pi_{2p_x^1}^* = \pi_{2p_y^0}^*$$

Here 1 unpaired electron present, so it is paramagnetic.

Question 109

The total number of acidic oxides from the following list is

NO, N_2O , B_2O_3 , N_2O_5 , CO, SO_3 , P_4O_{10}

[25-Jul-2022-Shift-2]

Options:

A. 3

B. 4

C. 5

D. 6

Answer: B

Solution:

NO, N₂O, CO - neutral oxides

B₂O₃, N₂O₅, SO₃, P₄O₁₀ – acidic oxides

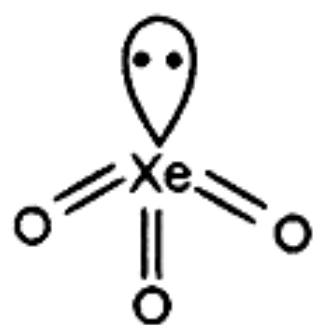
Question110

The sum of number of lone pairs of electrons present on the central atoms of XeO₃, XeOF₄ and XeF₆, is

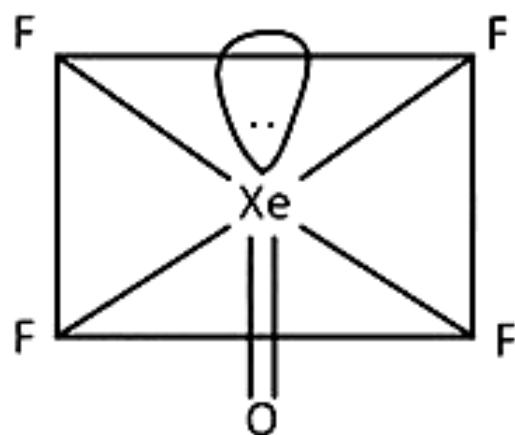
[25-Jul-2022-Shift-2]

Answer: 3

Solution:

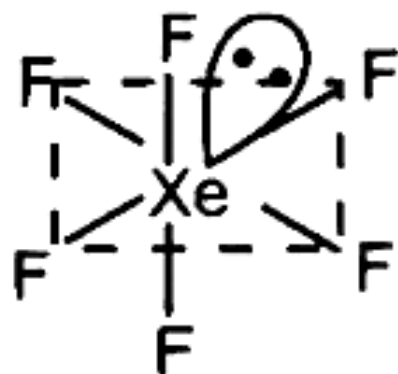


lone pair
on central atom = 1



sp^3d^2 hybridization

From structure, it is clear that it has five bond pairs and one lone pair.



lone pair
on central atom = 1

Question111

Match List-I with List-II :

List I (Compound)	List II (Shape)
(A) BrF_5	(I) bent
(B) $[\text{CrF}_6]^{3-}$	(II) square pyramidal
(C) O_3	(III) trigonal bipyramidal
(D) PCl_5	(IV) octahedral

**Choose the correct answer from the options given below :
[26-Jul-2022-Shift-1]**

Options:

- A. (A) – (I), (B) – (II), (C) – (II), (D) – (IV)
- B. (A) – (IV), (B) – (II), (C) – (II), (D) – (I)
- C. (A) – (II), (B) – (IV), (C) – (I), (D) – (III)
- D. (A) – (II), (B) – (IV), (C) – (II), (D) – (I)

Answer: C

Solution:

- (A) BrF_5 - square pyramidal
- (B) $[\text{CrF}_6]^{3-}$ - octahedral
- (C) O_3 – bent
- (D) PCl_5 - trigonal bipyramidal
-

Question112

Arrange the following in increasing order of their covalent character.

- A. CaF_2
- B. CaCl_2

C. CaBr_2

D. CaI_2

Choose the correct answer from the options given below.

[26-Jul-2022-Shift-2]

Options:

A. $B < A < C < D$

B. $A < B < C < D$

C. $A < B < D < C$

D. $A < C < B < D$

Answer: B

Solution:

From Fajan's rule, for a given metal ion, as the size of anion increases, polarizability of anion increases and hence covalent character of the given ionic compound increases.

Hence, the increasing order of covalent character is $\text{CaF}_2 < \text{CaCl}_2 < \text{CaBr}_2 < \text{CaI}_2$

Question113

Given below are two statements.

Statement I: I_2O_2 , Cu^{2+} , and Fe^{3+} are weakly attracted by magnetic field and are magnetized in the same direction as magnetic field.

Statement II: NaCl and H_2O are weakly magnetized in opposite direction to magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below.

[27-Jul-2022-Shift-1]

Options:

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

Answer: A

Solution:

O_2 , Cu^{2+} and Fe^{3+} have 2, 1 and 5 unpaired electrons respectively, so these are the paramagnetic species. Hence, they are attracted by magnetic field.

$NaCl$ and H_2O are the diamagnetic species so they are repelled by the magnetic field.

Question114

Amongst the following, the number of oxide(s) which are paramagnetic in nature is

Na_2O , KO_2 , NO_2 , N_2O , ClO_2 , NO , SO_2 , Cl_2O

[27-Jul-2022-Shift-1]

Answer: 4

Solution:

Paramagnetic species: KO_2 , NO_2 , ClO_2 , NO

Diamagnetic species are: Na_2O , N_2O , SO_2 , Cl_2O

Question115

According to MO theory, number of species/ions from the following having identical bond order is _____.

CN^- , NO^+ , O_2 , O_2^+ , O_2^{2+}

[27-Jul-2022-Shift-1]

Answer: 3

Solution:

CN^- , NO^+ and O_2^{2+} have bond order of 3

O_2 has bond order of 2

O_2^+ has bond order of 2.5

$\therefore$ 3 species have similar bond order.

Question 116

Match List - I with List - II.

List I	List II
(A) $\psi_{\text{MO}} = \psi_{\text{A}} - \psi_{\text{B}}$	(I) Dipole moment
(B) $\mu = Q \times r$	(II) Bonding molecular orbital
(C) $\frac{N_{\text{b}} - N_{\text{a}}}{2}$	(III) Anti-bonding molecular orbital
(D) $\psi_{\text{MO}} = \psi_{\text{A}} + \psi_{\text{B}}$	(IV) Bond order

Choose the correct answer from the options given below :
[27-Jul-2022-Shift-2]

Options:

- A. (A) – (II), (B) – (I), (C) – (IV), (D) – (III)
- B. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
- C. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)
- D. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)

Answer: C

Solution:

$\Psi_A - \Psi_B = \Psi_{MO}$ is anti-bonding molecular orbital

$\mu = Q \times r$ -is dipole moment

$\frac{N_b - N_a}{2} = \text{bond order}$

$\Psi_A + \Psi_B = \Psi_{MO}$ is bonding molecular orbital.

Question117

The number of molecule(s) or ion(s) from the following having non-planar structure is _____.

NO_3^- , H_2O_2 , BF_3 , PCl_3 , XeF_4 , SF_4 , XeO_3 , PH_4^+ , SO_3 , $[\text{Al}(\text{OH})_4]^-$

[27-Jul-2022-Shift-2]

Answer: 6

Solution:

$\text{NO}_3^- \rightarrow \text{Trigonal planar (Planar)}$

$\text{H}_2\text{O}_2 \rightarrow \text{Open book (Non-planar)}$

$\text{BF}_3 \rightarrow \text{Trigonal planar (Planar)}$

$\text{PCl}_3 \rightarrow \text{Pyramidal (Non-planar)}$

$\text{XeF}_4 \rightarrow \text{Square planar (Planar)}$

$\text{SF}_4 \rightarrow \text{See-Saw (Non-planar)}$ $\text{XeO}_3 \rightarrow \text{Pyramidal (Non-planar)}$

$\text{PH}_4^+ \rightarrow \text{Tetrahedral (Non-planar)}$

$\text{SO}_3 \rightarrow \text{Trigonal planar (Planar)}$

$[\text{Al}(\text{OH})_4]^- \rightarrow \text{Tetrahedral (Non-planar)}$

Question118

The number of paramagnetic species among the following is _____.

B_2 , Li_2 , C_2 , C_2^- , O_2^{2-} , O_2^+ and He_2^+

[28-Jul-2022-Shift-1]

Answer: 4

Solution:

Those species which have unpaired electrons are called paramagnetic species.
And those species which have no unpaired electrons are called diamagnetic species.

B_2 has 10 electrons.

Molecular orbital configuration of B_2 is

$$\sigma_{1s^2}\sigma_{1s^2}^* \sigma_{2s^2}\sigma_{2s^2}^* \pi_{2p_x^1} = \pi_{2p_y^1}$$

Here two unpaired electrons present. So it is paramagnetic.

O_2^{2-} has 18 electrons.

Molecular orbital configuration of O_2^{2-} is

$$\sigma_{1s^2}\sigma_{1s^2}^* \sigma_{2s^2}\sigma_{2s^2}^* \sigma_{2p_z^2}\pi_{2p_x^2} = \pi_{2p_y^2}\pi_{2p_x^2}^* = \pi_{2p_y^2}^*$$

Here is no unpaired electron so it is diamagnetic.

O_2^+ has 15 electrons.

Molecular orbital configuration of O_2^+ is

$$\sigma_{1s^2}\sigma_{1s^2}^* \sigma_{2s^2}\sigma_{2s^2}^* \sigma_{2p_z^2}\pi_{2p_x^2} = \pi_{2p_y^2}\pi_{2p_x^1}^* = \pi_{2p_y^0}^*$$

Here 1 unpaired electron present, so it is paramagnetic.

C_2 has 12 electrons.

Molecular orbital configuration of C_2

$$= \sigma_{1s^2}\sigma_{1s^2}^* \sigma_{2s^2}\sigma_{2s^2}^* \pi_{2p_x^2} = \pi_{2p_y^2}$$

Here no unpaired electron present, so it is diamagnetic.

C_2^- has 13 electrons.

Molecular orbital configuration of C_2^- is

$$\sigma_{1s^2}\sigma_{1s^2}^* \sigma_{2s^2}\sigma_{2s^2}^* \pi_{2p_x^2} = \pi_{2p_y^2}\sigma_{2p_z^1}$$

Here 1 unpaired electron present, so it is paramagnetic.

Li_2 has 6 electrons.

$$Li_2 = \sigma_{1s^2}\sigma_{1s^2}^* \sigma_{2s^2}$$

Here no unpaired electron present, so it is diamagnetic.

Configuration of He_2^+ (3 electrons) is $= \sigma_{1s^2}\sigma_{1s^1}^*$

Here 1 unpaired electron present, so it is paramagnetic.

Question 119

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Zero orbital overlap is an out of phase overlap.

Reason R: It results due to different orientation ∨ direction of approach of orbitals.

In the light of the above statements, choose the correct answer from the options given below

[28-Jul-2022-Shift-2]

Options:

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is NOT the correct explanation of A
- C. A is true but R is false
- D. A is false but R is true

Answer: A

Solution:

Zero overlapping is something in which there is no overlapping between two orbitals. The first condition is that the two orbitals should not be symmetrical and the second condition is that both orbitals should be in different planes.

Question120

Number of lone pairs of electrons in the central atom of SCl_2 , O_3 , ClF_3 and SF_6 , respectively, are :

[29-Jul-2022-Shift-1]

Options:

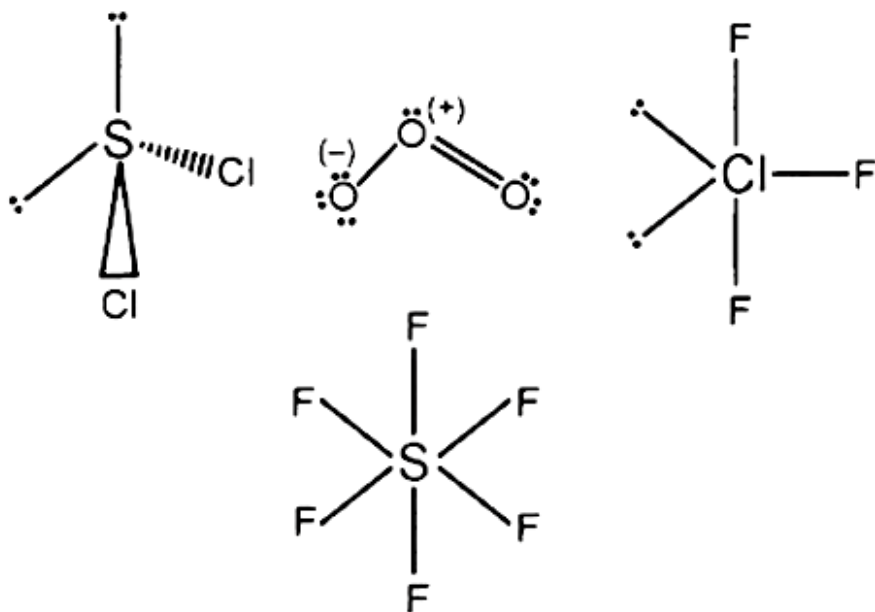
- A. 0, 1, 2 and 2
- B. 2, 1, 2 and 0
- C. 1, 2, 2 and 0
- D. 2, 1, 0 and 2

Answer: B

Solution:

The number of lone pair of electrons in the central atom of SCl_2 , O_3 , ClF_3 and SF_6 are 2, 1, 2 and 0 respectively

Their structures are as,



Question121

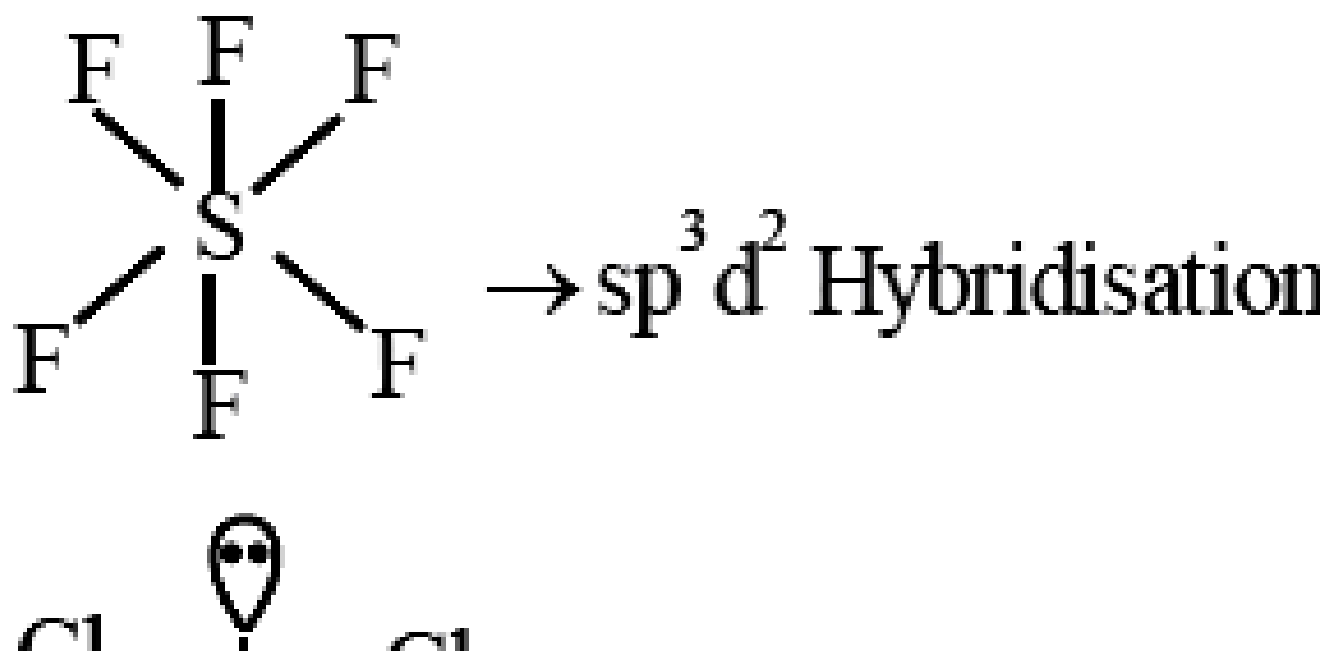
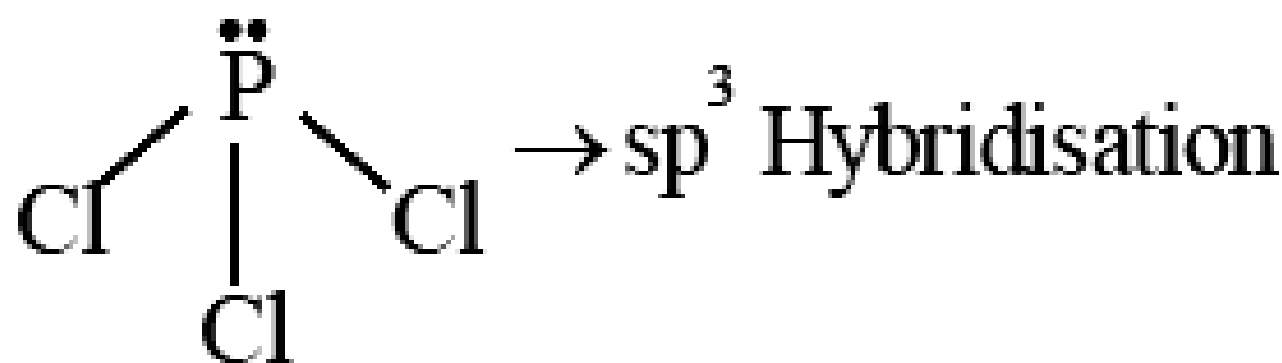
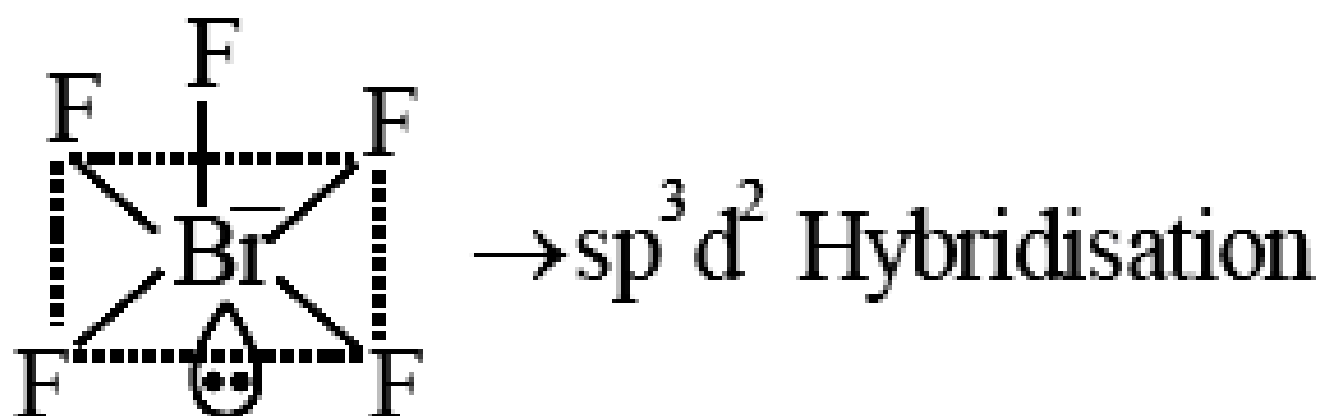
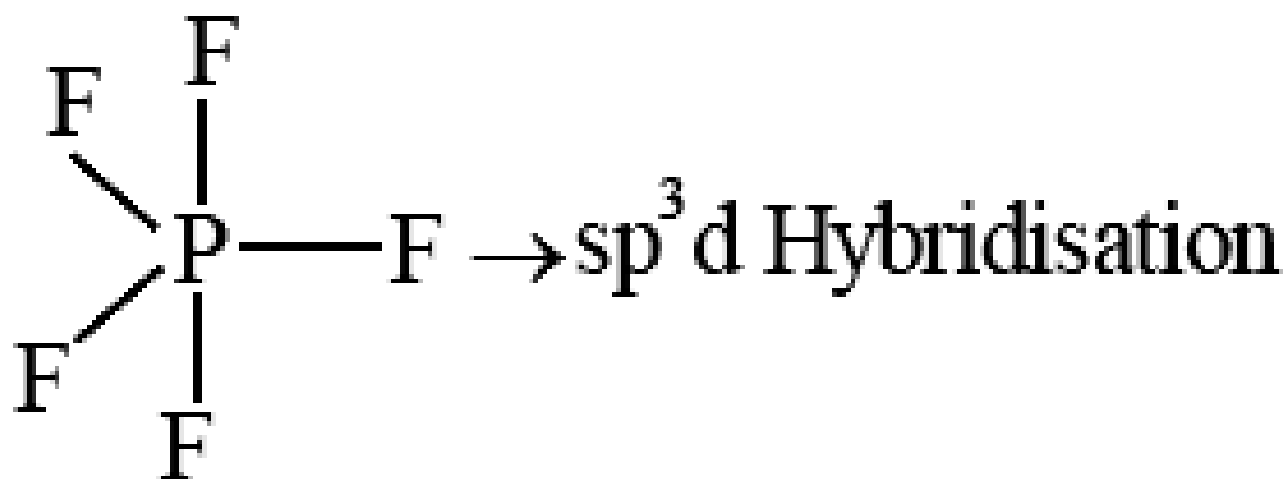
Consider, PF_5 , BrF_5 , PCl_3 , SF_6 , $[\text{ICl}_4]^-$, ClF_3 and IF_5 .

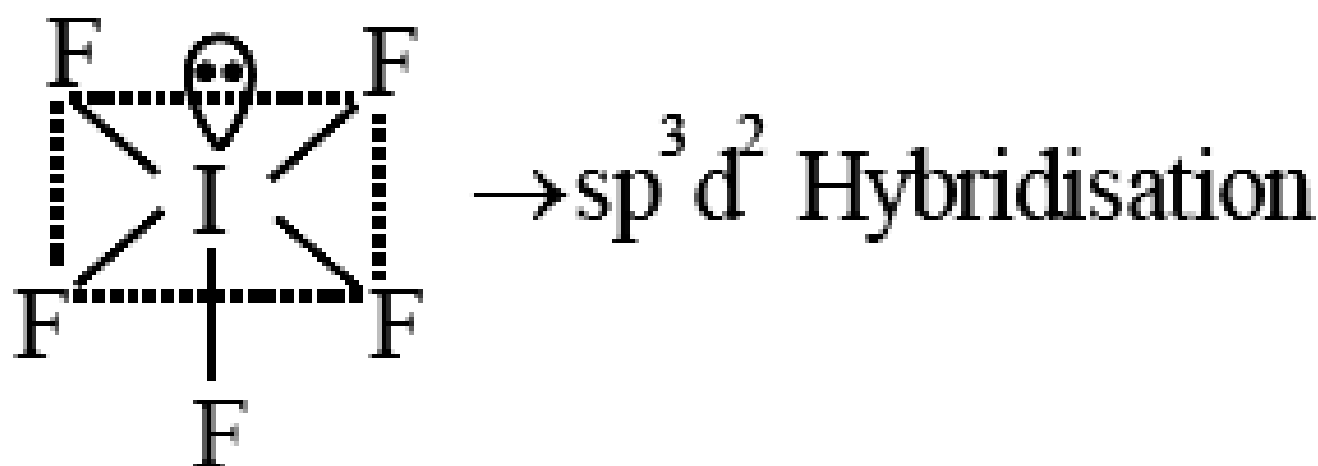
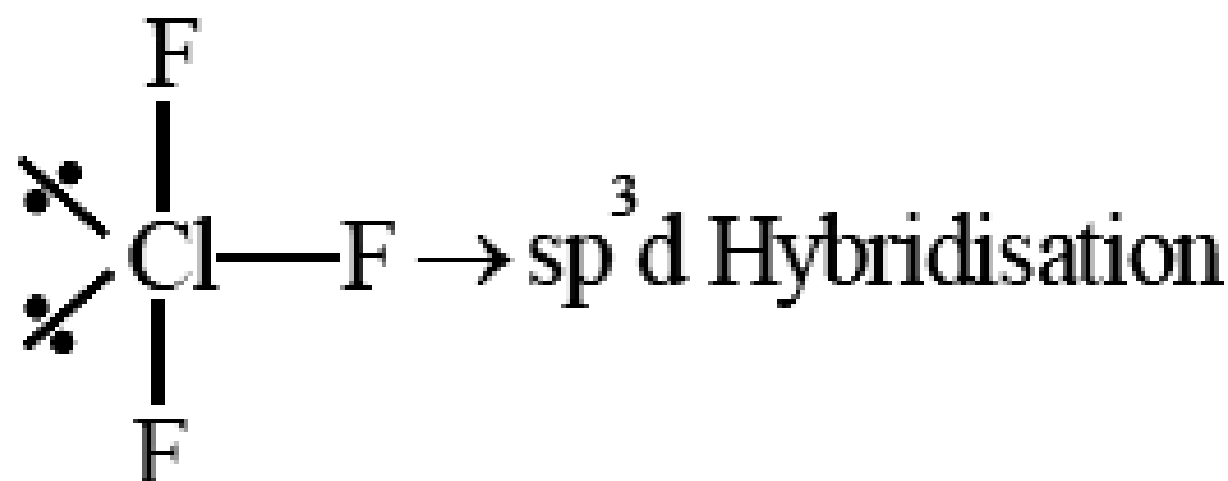
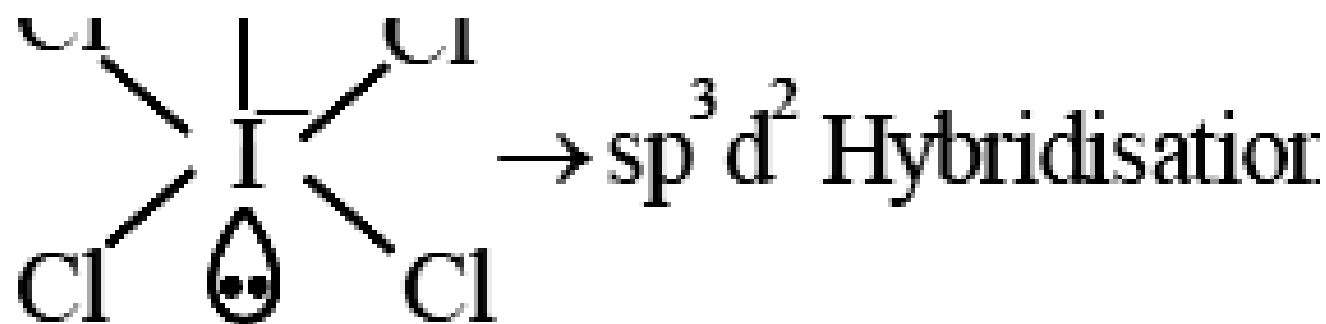
Amongst the above molecule(s)/ion(s), the number of molecule(s)/ion(s) having sp^3d^2 hybridisation is _____.

[29-Jul-2022-Shift-2]

Answer: 4

Solution:





Question122

Match List-I with List-II

	List-I(Molecule)		List-II(Bond order)
A.	Ne_2	(i)	1
B.	N_2	(ii)	2
C.	F_2	(iii)	0
D.	O_2	(iv)	3

Choose the correct answer from the options given below.

[26 Feb 2021 Shift 2]

Options:

A. (A-iii), (B-iv), (C-i), (D-ii)

B. (A-i), (B-ii), (C-iii), (D-iv)

C. (A-ii), (B-i), (C-iv), (D-iii)

D. (A-iv), (B-iii), (C-ii), (D-i)

Answer: A

Solution:

(A) $Ne_2(20e^-)$

$$\Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^2 \pi^* 2p_y^2 \sigma^* 2p_z^2$$

$$\text{Bond order} = \frac{10 - 10}{2} = 0 \Rightarrow \text{(iii) of List-II.}$$

(B) $N_2(14e^-)$

$$\Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2, \pi 2p_y^2 \sigma 2p_z^2$$

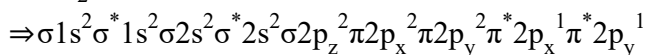
$$\text{Bond order} = \frac{10 - 4}{2} = 3 \Rightarrow \text{(iv) of List-II.}$$

(c) $F_2(18e^-)$

$$\Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^2 \pi^* 2p_y^2$$

$$\text{Bond order} = \frac{10 - 8}{2} = 1 \Rightarrow \text{(i) of List-II.}$$

(d) $O_2(16e^-)$



$$\text{Bond order} = \frac{10 - 6}{2} = 2 \Rightarrow \text{(ii) of List-II.}$$

The correct matching is option (a).

No. of electrons in antibonding

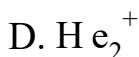
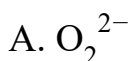
MO (Molecular Orbital)

$$\text{Note : Bond order} = \frac{\text{No. of electrons in bonding MO} - \text{No. of electrons in antibonding MO}}{2}$$

Question 123

According to molecular orbital theory, the species among the following that does not exist is
[25 Feb 2021 Shift 1]

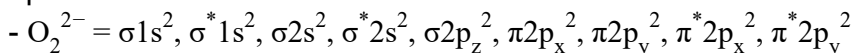
Options:



Answer: C

Solution:

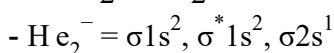
According to Molecular Orbital Theory (MOT), electronic configuration and their bond order of given options are as follows



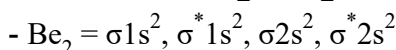
[No. of bonding electrons

$$\text{Bond order} = \frac{\text{No. of bonding electrons} - \text{No. of antibonding electrons}}{2}$$

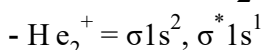
$$= \frac{10 - 8}{2} = \frac{2}{2} = 1$$



$$\text{Bond order} = \frac{3 - 2}{2} = \frac{1}{2} = 0.5$$



$$\text{Bond order} = \frac{4 - 4}{2} = 0$$



$$\text{Bond order} = \frac{2-1}{2} = \frac{1}{2} = 0.5$$

If bond order of chemical species is zero then that chemical species does not exist. Therefore, Be_2 does not exist.

Question124

Which of the following are isostructural pairs?



[24feb2021shift1]

Options:

A. C and D only

B. A and B only

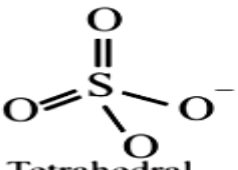
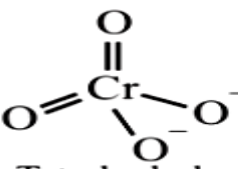
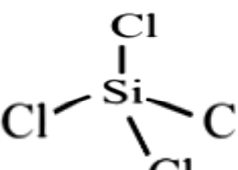
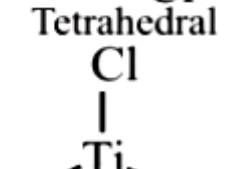
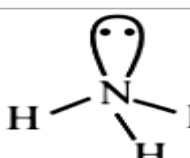
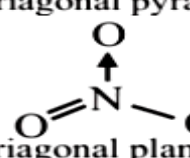
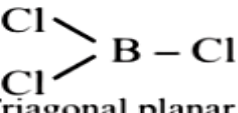
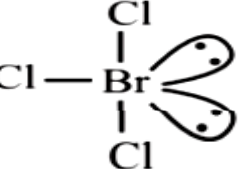
C. A and C only

D. B and C only

Answer: B

Solution:

Isostructural means same structure.

Species	Structure
(A) SO_4^{2-} CrO_4^{2-}	 Tetrahedral  Tetrahedral
(B) $SiCl_4$ $TiCl_4$	 Tetrahedral  Tetrahedral
(C) NH_3 NO_3^-	 Triagonal pyramidal  Triagonal planar
(D) BCl_3 $BrCl_3$	 Triagonal planar  T-shape

Question125

Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond formation.

Reason (R) Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.

In the light of the above statements, choose the most appropriate answer from the options given below.

[26 Feb 2021 Shift 1]

Options:

- A. A is false but R is true.
- B. Both A and R are true and R is the correct explanation of A.
- C. A is true R is false.
- D. Both A and R are true but R is not the correct explanation of A.

Answer: A

Solution:

Assertion is false but Reason is true.

Corrected Assertion Dipole-dipole interactions are purely covalent interactions of dipolar covalent bonds.

Ends of dipoles of a polar covalent bonds possess partial charges (δ^+ and δ^-) which are less than unit electronic charge, $e = 1.6 \times 10^{-19} \text{C}$.

Question126

The correct shape and I-I-I bond angles respectively in I_3^- , ion are
[24 Feb 2021 Shift 2]

Options:

- A. distorted trigonal planar, 135° and 90°
- B. T-shaped, 180° and 90°
- C. Trigonal planar, 120°
- D. Linear, 180°

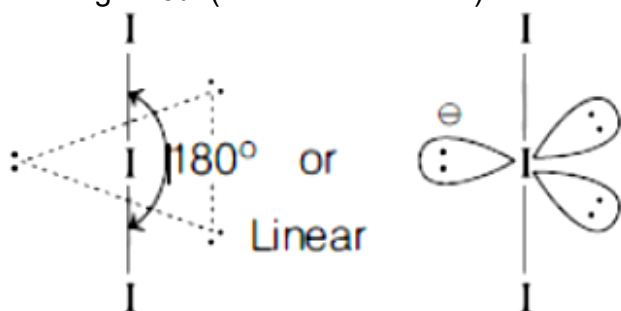
Answer: D

Solution:

Hybridisation of central I in I_3^- is sp^3d with 3 lone pair and 2 bond pair.

Shape Linear Lone pair 3 lone pair

Bond angle 180° (for linear molecule)



Question127

A hard substance melts at high temperature and is an insulator in both solid and in molten state.

This solid is most likely to be a
/an

[18 Mar 2021 Shift 2]

Options:

- A. ionic solid
- B. molecular solid

C. metallic solid

D. covalent solid

Answer: D

Solution:

Covalent or network solid have very high melting point and they are insulators in their solid and molten form.

If a substance is an insulator in both solid and molten phases, then it cannot be ionic or metallic solid.

If the melting point is higher, then it cannot be a molecular solid.

∴ It should be covalent network solid.

Question128

The number of species below that have two lone pairs of electrons in their central atom is (Round off to the nearest integer)

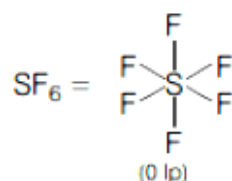
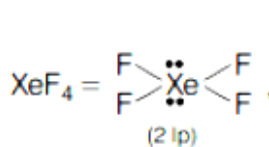
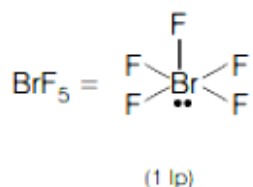
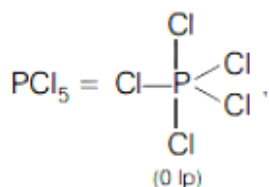
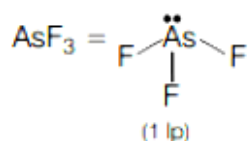
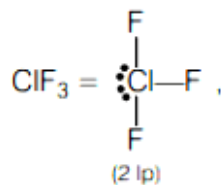
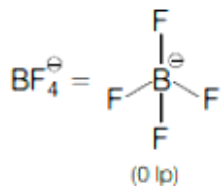
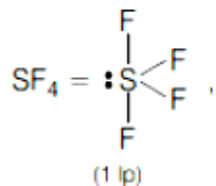
SF_4 , BF_4^- , ClF_3 , AsF_3 , PCl_5 , BrF_5 , XeF_4 , SF_6

[18 Mar 2021 Shift 2]

Answer: 2

Solution:

Number of lone pairs on central atom in given compounds are as follows



Hence, two (lone pair) lp on central atom is ClF_3 and XeF_4 .

Question129

Amongst the following, the linear species is
[17 Mar 2021 Shift 2]

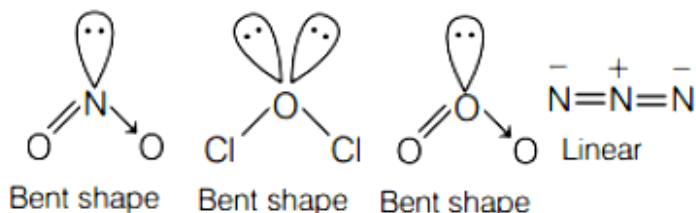
Options:



Answer: D

Solution:

N_3^- is linear species.



Hybridisation = Number of sigma bond + number of lone pairs + number of coordinate bonds.

Hybridisation of N in NO = $1 + 1 + 1 = 3$. So, hybridisation is sp^2 and structure is bent or V-shape.

Hybridisation of O in $\text{OCl}_2 = 2 + 2 = 4$. So, hybridisation is sp^3 and structure is bent or V-shape.

Hybridisation of O in $\text{O}_3 = 1 + 1 + 1 = 3$. So, hybridisation is sp^2 and structure is bent or V-shape. linear.

Question130

A central atom in a molecule has two lone pairs of electrons and forms three single bonds. The shape of this molecule is
[17 Mar 2021 Shift 1]

Options:

- A. see-saw
- B. planar triangular
- C. T-shaped
- D. trigonal pyramidal

Answer: C

Solution:

2 lone pair + 3 bond pair

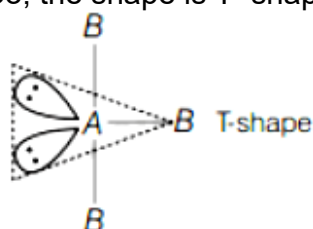
Stearic number = 5

So, general formula is AB_3L_2 .

(Here, L = number of lone pair.)

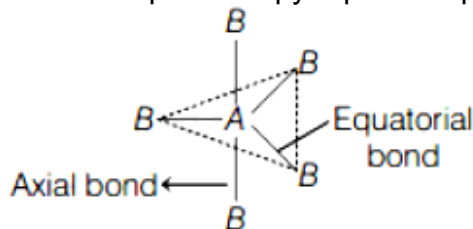
Hybridisation = sp^3d .

So, the shape is T-shape



Note Geometry of the compound having sp^3d hybridisation should to tetragonal bipyramidal (TBP) according to VSEPR theory. But due to two lone pair, geometry is distorted.

In TBP lone pair occupy equatorial position.



Question131

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) The H – O – H bond angle in water molecule is 104.5° .

Reason (R) The lone pair - lone pair repulsion of electrons is higher than the bond pair - bond pair repulsion.

[16 Mar 2021 Shift 1]

Options:

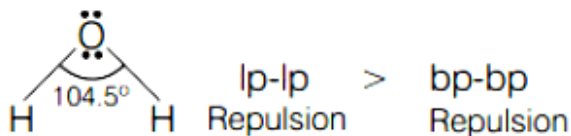
- A. A is false but R is true.
- B. Both A and R are true, but R is not the correct correct explanation of A.
- C. A is true but R is false.
- D. Both A and R are true, and R is the correct explanation of A.

Answer: D

Solution:

Hybridisation of 'O' in H_2O is sp^3 as it is having two lone pair and two sigma bond. Bond angle of sp^3 hybridised compound should be 109.5° .

But in case of water, it is only 104.5° due to repulsion between the lone pairs. Lone pair-lone pair repulsion is higher than bond pair-bond pair repulsion. (VSEPR Theory)



So, both A and R are true and R is the correct explanation of A.

Question132

The total number of electrons in all bonding molecular orbitals of O_2^{2-} is
 (Round off to the nearest integer)
 [27 Jul 2021 Shift 2]

Answer: 10

Solution:

M. O. Configuration of O_2^{2-} ($18e^-$)
 $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2$
 $\pi 2p_x^2 = \pi 2p_y^2$
 Total B.M.O electrons = 10

Question 133

The difference between bond orders of CO and NO^+ is $\frac{x}{2}$ where x = _____.
 (Round off to the Nearest Integer)
 [27 Jul 2021 Shift 1]

Answer: 0

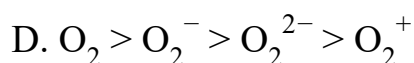
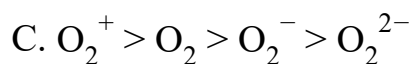
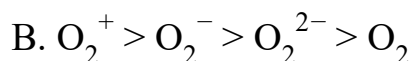
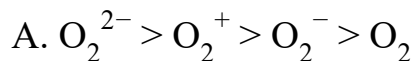
Solution:

Bond order of CO = 3
 $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2$
 B. O. = $\frac{10-4}{2} = 3$
 NO^+ :
 B. O. = $\frac{8-2}{2} = 3$
 Difference = $3 - 3 = 0$

Question134

In the following the correct bond order sequence is:
[25 Jul 2021 Shift 2]

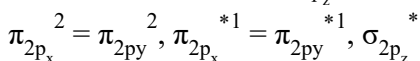
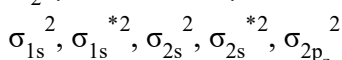
Options:



Answer: C

Solution:

O_2 (16 electrons)



Bond order of $O_2 \Rightarrow 2$

Bond order of $O_2^- \Rightarrow 1.5$

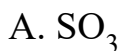
Bond order of $O_2^{2-} \Rightarrow 1$

Bond order of $O_2^+ \Rightarrow 2.5$

Question135

Identify the species having one π -bond and maximum number of canonical forms from the following:
[25 Jul 2021 Shift 2]

Options:



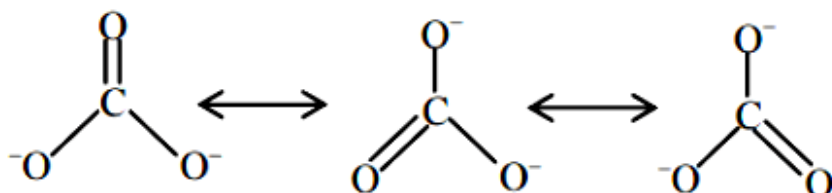
C. SO_2

D. CO_3^{2-}

Answer: D

Solution:

Among SO_3 , O_2 , SO_2 and CO_3^{2-} , only O_2 and CO_3^{2-} has only one π -bond



Question136

**The set in which compounds have different nature is :
[20 Jul 2021 Shift 1]**

Options:

A. B(OH)_3 and H_3PO_3

B. B(OH)_3 and Al(OH)_3

C. NaOH and Ca(OH)_2

D. Be(OH)_2 and Al(OH)_3

Answer: B

Solution:

1) B(OH)_3 acidic and H_3PO_3 acidic

2) B(OH)_3 acidic and Al(OH)_3 amphoteric

3) NaOH basic and Ca(OH)_2 basic

4) Be(OH)_2 amphoteric and Al(OH)_3 amphoteric

Question137

Match List-I with List-II :

List-I (Species)	List-II (Hybrid Orbitals)
(a) SF_4	(i) sp^3d^2
(b) IF_5	(ii) d^2sp^3
(c) NO_2^+	(iii) sp^3d
(d) NH_4^+	(iv) sp^3
	(v) sp

Choose the correct answer from the options given below :
[22 Jul 2021 Shift 2]

Options:

- A. (a)-(i), (b)-(ii), (c)-(v) and (d)-(iii)
- B. (a)-(ii), (b)-(i), (c)-(iv) and (d)-(v)
- C. (a)-(iii), (b)-(i), (c)-(v) and (d)-(iv)
- D. (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(v)

Answer: C

Solution:

(a) SF_4 – sp^3d hybridisation

(b) IF_5 – sp^3d^2 hybridisation

(c) NO_2^+ – sp hybridisation

(d) NH_4^+ – sp^3 hybridisation

Question138

The hybridisations of the atomic orbitals of nitrogen in NO_2^- , NO_2^+ and NH_4^+ respectively are.

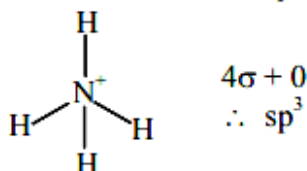
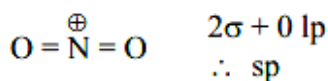
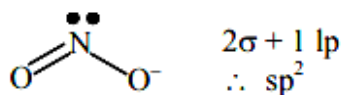
[20 Jul 2021 Shift 2]

Options:

- A. sp^3 , sp^2 and sp
- B. sp , sp^2 and sp^3
- C. sp^3 , sp and sp^2
- D. sp^2 , sp and sp^3

Answer: D

Solution:



Question139

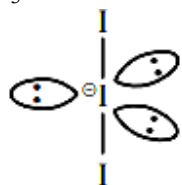
The number of lone pairs of electrons on the central I atom in I_3^- is

[20 Jul 2021 Shift 1]

Answer: 3

Solution:

I_3^- :



The number of lone pairs of electron on the central atom is 3.

Question140

The bond order and magnetic behaviour of O_2^- ion are, respectively
[26 Aug 2021 Shift 2]

Options:

- A. 1.5 and paramagnetic
- B. 1.5 and diamagnetic
- C. 2 and diamagnetic
- D. 1 and paramagnetic

Answer: A

Solution:

O_2^- (Total electrons = 17) =

$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2$

$\pi^* 2p_x^2 = \pi^* 2p_y^1, \sigma^* 2p_z^0$

Number of unpaired electron = 1

So, it is paramagnetic.

Bond order = $\frac{n_b - n_a}{2}$

$= \frac{10 - 7}{2} = 1.5$

Question141

The interaction energy of London forces between two particles is proportional to r^x , where r is the distance between the particles. The

value of x is
[26 Aug 2021 Shift 2]

Options:

- A. 3
- B. -3
- C. -6
- D. 6

Answer: C

Solution:

In London forces, interaction energy is inversely proportional to the sixth power of the distance between two interacting particles

$$\left(\frac{1}{r^6} \text{ or } r^{-6} \right)$$

∴ x is -6.

Question 142

The number of species having non-pyramidal shape among the following is

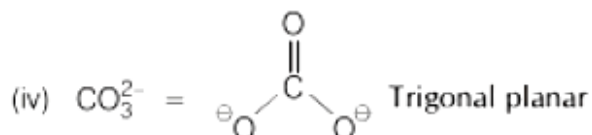
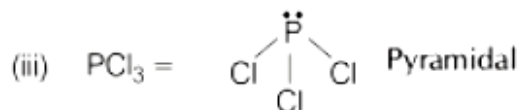
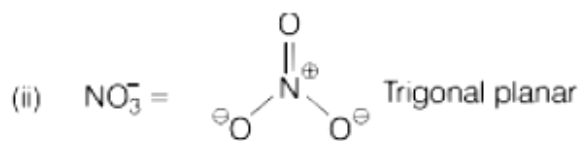
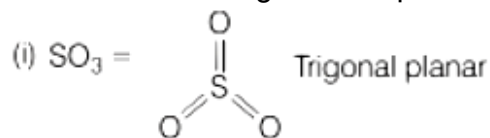
(i) SO_3 (ii) NO_3^- (iii) PCl_3 (iv) CO_3^{2-}

[27 Aug 2021 Shift 2]

Answer: 3

Solution:

The structure of the given compounds are as follows :



3 molecules/ions i.e. (i), (ii) and (iv) have non-pyramidal shape.

Question143

According to molecular orbital theory, the number of unpaired electron(s) in O_2^{2-} is

[31 Aug 2021 Shift 2]

Answer: 0

Solution:

In O_2^{2-} , 18 electrons are present.

Configuration is given by

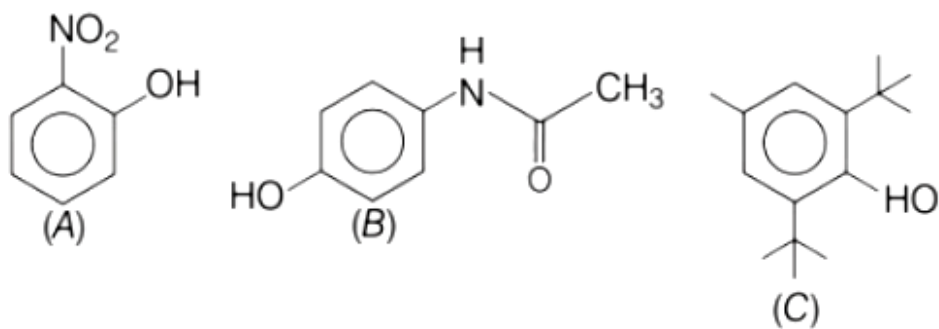
$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2$$

$$\pi^* 2p_x^2 = \pi^* 2p_y^2, \sigma^* 2p_z^0$$

So, number of unpaired electrons in $\text{O}_2^{2-} = 0$.

Question144

The compound/s which will show significant intermolecular H-bonding is/are



[27 Aug 2021 Shift 2]

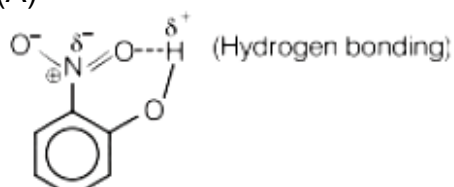
Options:

- A. (B) only
- B. (C) only
- C. (A) and (B)
- D. (A),(B) and (C)

Answer: A

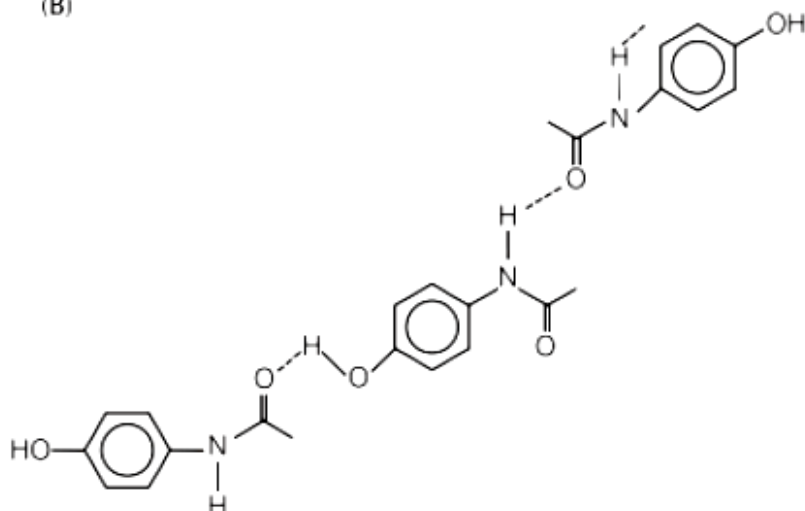
Solution:

(A)



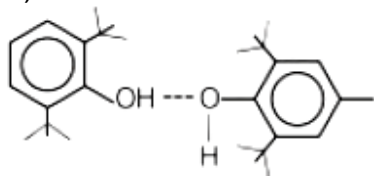
In o-nitrophenol intramolecular hydrogen bonding takes place.

(B)



Extensive H-bonding is observed in this case as two H-atoms bonded to strongly electronegative atoms (nitrogen, oxygen) are present. H-bonding occurs when H-bonded to strongly electronegative atoms exists in the vicinity of other electronegative atoms possessing lone pairs of electron.

(C)



Here, only one hydrogen atom bonded to electronegative oxygen atom is present. So, intermolecular H-bonding is present but not so extensive.
So, option (a) is correct.

Question145

The spin-only magnetic moment value of B_2^+ species is

$\times 10^{-2}$ BM. (Nearest integer)

[Given, $\sqrt{3} = 1.73$]

[1 Sep 2021 Shift 2]

Answer: 173

Solution:

Spin only magnetic moment is given as

$$\mu_s = \sqrt{n(n+2)} \text{ BM}$$

where, n = number of unpaired electrons

Electronic configuration of B_2^+ is as follows

$$B_2^+ = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_y^1 = \pi 2p_z^0$$

$\therefore B_2^+$ has one unpaired electron,

$$\mu = \sqrt{1(1+2)} = \sqrt{3} = 1.73 \text{ BM}$$

$$= 173 \times 10^{-2} \text{ BM}$$

$$\therefore x \times 10^{-2} = 173 \times 10^{-2}$$

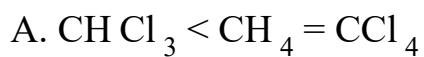
$$\therefore x = 173$$

Hence, answer is 173.

Question146

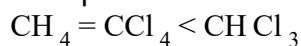
The dipole moments of CCl_4 , $CHCl_3$ and CH_4 are in the order:

[Jan. 07, 2020 (I)]

Options:**Answer: D****Solution:**

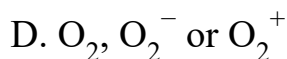
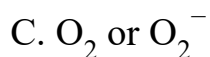
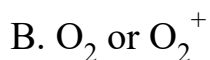
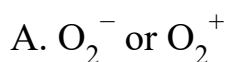
Hint: Dipole moment is the separation of two charges by a distance. When one atom is more electronegative than the other, it tries to pull the shared electrons towards itself. Hence, greater the electronegativity of one atom than the other, greater will be the length and more dipole moment.

So dipole moment order is:



Question 147

If the magnetic moment of a dioxygen species is 1.73 B.M. , it may be:
[Jan. 09, 2020 (I)]

Options:**Answer: A****Solution:**

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

$$1.73 = \sqrt{n(n+2)}$$

$$n = 1$$

$$\begin{aligned}
 O_2^+ &= \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \\
 &= \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^0 \\
 O_2^- &= \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \\
 &= \pi 2p_y^2 \pi^* 2p_x^2 = \pi^* 2p_y^1
 \end{aligned}$$

Question148

**The bond order and the magnetic characteristics of CN^- are:
[Jan. 07, 2020 (II)]**

Options:

- A. $2 \frac{1}{2}$, diamagnetic
- B. 3 , diamagnetic
- C. 3 , paramagnetic
- D. $2 \frac{1}{2}$, paramagnetic

Answer: B

Solution:

Total number of electrons in $CN^- = 6 + 7 + 1 = 14$
 $\therefore$ Molecular orbital distribution

$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \left[\begin{array}{c} \pi 2p_x^2 \\ \pi 2p_y^2 \end{array} \right] \sigma 2p_z^2$$

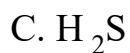
$$\therefore \text{Bond order} = \frac{10 - 4}{2} = 3$$

CN is diamagnetic because all electrons are paired.

Question149

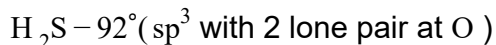
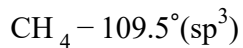
**The compound that has the largest $H - M - H$ bond angle
($M = N, S, C$), is
[Sep. 05, 2020 (II)]**

Options:



Answer: D

Solution:



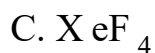
Lone pair-bond pair repulsion in H_2S will increase because 'S' has lower electronegativity than 'O'. So there will be lesser electron density on 'S' and thus $\text{H} - \text{S} - \text{H}$ bond angle will be smaller than H_2O .

Question150

The molecule in which hybrid MOs involve only one d-orbital of the central atom is:

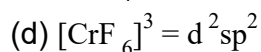
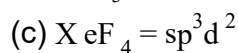
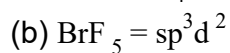
[Sep. 04, 2020 (II)]

Options:



Answer: A

Solution:



Question151

If AB_4 molecule is a polar molecule, a possible geometry of AB_4 is:
[Sep. 02,2020(I)]

Options:

- A. Square pyramidal
- B. Tetrahedral
- C. Rectangular planar
- D. Squareplanar

Answer: A

Solution:

For AB_4 compound possible geometry are

No. of Bond pair

4

4

4

Structure with sp^3d^2 hybridisation is polar due to lone pair moment while in other possibilities molecules is non-polar. Square pyramidal can be polar due to lone pair moment as the bond pair moments will get cancelled out.

Question152

The shape/ structure of $[\text{XeF}_5]^-$ and XeO_3F_2 , respectively, are:
[Sep. 02,2020 (II)]

Options:

- A. pentagonal planar and trigonal bipyramidal

- B. octahedral and square pyramidal
- C. trigonal bipyramidal and pentagonal planar
- D. trigonal bipyramidal and trigonal bipyramidal

Answer: A

Solution:

(i) XeF_5^- St. No. = Bond pair + Lone Pair
 $= (5 + 2) = 7$

So, hybridisation is $= \text{sp}^3\text{d}^2$ and structure is pentagonal planar.

(ii) XeO_3F_2 St. No. = 5

So, hybridisation is $= \text{sp}^3\text{d}^2$ and structure is trigonal bipyramidal.

Question 153

The molecular geometry of SF_6 is octahedral. What is the geometry of SF_4 (including lone pair(s) of electrons, if any)?

[Sep. 02, 2020 (II)]

Options:

- A. Tetrahedral
- B. Trigonal bipyramidal
- C. Pyramidal
- D. Square planar

Answer: B

Solution:

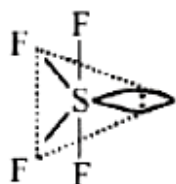
SF_4

Bond pair = 4

Lone pair = 1

Steric number = 5,,

So, hybridisation is sp^3d .



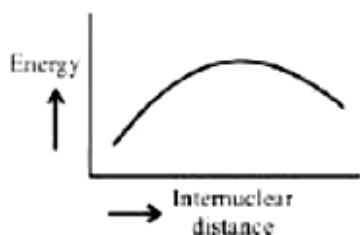
Geometry is trigonal bipyramidal but shape is "See Saw".

Question154

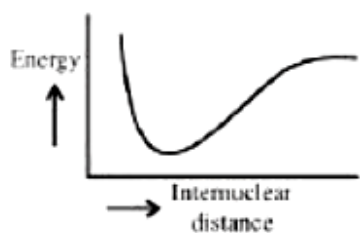
The potential energy curve for the H_2 molecule as a function of internuclear distance is :
[Sep. 05, 2020 (I)]

Options:

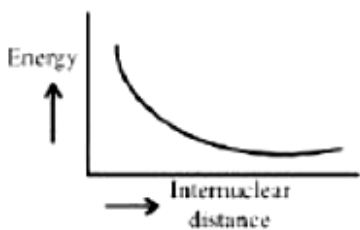
A.



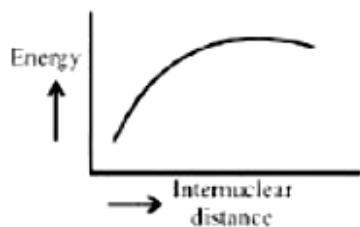
B.



C.



D.



Answer: B

Solution:

When two H-atoms come closer then initially due to attraction P.E. is $-ve$, which decreases more as atoms come closer and after reacting to a minimum value as repulsion starts dominating so, P.E. increases then.

Question155

The structure of PCl_5 in the solid state is:

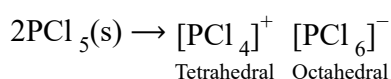
[Sep.05,2020(I)]

Options:

- A. tetrahedral $[\text{PCl}_4]^+$ and octahedral $[\text{PCl}_6]^-$
- B. square planar $[\text{PCl}_4]^+$ and octahedral $[\text{PCl}_6]^-$
- C. square pyramidal
- D. trigonal bipyramidal

Answer: A

Solution:



Question156

Of the species, N O , N O^+ , N O^{2+} and N O^- , the one with minimum bond strength is:

[Sep. 03,2020 (I)]

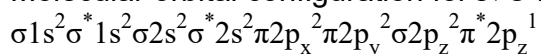
Options:



Answer: D

Solution:

Molecular orbital configuration for NO is



Species	Bond order
NO^+	3
NO^{2+}	2.5
NO^-	2
NO	2.5

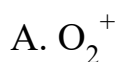
Bond strength is directly proportional to the bond order, so NO^+ has minimum bond strength.

Question157

Two pi and half sigma bonds are present in:

[Jan. 10,2019(I)]

Options:



C. O_2

D. N_2^+

Answer: D

Solution:

$$N_2^+ = 13e^- = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \sigma 2p_z^1$$

$$B.O = \frac{\text{Bonding electrons} - \text{Antibonding electrons}}{2}$$

$$B.O. = \frac{9-4}{2} = 2.5 = 2\pi \text{ bond} + 0.5\sigma \text{ bond}$$

Question 158

According to molecular orbital theory, which of the following is true with respect to Li_2^+ and Li_2^- ?

[Jan .9,2019(I)]

Options:

A. Li_2^+ is unstable and Li_2^- is stable

B. Li_2^+ is stable and Li_2^- is unstable

C. Both are stable

D. Both are unstable

Answer: C

Solution:

Electronic configurations of Li_2^+ and Li_2^- :

$$Li_2^+ : \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^1$$

$$Li_2^- : \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^1$$

Now,

$$\text{Bond order of } Li_2^+ = \frac{1}{2}(3-2) = \frac{1}{2}$$

$$\text{Bond order of Li}_2^- = \frac{1}{2}(4 - 3) = \frac{1}{2}$$

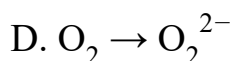
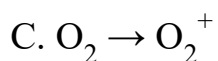
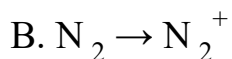
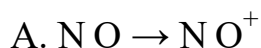
Here, both Li_2^+ and Li_2^- have positive bond order, thus both are stable.

Question 159

In which of the following processes, the bond order has increased and paramagnetic character has changed to diamagnetic?

[Jan. 9, 2019 (II)]

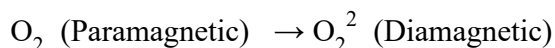
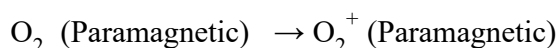
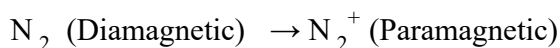
Options:



Answer: A

Solution:

In case of NO (paramagnetic) $\rightarrow$ NO (diamagnetic) the bond order has increased from 2.5 to 3 .
For other cases:



Question 160

The correct statement about ICl_5 and ICl_4^- is :

[April 8, 2019 (II)]

Options:

A. both are isostructural.

B. ICl_5 is trigonal bipyramidal and ICl_4^- is tetrahedral.

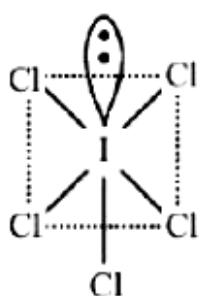
C. ICl_5 is square pyramidal and ICl_4^- is tetrahedral.

D. ICl_5 is square pyramidal and ICl_4^- is square planar.

Answer: D

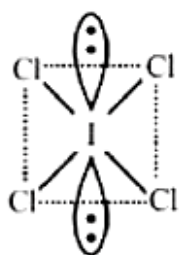
Solution:

ICl_5 is sp^3d^2 hybridised (5bp, 1lp)



Square pyramidal

ICl_4^- is sp^3d^2 hybridised (4bp, 2lp)



Square planar

Question161

The ion that has sp^3d^2 hybridisation for the central atom, is:
[April 8, 2019 (II)]

Options:

A. $[\text{ICl}_4]^-$

B. $[\text{ICl}_2]^-$

C. $[\text{IF}_6]^-$

D. $[\text{BrF}_2]^-$

Answer: A

Solution:

Species	Hybridisation
ICl_2^-	sp^3d
ICl_4^-	sp^3d^2
BrF_2	sp^3d
IF_6^-	sp^3d^3

Question162

During the change of O_2 to O_2^- , the incoming electron goes to the orbital :

[April 10, 2019 (I)]

Options:

A. $\pi 2p_y$

B. $\sigma^* 2p_z$

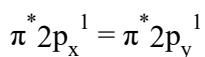
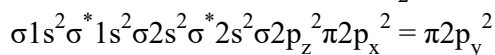
C. $\pi^* 2p_x$

D. $\pi 2p_x$

Answer: C

Solution:

Electronic configuration of O_2 is



When an electron is added in O_2 to form O_2^- , the incoming electron goes to $\pi^* 2p_x$ or $\pi^* 2p_y$ orbital.

Question 163

Among the following, the molecule expected to be stabilised by anion formation is:



[April 9, 2019 (I)]

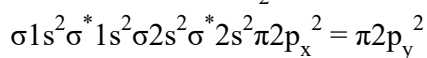
Options:



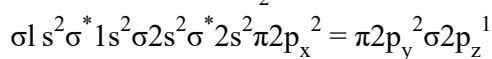
Answer: A

Solution:

Configuration of C_2



Configuration of C_2^-



Bond order

$$= \frac{\text{No. of bonding } e^- - \text{No. of antibonding } e^-}{2}$$

C_2 has s-p mixing and the HOMO is $\pi 2p_x = \pi 2p_y$ and LUMO is $\sigma 2p_z$.

So, the extra electron will occupy bonding molecular orbital and this will lead to increase in bond order.

So, C_2^- has more bond order than C_2 .

Question 164

Among the following molecules/ions, C_2^{2-} , N_2^{2-} , O_2^{2-} , O_2 Which one is diamagnetic and has the shortest bond length?
[April 8, 2019(II)]

Options:

A. O_2

B. N_2^{2-}

C. O_2^{2-}

D. C_2^{2-}

Answer: D

Solution:

Bond length $\propto \frac{1}{\text{Bond order}}$ and diamagnetic species has no unpaired electron in their molecular orbitals.

	No. of unpaired electrons	Bond order	Magnetic character
C_2^{2-}	0	3	diamagnetic
N_2^{2-}	2	2	paramagnetic
O_2^{2-}	0	1	diamagnetic
O_2	1	2	paramagnetic

$\therefore C_2^{2-}$ has least bond length and is diamagnetic.

Question 165

Which of the following compounds contain(s) no covalent bond(s)?
 KCl , PH_3 , O_2 , B_2H_6 , H_2SO_4

[2018]

Options:

A. KCl, B_2H_6 , PH_3

B. KCl, H_2SO_4

C. KCl

D. KCl, B_2H_6

Answer: C

Solution:

KCl is an ionic compound while others (PH_3 , O_2 , B_2H_6 , and H_2SO_4) are covalent compounds.

Question 166

Total number of lone pair of electrons in I_3 ion is:

[2018]

Options:

A. 3

B. 6

C. 9

D. 12

Answer: C

Solution:



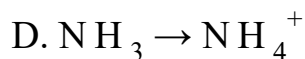
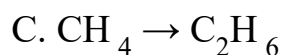
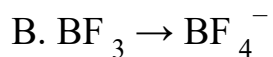
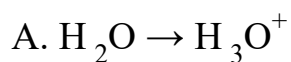
∴ Total number of lone pair of electrons is 9.

Question167

Which of the following conversions involves change in both shape and hybridisation?

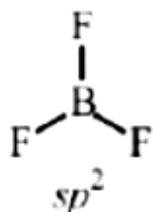
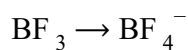
[Online April 16, 2018]

Options:

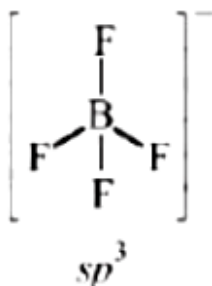


Answer: B

Solution:



Trigonal planar



Tetrahedral

Question168

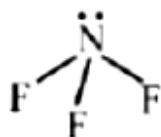
**The incorrect geometry is represented by
[Online April 16, 2018]**

Options:

- A. NF_3 – trigonal planar
- B. BF_3 – trigonal planar
- C. AsF_5 – trigonal bipyramidal
- D. H_2O – bent

Answer: A

Solution:



NF_3 has trigonal pyramidal geometry. N atom has one lone pair and three bond pairs of electrons. The electron pair geometry is tetrahedral and molecular geometry is trigonal pyramidal. The bond angles are lower than tetrahedral bond angles due to lone pair - lone pair and lone pair - bond pair repulsions. N atom is sp^3 hybridised.

Question169

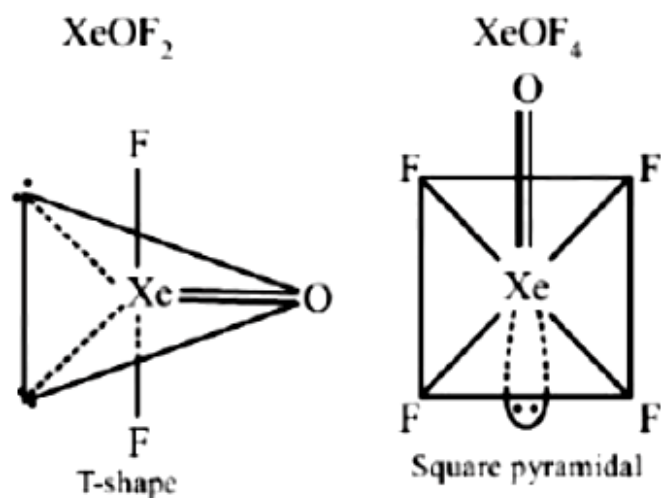
**Identify the pair in which the geometry of the species is Tshape and square pyramidal, respectively
[Online April 15, 2018(I)]**

Options:

- A. ICl_2^- and ICl_5
- B. IO_3^- and IO_2F_2^-
- C. ClF_3 and IO_4
- D. XeOF_2 and XeOF_4

Answer: D

Solution:



Question170

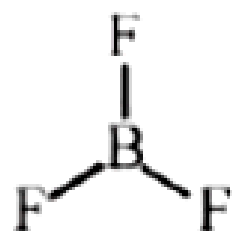
The decreasing order of bond angles in BF_3 , NH_3 , PF_3 and I_3^- is:
[Online April 15, 2018 (I)]

Options:

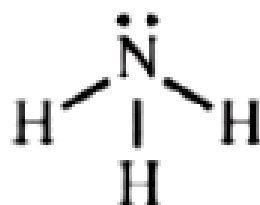
- A. $\text{I}_3^- > \text{BF}_3 > \text{NH}_3 > \text{PF}_3$
- B. $\text{BF}_3 > \text{I}_3^- > \text{PF}_3 > \text{NH}_3$
- C. $\text{BF}_3 > \text{NH}_3 > \text{PF}_3 > \text{I}_3^-$
- D. $\text{I}_3^- > \text{NH}_3 > \text{PF}_3 > \text{BF}_3$

Answer: A

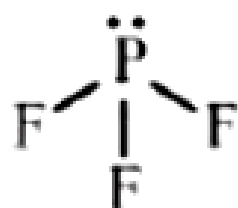
Solution:



B is sp^2 , bond angle $= 120^\circ$

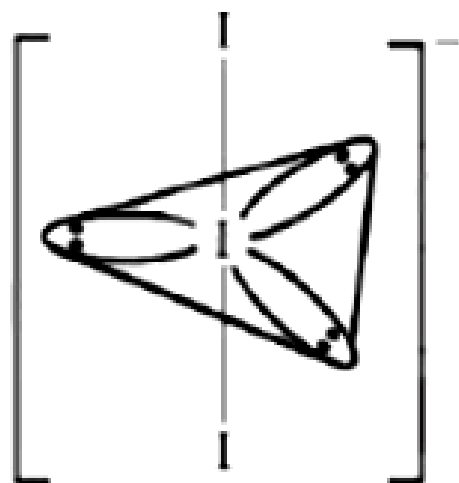


N is sp^3 with $1l p$, bond angle $= 107^\circ$



P is sp^3 with $1l p$, when central atom

size $\uparrow$, bond angle $\downarrow$, $\therefore NH_3 > PF_3$



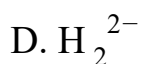
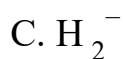
I is sp^3d (linear), bond angle $= 180^\circ$

$\therefore$ Decreasing order of bond angle is $I_3^- > BF_3 > NH_3 > PF_3$

Question171

According to molecular orbital theory, which of the following will not be a viable molecule?
[2018]

Options:



Answer: D

Solution:

Species	No. of e^- s	Elec. conf.	Bond order
He_2^+	$(4 - 1 = 3)$	$\sigma_{1s}^2 \sigma_{1s}^{*1}$	$\frac{2 - 1}{2} = 0.5$
H_2^-	$(2 + 1 = 3)$	$\sigma_{1s}^2 \sigma_{1s}^{*1}$	$\frac{2 - 1}{2} = 0.5$
H_2^{2-}	$(2 + 2 = 4)$	$\sigma_{1s}^2 \sigma_{1s}^{*2}$	$\frac{2 - 2}{2} = 0$
He_2^{2+}	$(4 - 2 = 2)$	σ_{1s}^2	$\frac{2 - 0}{2} = 1$

Molecule having zero bond order will not be a viable molecule.

Question 172

$\text{H} - \text{N}^{\text{I}} - \text{N}^{\text{II}} - \text{N}$ In hydrogen azide, the bond orders of bonds (I) and (II) are
[Online April 15, 2018(I)]

Options:

A. I < 2, II > 2

B. I > 2, II > 2

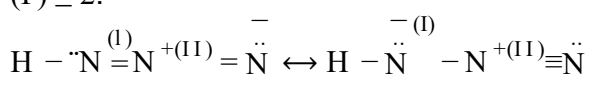
C. I > 2, II < 2

D. I < 2, II < 2

Answer: A

Solution:

As in the resonance structure of hydrogen azide, it can be seen that number of N – N bond for bond (I) ≤ 2 .



Hence for bond (I), bond order will be < 2 whereas for bond (II), number of bond ≥ 2 . Thus its bond order will be > 2 .

Question 173

Which of the following best describes the diagram of molecular orbital?



[Online April 15, 2018 (II)]

Options:

A. A bonding π orbital

B. A non-bonding orbital

C. An antibonding σ orbital

D. An antibonding π orbital

Answer: D

Solution:

An antibonding π orbital best describes the given diagram of a molecular orbital. Two orbitals laterally overlap to form π bond. Out of phase combination of these two p orbitals give π^* MO.

Question174

In the molecular orbital diagram for the molecular ion, N_2^+ , the number of electrons in the σ_{2p} molecular orbital is:
[Online April 15,2018(I)]

Options:

- A. 0
- B. 2
- C. 3
- D. 1

Answer: D

Solution:

Total electrons in $N_2^+ = (7 \times 2) - 1 = 13$

$N_2^+ \rightarrow KK \sigma(2s)^2 \sigma^*(2s)^2 \pi(2p_x)^2 = \pi(2p_y)^2 \sigma(2p_z)^1$

Number of electron in σ_{2p_z} is 1

Question175

sp^3d^2 Hybridisation is not displayed by:
[Online April 8, 2017]

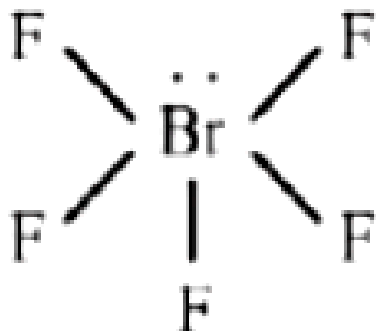
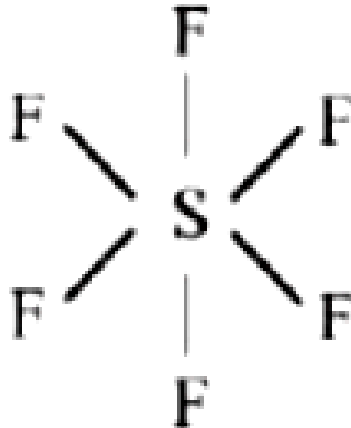
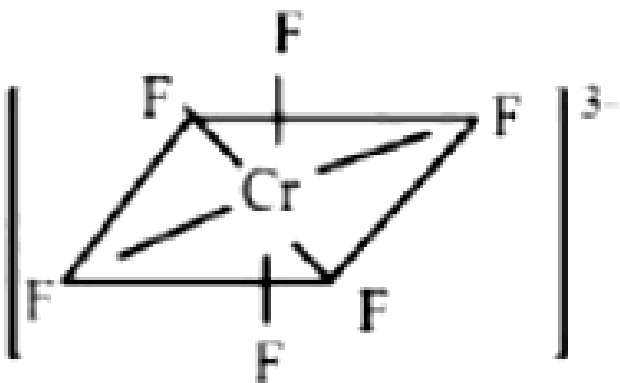
Options:

- A. BrF_5
- B. SF_6
- C. $[CrF_6]^{3-}$

D. PF_5

Answer: D

Solution:

(a) BrF_5		sp^3d^2
(b) SF_6		sp^3d^2
(c) $[CrF_6]^{3-}$		sp^3d^2
(d) PF_5		sp^3d

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Question 176

The group having triangular planar structures is :
[Online April 9, 2017]

Options:

- A. BF_3 , NF_3 , CO_3^{2-}
- B. CO_3^{2-} , NO_3^- , SO_3
- C. NH_3 , SO_3 , CO_3^{2-}
- D. NCl_3 , BCl_3 , SO_3

Answer: B

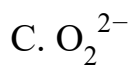
Solution:

Group	Hybridisation	Shape
(a) BF_3	sp^2	Triangular Planar (T.P.)
NF_3	sp^3	Tetrahedral (T)
CO_3^{2-}	sp^2	T.P.
(b) CO_3^{2-}	sp^2	T.P.
NO_3^-	sp^2	T.P.} (All have same hybridisation)
SO_3	sp^2	T.P.
(c) NH_3	sp^3	T
SO_2	sp^3	T.P.
CO_3^{2-}	sp^2	T.P.
(d) NCl_3	sp^3	T
BCl_3	sp^2	T.P.
SO_3	sp^2	T.P.

Question177

Which of the following is paramagnetic?
[Online April 8, 2017]

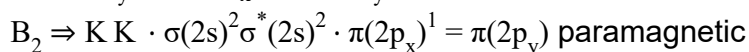
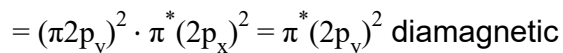
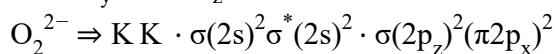
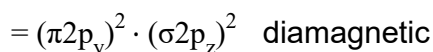
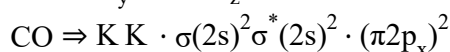
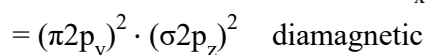
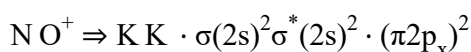
Options:



Answer: D

Solution:

Total electron	
NO^+	14
CO	14
O_2^{2-}	18
B_2	10



Question 178

**Which of the following species is not paramagnetic?
[2017]**

Options:



B. CO

C. O₂

D. B₂

Answer: B

Solution:

NO → One unpaired electron is present in π^* molecular orbit, hence paramagnetic.

(b) CO(14) → $\text{K K } \sigma 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \sigma 2p_z^2 \sigma^* 2s^2$

No unpaired electron, hence diamagnetic.

(c) O₂(16) →

$\text{K K } \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^1$

Two unpaired electrons, hence paramagnetic.

(d) B₂(10) → $\text{K K } \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^1 = \pi 2p_y^1$

B₂ contains two unpaired electrons, hence paramagnetic.

Question 179

**The species in which the N atom is in a state of sp hybridisation is :
[2016]**

Options:

A. NO₃⁻

B. NO₂

C. NO₂⁺

D. NO₂⁻

Answer: C

Solution:

Hybridisation (H) = [No. of valence electrons of central atom + No. of monovalent atoms attached to it + (-ve charge if any - (+ve charge if any))]

$\text{NO}_2^+ = \text{i.e. } sp \text{ hybridisation}$

$\text{NO}_2^- = \text{i.e. } sp^2 \text{ hybridisation}$

$\text{NO}_3^- = \text{i.e. } sp^2 \text{ hybridisation}$

The Lewis structure of NO_2 shows a bent molecular geometry with trigonal planar electron pair geometry hence the hybridization will be sp^2 .

Question180

**The group of molecules having identical shape is:
[Online April 9, 2016]**

Options:

A. PCl_5 , IF_5 , XeO_2F_2

B. BF_3 , PCl_3 , XeO_3

C. SF_4 , XeF_4 , CCl_4

D. ClF_3 , XeOF_2 , XeF_3^+

Answer: D

Solution:

$\text{ClF}_3 \rightarrow \text{Hybridisation} = 3 + \frac{1}{2}[7 - 3] = 5(sp^3d)$

$\text{XeOF}_2 \rightarrow \text{Hybridisation} = 3 + \frac{1}{2}[8 - 4] = 5(sp^3d)$

$\text{XeF}_3^+ \rightarrow \text{Hybridisation} = 3 + \frac{1}{2}[8 - 3 - 1] = 5(sp^3d)$

All molecules have sp^3d hybridisation and 2 lone pairs. Hence all have identical (T-shape).

Question181

**The bond angle $\text{H} - \text{X} - \text{H}$ is the greatest in the compound:
[Online April 10, 2016]**

Options:

A. PH_3

B. CH_4

C. NH_3

D. H_2O

Answer: B

Solution:

More the number of lone pairs on central atom, the greater is the contraction caused in the angle between bond pairs. In CH_4 there is no lone pair of electrons, hence bond angle is greatest.

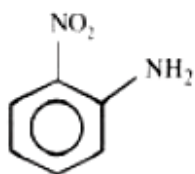
Question182

Which compound exhibits maximum dipole moment among the following ?

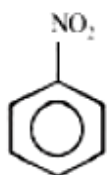
[Online April 11, 2015]

Options:

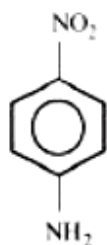
A.



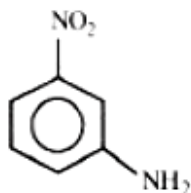
B.



C.

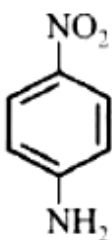


D.



Answer: C

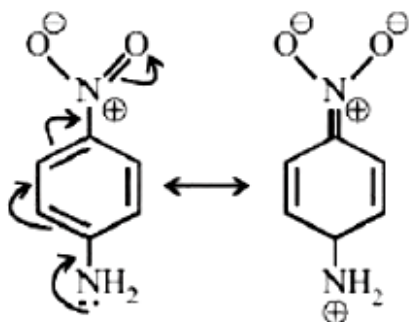
Solution:



Dipole moment = (Distance between opposite charges) x (charge, q)

$$\mu = q \times d$$

So, greater the distance between the opposite charges higher the dipole. Due to the resonance the greater charge separation occurs between charges due to linearity.



Question183

Molecule AB has a bond length of 1.617Å and a dipole moment of 0.38D. The fractional charge on each atom (absolute magnitude) is :

($e_0 = 4.802 \times 10^{-10}$ esu)

[Online April 11, 2015]

Options:

A. 0.5

B. 0.05

C. 0

D. 1.0

Answer: B

Solution:

Dipole moment (μ) = $q \times d$

$$\Rightarrow 1D \approx 10^{-18} \text{ esu cm}$$

$$0.38 \times 10^{-18} \text{ esu cm} = q \times (1.617 \times 10^{-8} \text{ cm})$$

$$q = 2.35 \times 10^{-11} \text{ esu}$$

$$\text{So, fractional charge} = \frac{\text{Partial charge}}{\text{Total charge}} = \frac{q}{Q}$$

$$= \frac{2.35 \times 10^{-11} \text{ esu}}{4.802 \times 10^{-10} \text{ esu}} = 0.049 \approx 0.05$$

Question 184

After understanding the assertion and reason, choose the correct option.

Assertion : In the bonding molecular orbital (MO) of H_2 , electron density is increased between the nuclei.

Reason : The bonding MO is $\Psi_A + \Psi_B$, which shows destructive interference of the combining electron waves.

[Online April 10, 2015]

Options:

A. Assertion is incorrect, reason is correct.

B. Assertion is correct, reason is incorrect.

C. Assertion and reason are correct and reason is the correct explanation for the assertion.

D. Assertion and reason are correct, but reason is not the correct explanation for the assertion.

Answer: B

Solution:

Assertion is correct but reason is incorrect. Bonding MO shows constructive interference of the combining electron waves.

Question185

Amongst LiCl , RbCl , BeCl_2 and MgCl_2 the compounds with the greatest and the least ionic character, respectively are:
[Online April 19, 2014]

Options:

- A. LiCl and RbCl
- B. RbCl and BeCl_2
- C. MgCl_2 and BeCl_2
- D. RbCl and MgCl_2

Answer: B

Solution:

According to Fajan's rules smaller, highly charged cation has greatest covalent character while large cation with smaller charge has greatest ionic character.

Question186

Which of these statements is not true?
[Online April 19, 2014]

Options:

- A. NO^+ is not isoelectronic with O_2
- B. B is always covalent in its compounds
- C. In aqueous solution, the Tl^+ ion is much more stable than Tl^{+3}
- D. LiAlH_4 is a versatile reducing agent in organic synthesis.

Answer: A

Solution:

NO^+ has 14 electrons and O_2 has 16 electrons. Therefore, they are not isoelectronic species.

Boron forms covalent bond in its compound.

Tl^+ is much more stable than Tl^{3+} as going down the group the stability of lower oxidation number increases due to inert pair effect.

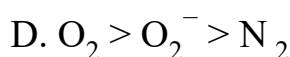
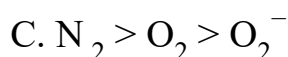
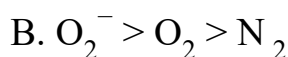
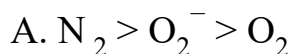
LiAlH_4 is a selective reducing agent in organic synthesis.

Hence, option A is not true.

Question187

The correct order of bond dissociation energy among N_2 , O_2 , O_2^- is shown in which of the following arrangements?
[Online April 11, 2014]

Options:



Answer: C

Solution:

The bond order of N_2 , O_2 , and O_2^- are 3, 2 and 1.5 respectively.

Since higher bond order implies higher bond dissociation energy, hence the correct order will be $\text{N}_2 > \text{O}_2 > \text{O}_2^-$

Question188

The number and type of bonds in C_2^{2-} ion in CaC_2 are:
[Online April 9, 2014]

Options:

- A. One σ bond and one π -bond
- B. One σ bond and two π -bond
- C. Two σ bond and two π -bond
- D. Two σ bond and one π -bond

Answer: B

Solution:

C_2^{2-} ion $\rightarrow [\text{C} \equiv \text{C}]^{2-}$ Dicarbide ion C_2^{2-} has 1 sigma and 2 pi bonds.
There are two π bonds and one σ bond in a triple bond.

Question189

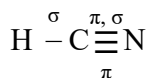
Which of the following molecules has two sigma (σ) and two pi (π) bonds?
[Online April 12, 2014]

Options:

- A. C_2H_4
- B. N_2F_2
- C. $\text{C}_2\text{H}_2\text{Cl}_2$
- D. HCN

Answer: D

Solution:



Therefore, HCN has 2π and 2σ bonds.

Question 190

Which one of the following properties is not shown by N O?
[2014]

Options:

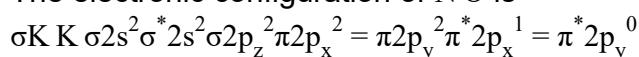
- A. It is diamagnetic in gaseous state
- B. It is neutral oxide
- C. It combines with oxygen to form nitrogen dioxide
- D. Its bond order is 2.5

Answer: A

Solution:

Nitric oxide is paramagnetic in the gaseous state because of the presence of one unpaired electron in its outermost shell.

The electronic configuration of N O is



Question 191

Which one of the following molecules is paramagnetic?
[Online April 19, 2014]

Options:

- A. N_2
- B. NO
- C. CO
- D. O_3

Answer: B

Solution:

The molecular orbital configuration of the molecules given is

Total no. of electrons in $\text{NO} = 7(\text{N}) + 8(\text{O}) = 15$

Hence E.C. of $\text{NO} = \text{KK} \sigma(2s)^2 \sigma^*(2s)^2 \sigma 2p_z^2$

$\pi(2p_x)^2 \pi(2p_y)^2 \pi^*(2p_x)^1$

Due to presence of one unpaired electron NO is paramagnetic. Except NO all are diamagnetic due to absence of unpaired electrons.

Question192

Which of the following has unpaired electron(s)?
[Online April 9, 2014]

Options:

A. N_2

B. O_2^-

C. N_2^{2+}

D. O_2^{2-}

Answer: B

Solution:

$\text{O}_2^-(17) = \text{KK} (\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2p_z)^2$

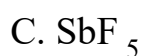
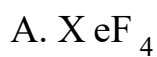
$(\pi 2p_x)^2 = (\pi 2p_y)^2 (\pi^* 2p_x)^2 = (\pi^* 2p_y)^1$

One unpaired electron - paramagnetic.

Question193

Which one of the following molecules is polar?
[Online April 9, 2013]

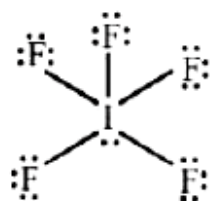
Options:



Answer: B

Solution:

The geometry of IF_5 is square pyramide with an unsymmetric charge distribution, therefore this molecule is polar.



Question194

Bond distance in HF is 9.17×10^{-11} m. Dipole moment of HF is 6.104×10^{-30} Cm. The percentage ionic character in HF will be : (electron charge = 1.60×10^{-19} C) [Online April 23, 2013]

Options:

A. 61.0%

B. 38.0%

C. 35.5%

D. 41.5%

Answer: D

Solution:

Given $e = 1.60 \times 10^{-19} \text{C}$

$d = 9.17 \times 10^{-11} \text{m}$

From $\mu = e \times d$

$\mu = 1.60 \times 10^{-19} \times 9.17 \times 10^{-11}$

$= 14.672 \times 10^{-30}$

% ionic character

$= \frac{\text{Observed dipole moment}}{\text{Dipole moment for } 100\% \text{ ionic bond}}$

$= \frac{6.104 \times 10^{-30}}{14.672 \times 10^{-30}} \times 100$

$= 41.5\%$

Question195

The shape of IF_6^- is:

[Online April 23, 2013]

Options:

A. Trigonally distorted octahedron

B. Pyramidal

C. Octahedral

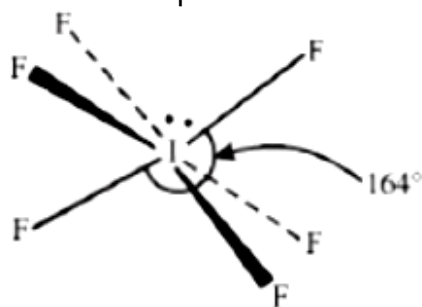
D. Square antiprism

Answer: A

Solution:

The structure of IF_6^- is distorted octahedral.

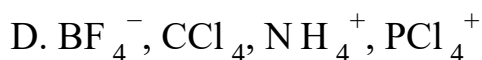
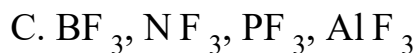
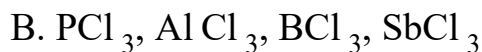
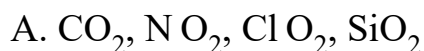
This is due to presence of a "weak" lone pair.



Question196

**In which of the following sets, all the given species are isostructural?
[Online April 25, 2013]**

Options:



Answer: D

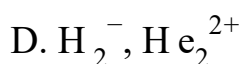
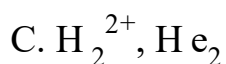
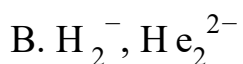
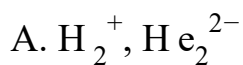
Solution:

All have tetrahedral structure.

Question 197

**In which of the following pairs of molecules\ions, both the species are not likely to exist?
[2013]**

Options:



Answer: C

Solution:

$$H_2^{2+} = \sigma 1s^0 \sigma^* 1s^0$$

$$\text{Bond order for } H_2^{2+} = \frac{1}{2}(0 - 0) = 0$$

$$H e_2 = \sigma 1s^2 \sigma^* 1s^2$$

$$\text{Bond order for } H e_2 = \frac{1}{2}(2 - 2) = 0$$

So both H_2^{2+} and $H e_2$ do not exist.

Question198

Which one of the following molecules is expected to exhibit diamagnetic behaviour?
[2013]

Options:

A. C_2

B. N_2

C. O_2

D. S_2

Answer: 0

Solution:

The molecular orbital structures of C_2 and N_2 are

$$N_2 = K K \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \sigma 2p_z^2$$

$$C_2 = K K \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2$$

Both N_2 and C_2 have paired electrons, hence they are diamagnetic.

Question199

Which of the following is the wrong statement
[2013]

Options:

A. ONCl and ONO - are not isoelectronic.

B. O₃ molecule is bent

C. Ozone is violet-black in solid state

D. Ozone is diamagnetic gas.

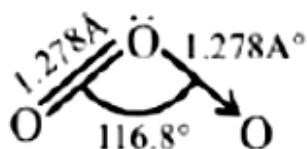
Answer: 0

Solution:

All options are correct,

(a) ONCl = $8 + 7 + 17 = 32e^-$

ON O⁻ = $8 + 7 + 8 + 1 = 24e^-$ not isoelectronic



(b) The central atom is sp^2 hybridised with one lone pair.

(c) It is a pale blue gas. At -249.7° , it forms violet black crystals.

(d) It is diamagnetic in nature due to absence of unpaired electrons.

Question200

Stability of the species Li_2 , Li_2^- and Li_2^+ increases in the order of :
[2013]

Options:

A. $Li_2 < Li_2^+ < Li_2^-$

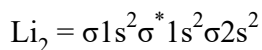
B. $Li_2^- < Li_2^+ < Li_2$

C. $Li_2 < Li_2^- < Li_2^+$

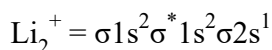
D. $Li_2^- < Li_2 < Li_2^+$

Answer: B

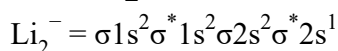
Solution:



$$\therefore \text{Bond order} = \frac{1}{2}(4 - 2) = 1$$



$$\text{B.O.} = \frac{1}{2}(3 - 2) = 0.5$$



$$\text{B.O.} = \frac{1}{2}(4 - 3) = 0.5$$

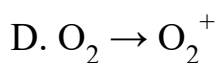
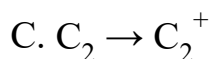
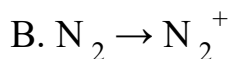
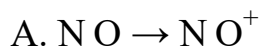
The bond order of Li_2^+ and Li_2^- is same but Li_2^+ is more stable than Li_2^- because Li_2^+ is smaller in size and has 2 electrons in antibonding orbitals whereas Li_2^- has 3 electrons in antibonding orbitals. Hence Li_2^+ is more stable than Li_2^-

Question 201

In which of the following ionization processes the bond energy has increased and also the magnetic behaviour has changed from paramagnetic to diamagnetic?

[Online April 9, 2013]

Options:



Answer: A

Solution:

For NO Total no. of electrons = 15

B.O. = 2.5

Mag. behaviour = paramagnetic

For NO^+

Total no. of electrons = 14

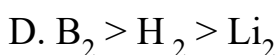
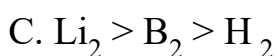
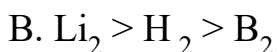
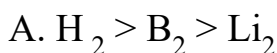
B.O. = 3

Mag. behaviour = diamagnetic

Question202

Bond order normally gives idea of stability of a molecular species. All the molecules viz. H_2 , Li_2 and B_2 have the same bond order yet they are not equally stable. Their stability order is
[Online April 22, 2013]

Options:



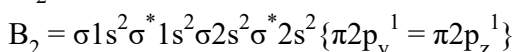
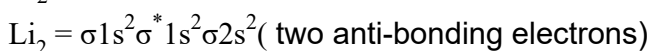
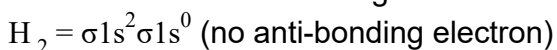
E. None of above

Answer: E

Solution:

(N) None of the given option is correct.

The molecular orbital configuration of the given molecules is



(4 anti-bonding electrons)

Though the bond order of all the species are same ($B \cdot O = 1$) but stability is different. This is due to difference in the presence of no. of anti-bonding electron. Higher the no. of anti-bonding electron lower is the stability hence the correct order is $H_2 > Li_2 > B_2$

Question203

The internuclear distances in O – O bonds for O_2^+ , O_2 , O_2^- and O_2^{2-} respectively are:
[Online April 25, 2013]

Options:

A. $1.30\overset{\circ}{\text{\AA}}$, $1.49\overset{\circ}{\text{\AA}}$, $1.12\overset{\circ}{\text{\AA}}$, $1.21\overset{\circ}{\text{\AA}}$

B. $1.49\overset{\circ}{\text{\AA}}$, $1.21\overset{\circ}{\text{\AA}}$, $1.12\overset{\circ}{\text{\AA}}$, $1.30\overset{\circ}{\text{\AA}}$

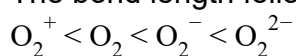
C. $1.21\overset{\circ}{\text{\AA}}$, $1.12\overset{\circ}{\text{\AA}}$, $1.49\overset{\circ}{\text{\AA}}$, $1.30\overset{\circ}{\text{\AA}}$

D. $1.12\overset{\circ}{\text{\AA}}$, $1.21\overset{\circ}{\text{\AA}}$, $1.30\overset{\circ}{\text{\AA}}$, $1.49\overset{\circ}{\text{\AA}}$

Answer: D

Solution:

The bond length follows the order



According to this the possible values are

$1.12\overset{\circ}{\text{\AA}}$, $1.21\overset{\circ}{\text{\AA}}$, $1.30\overset{\circ}{\text{\AA}}$, $1.49\overset{\circ}{\text{\AA}}$

Question204

ortho-Nitrophenol is less soluble in water than p - and m - nitrophenols because :

[2012]

Options:

A. o -nitrophenol is more volatile steam than those of m and p -isomers.

B. o -nitrophenol shows intramolecular H-bonding

C. o-nitrophenol shows intermolecular H-bonding

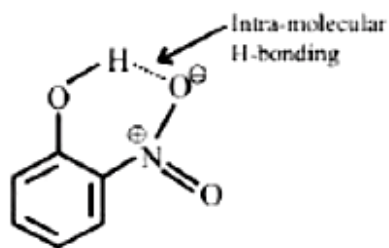
D. Melting point of o -nitrophenol is lower than those of m - and p -isomers.

Answer: B

Solution:

Compounds involved in chelation become non-polar. Consequently such compounds are soluble in non-polar solvents like ether, benzene etc. and are only sparingly soluble in water, whereas meta and

para isomers are more soluble in water & less soluble in non-polar solvents.



Question205

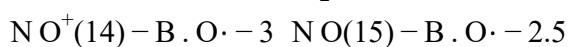
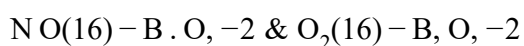
Among the following, the species having the smallest bond length is
[Online May 7,2012]

Options:



Answer: B

Solution:



Higher the bond order lower is the bond length. Hence NO^+ will have smallest bond.

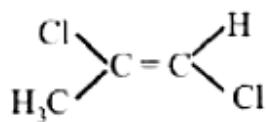
Question206

Among the following chloro-compound having the lowest dipole moment is
[Online May 12, 2012]

Options:

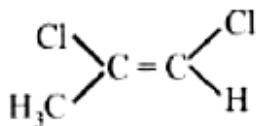


B.



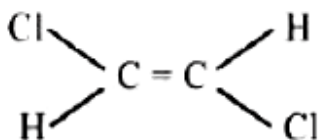
C. CH_2Cl_2

D.



Answer: C

Solution:



Dipole moment (μ) = 0

Question207

Although CN^- ion and N_2 molecule are isoelectronic, yet N_2 molecule is chemically inert because of [Online May 12, 2012]

Options:

- A. presence of more number of electrons in bonding orbitals
- B. low bond energy
- C. absence of bond polarity
- D. uneven electron distribution.

Answer: C

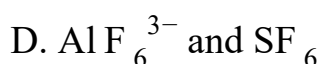
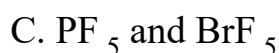
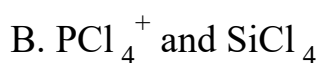
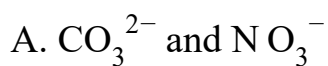
Solution:

In nitrogen molecule, both the nitrogen atoms have same electronegativity. So it has zero polarity and hence less tendency to break away and forms ions.

Question208

**In which of the following pairs, the two species are not isostructural ?
[2012]**

Options:



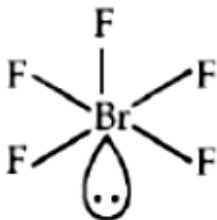
Answer: C

Solution:

PF_5 is trigonal bipyramidal



BrF_5 is square pyramidal (distorted)



Question209

The formation of molecular complex $\text{BF}_3 - \text{NH}_3$ results in a change in hybridisation of boron

[Online May 12, 2012]

Options:

A. from sp^2 to $d\ sp^2$

B. from sp^2 to sp^3

C. from sp^3 to sp^2

D. from sp^3 to sp^3d

Answer: B

Solution:

In BF_3 , B is sp^2 hybridized with one empty p_z orbital. The empty p_z orbital of BF_3 can be filled by lone pair of molecules such as NH_3 . When this occurs a tetrahedral molecule or ion is formed which is sp^3 hybridized.

Question210

Which of the following has the square planar structure?

[Online May 19, 2012]

Options:

A. XeF_4

B. NH_4^+

C. BF_4^-

D. CCl_4

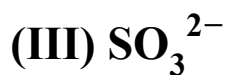
Answer: A

Solution:

XeF_4 has square pyramidal structure, while NH_4^+ , BF_4^- and CCl_4 have tetrahedral structure.

Question211

Among the following species which two have trigonal bipyramidal shape?



[Online May 26, 2012]

Options:

A. I and III

B. III and IV

C. I and IV

D. II and III

Answer: A

Solution:

Specise	Hybridisation	Shape
NI_3	sp^3	Trigonal pyramidal
I_3^-	sp^3d	Linear
SO_3^{2-}	sp^3	Trigonal pyramidal
NO_3^-	sp^2	Trigonal planer

Hence, NI_3 and SO_3^{2-} have same shape.

Question212

The number of types of bonds between two carbon atoms in calcium carbide is :
[2011RS]

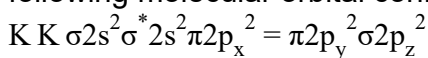
Options:

- A. One sigma, one pi
- B. Two sigma, one pi
- C. Two sigma, two pi
- D. One sigma, two pi

Answer: D

Solution:

Calcium carbide exists as Ca^{2+} and C_2^{2-} . According to the molecular orbital model, C_2^{2-} should have following molecular orbital configuration:



Thus M.O. configuration suggests that it contains one σ & two π bonds.

Question213

Which one of the following pairs of species have the same bond order?
[2008]

Options:

- A. CN^- and NO^+
- B. CN^- and CN^+
- C. O^- , and CN^-
- D. NO^+ and CN^+

Answer: A

Solution:

For any species to have same bond order we can expect them to have same number of electrons.
Calculating the number of electrons in various species.

O_2^- ($8 + 8 + 1 = 17$); CN^- ($6 + 7 + 1 = 14$)

NO^+ ($7 + 8 - 1 = 14$); CN^+ ($6 + 7 - 1 = 12$)

We find CN^- and NO^+ both have 14 electrons, so they have same bond order.

Question 214

The bond dissociation energy of B – F in BF_3 is 646 kJ mol^{-1} , whereas that of C – F in CF_4 is 515 kJ mol^{-1} . The correct reason for higher B – F bond dissociation energy as compared to that of C – F bond is [2008]

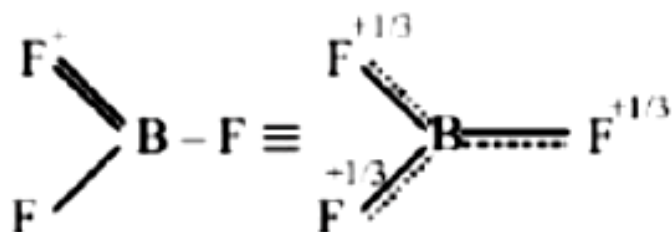
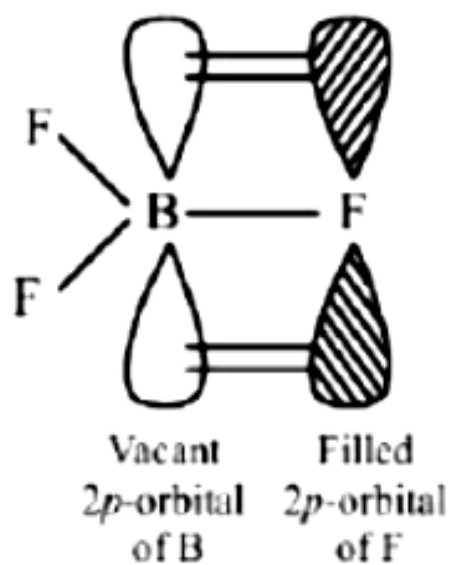
Options:

- A. stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4 .
- B. significant $p\pi - p\pi$ interaction between B and F in BF_3 , whereas there is no possibility of such interaction between C and F in CF_4 .
- C. lower degree of $p\pi - p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4 .
- D. smaller size of B -atom as compared to that of C -atom.

Answer: B

Solution:

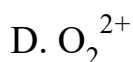
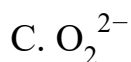
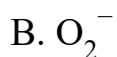
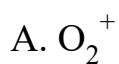
Note: The delocalised $p\pi - p\pi$ bonding between filled p -orbital of F and vacant p -orbital of B leads to shortening of B – F bond length which results in higher bond dissociation energy of the B – F bond.



Question215

Using MO theory, predict which of the following species has the shortest bond length?
[2008]

Options:



Answer: D

Solution:

Bond order

$$= \frac{\text{No. of bonding electrons} - \text{No. of antibonding electrons}}{2}$$

$$\text{Bond order in } \text{O}_2^+ = \frac{10 - 5}{2} = 2.5$$

$$\text{Bond order in } \text{O}_2^- = \frac{10 - 7}{2} = 1.5$$

$$\text{Bond order in } \text{O}_2^{2-} = \frac{10 - 8}{2} = 1$$

$$\text{Bond order in } \text{O}_2^{2+} = \frac{10 - 4}{2} = 3$$

$$\text{Since, bond order} \propto \frac{1}{\text{Bond length}}$$

$\therefore$ Bond length is shortest in O_2^{2+} .

Question 216

Which of the following hydrogen bonds is strongest?
[2007]

Options:





Answer: C

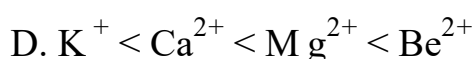
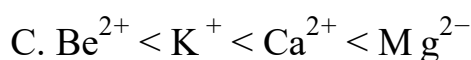
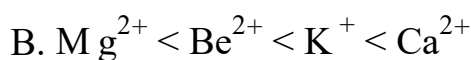
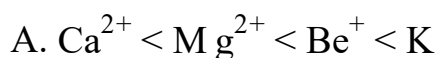
Solution:

Note: Greater the difference between electronegativity of bonded atoms, stronger will be bond. Since F is most electronegative, hence $\text{F} - \text{H} \dots \text{F}$ is the strongest bond.

Question217

The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+ , Ca^{2+} , Mg^{2+} , Be^{2+} ? [2007]

Options:



Answer: D

Solution:

Smaller the size and higher the charge, more will be the polarising power of the cation. Since the order of the size of cation is $\text{K}^+ > \text{Ca}^{2+} > \text{Mg}^{2+} > \text{Be}^{2+}$, so the correct order of polarising power is $\text{K}^+ < \text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$

Question218

Which of the following species exhibits the diamagnetic behaviour? [2007]

Options:

A. NO

B. O_2^{2-}

C. O_2^+

D. O_2

Answer: B

Solution:

Diamagnetic species have no unpaired electrons whereas paramagnetic species has one or more unpaired electrons.

For electronic configuration of O_2^+ , O_2 and O_2^{2-} , consult Q.2.

O_2 and O_2^+ have 2 and 1 unpaired electron respectively, while O_2^{2-} has no unpaired electron

$NO \rightarrow KK\sigma 2s^2\sigma^* 2s^2\sigma 2p_z^2\pi 2p_x^2 = \pi 2p_y^2$

$\pi^* 2p_x^1 = \pi^* 2p_y^0$

Thus NO has one unpaired electron

Question 219

In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed?

[2007]

Options:

A. $N_2 \rightarrow N_2^+$

B. $C_2 \rightarrow C_2^+$

C. $NO \rightarrow NO^+$

D. $O_2 \rightarrow O_2^+$

Answer: C

Solution:

- (a) N_2 : bond order 3 , diamagnetic
 N_2^+ : bond order 2.5, paramagnetic
(b) C_2 : bond order 2 , diamagnetic
 C_2^+ : bond order 1.5, paramagnetic
(c) NO : bond order 2.5, paramagnetic
 NO^+ : bond order 3, diamagnetic
(d) O_2 : bond order 2, paramagnetic
 O_2^+ : bond order 2.5, paramagnetic
-

Question220

In which of the following molecules\ions are all the bonds not equal?
[2006]

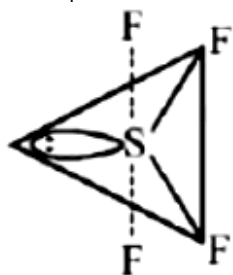
Options:

- A. XeF_4
B. BF_4
C. SF_4
D. SiF_4

Answer: C

Solution:

In SF_4 the hybridisation is sp^3d and the shape of molecule is



It contains two different bonds i.e., axial and equatorial.

Question221

The decreasing values of bond angles from $\text{NH}_3(106^\circ)$ to $\text{SbH}_3(101^\circ)$ down group- 15 of the periodic table is due to [2006]

Options:

- A. decreasing l p – bp repulsion
- B. decreasing electronegativity
- C. increasing bp – bp repulsion
- D. increasing p -orbital character in sp^3

Answer: B

Solution:

The bond angle decreases on moving down the group due to decrease in bond pair-bond pair repulsion.

NH_3 107°

PH_3 94°

AsH_3 92°

SbH_3 91°

BiH_3 90°

Note: This can also be explained due to decrease in electronegativity from N to Bi.

Question222

Which of the following molecules/ions does not contain unpaired electrons? [2006]

Options:

A. N_2^+

B. O_2

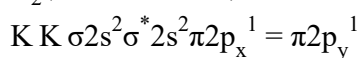
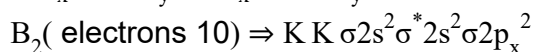
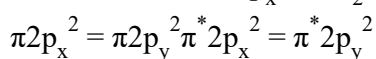
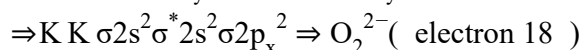
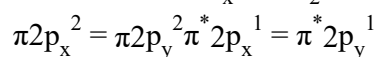
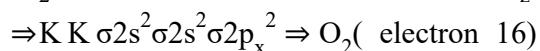
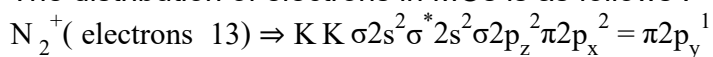
C. O_2^{2-}

D. B₂

Answer: C

Solution:

The distribution of electrons in MOs is as follows :



Only O₂²⁻ does not contain any unpaired electron.

Question223

Lattice energy of an ionic compound depends upon [2005]

Options:

A. Charge on the ion and size of the ion

B. Packing of ions only

C. Size of the ion only

D. Charge on the ion only

Answer: A

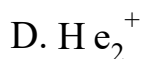
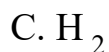
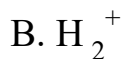
Solution:

The value of lattice energy depends on the charge present on the two ions and the distance between them.

Question224

Which of the following species is diamagnetic in nature?
[2005]

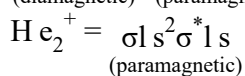
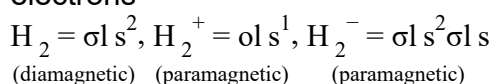
Options:



Answer: C

Solution:

A diamagnetic substance contains no unpaired electron. H_2 is diamagnetic as it contains all paired electrons



Question 225

The bond order in NO is 2.5 while that in NO^+ is 3 . Which of the following statements is true for these two species?
[2004]

Options:

A. Bond length in NO^+ is equal to that in NO

B. Bond length in NO is greater than in NO^+

C. Bond length in NO^+ is greater than in NO

D. Bond length is unpredictable

Answer: B

Solution:

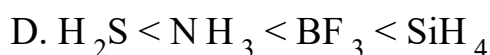
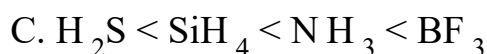
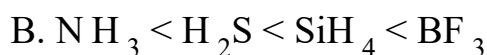
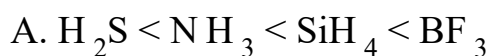
Now since bond order of NO^+ (3) is higher than that of NO (2.5). Thus bond length of NO^+ will be shorter.

Question 226

The correct order of bond angles (smallest first) in H_2S , NH_3 , BF_3 and SiH_4 is

[2004]

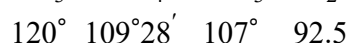
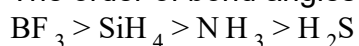
Options:



Answer: A

Solution:

The order of bond angles

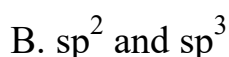
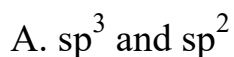


Question 227

The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively

[2004]

Options:

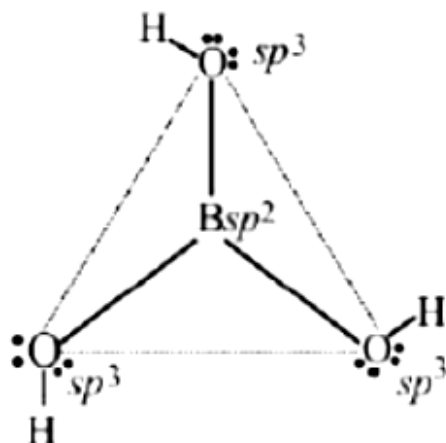


C. sp^2 and sp^2

D. sp^3 and sp^3

Answer: B

Solution:



Question228

**Which one of the following has the regular tetrahedral structure?
(Atomic nos. : B = 5, S = 16, N i = 28, X e = 54)
[2004]**

Options:

A. BF_4^-

B. SF_4

C. XeF_4

D. $[Ni(CN)_4]^{2-}$

Answer: A

Solution:

XeF_4 (sp^3d^2 , square planar),

$[\text{Ni}(\text{CN})_4]^{2-}$ (dsp^2 , square planar)

BF_4^- (sp^3 , tetrahedral), SF_4 (sp^3d , see saw shaped)

Question 229

The maximum number of 90° angles between bond pair bond pair of electrons is observed in
[2004]

Options:

A. dsp^2 hybridization

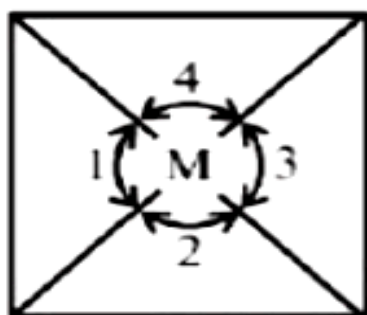
B. sp^3d hybridization

C. dsp^3 hybridization

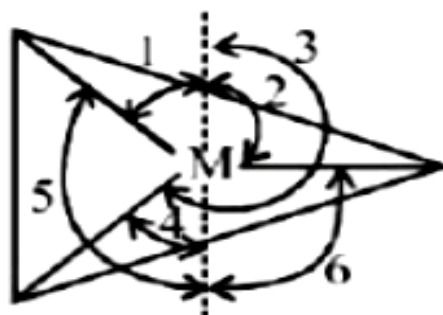
D. sp^3d^2 hybridization

Answer: D

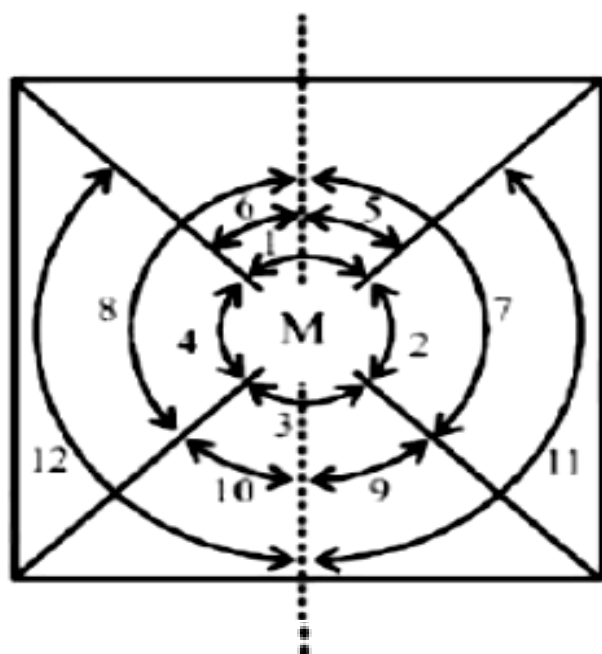
Solution:



dsp^2 hybridisation
Number of 90° angle
between bonds = 4



sp^3d or dsp^3 hybridisation
Number of 90° angle
between bonds = 6



sp^3d^2 Hybridisation

Number of 90° angle between bonds = 12

Question230

An ether is more volatile than an alcohol having the same molecular formula. This is due to
[2003]

Options:

- A. alcohols having resonance structures
- B. intermolecular hydrogen bonding in ethers
- C. intermolecular hydrogen bonding in alcohols
- D. dipolar character of ethers

Answer: C

Solution:

In ether, there is no H-bonding while alcohols have intermolecular H-bonding.

Question231

Which one of the following pairs of molecules will have permanent dipole moments for both members?
[2003]

Options:

- A. N O_2 and CO_2
- B. N O_2 and O_3
- C. SiF_4 and CO_2
- D. SiF_4 and N O_2

Answer: B

Solution:

Both N O_2 and O_3 have angular shape and hence will have net dipole moment.

Question232

Which one of the following compounds has the smallest bond angle in its molecule ?

[2003]

Options:

- A. OH_2
- B. SH_2
- C. NH_3
- D. SO_2

Answer: B

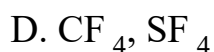
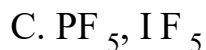
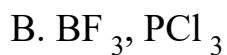
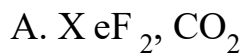
Solution:

In H_2S , due to low electronegativity of sulphur the l.p. – l.p. repulsion is more than b . p. – b . p. repulsion and hence the bond angle is minimum.

Question233

The pair of species having identical shapes for molecules of both species is
[2003]

Options:



Answer: A

Solution:

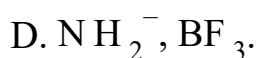
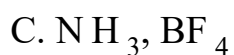
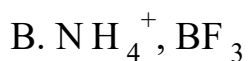
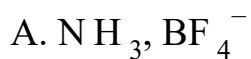
Both XeF_2 and CO_2 have a linear structure.



Question 234

In which of the following species the interatomic bond angle is $109^\circ 28''$?
[2002]

Options:



Answer: A

Solution:

In NH_3 and BF_4^- , the hybridisation is sp^3 and the bond angle is almost $109^\circ 28'$.

Question 235

Hybridisation of the underline atom changes in:
[2002]

Options:

A. AlH_3 changes to AlH_4^-

B. H_2O changes to H_3O^+

C. NH_3 changes to NH_4^+

D. in all cases

Answer: A

Solution:

$$\text{Hybridisation} = \frac{1}{2} \left[\left(\begin{array}{c} \text{No. of electrons} \\ \text{in valence} \\ \text{shell of atom} \end{array} \right) + \left(\begin{array}{c} \text{No. of monovalent} \\ \text{atoms around it} \end{array} \right) - \left(\begin{array}{c} \text{Charge on} \\ \text{cation} \end{array} \right) + \left(\begin{array}{c} \text{Charge on} \\ \text{anion} \end{array} \right) \right]$$

(a) For AlH_3

$$\text{Hybridisation of Al atom} = \frac{1}{2}[3 + 3 - 0 + 0] = 3 = \text{sp}^2$$

For AlH_4^- ,

$$\text{Hybridisation of Al atom} = \frac{1}{2}[3 + 4 - 0 + 1] = 4 = \text{sp}^3$$

(b) For H_2O ,

Hybridisation of O atom

$$= \frac{1}{2}[6 + 2 - 0 + 0] = 4 = \text{sp}^3$$

For H_3O^+ ,

Hybridisation of O atom

$$= \frac{1}{2}[6 + 3 - 1 + 0] = 4 = \text{sp}^3$$

(c) For NH_3 ,

Hybridisation of N atom

$$= \frac{1}{2}[5 + 3 - 0 + 0] = 4 = sp^3$$

For NH_4^+ ,

Hybridisation of N atom

$$= \frac{1}{2}[5 + 4 - 1 + 0] = 4 = sp^3$$

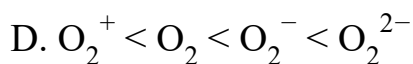
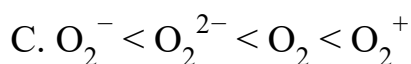
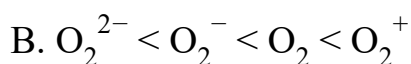
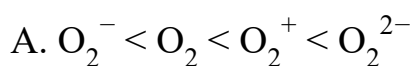
Thus hybridisation changes only in option (a).

Question 236

Which of the following are arranged in an increasing order of their bond strength?

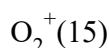
[2002]

Options:



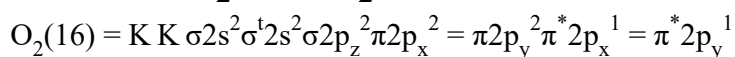
Answer: B

Solution:

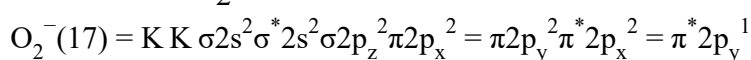


$$= KK \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1 = \pi 2p_y^0$$

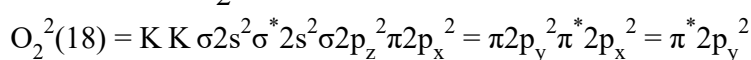
$$\text{Bond order} = \frac{1}{2}(8 - 3) = \frac{5}{2} = 2.5$$



$$\text{Bond order} = \frac{1}{2}(8 - 4) = 2$$



$$\text{Bond order} = \frac{1}{2}(8 - 5) = 1.5$$



$$\text{Bond order} = \frac{1}{2}(8 - 6) = 1$$

Note: As we know that as the bond order decreases, stability also decreases and hence the bond strength also decreases. Therefore the correct order of their increasing bond strength is

